GE VERNOVA

ADMS

16.25.1

Switching Operations User Guide

Change History

Date	Change Description
May 8, 2023	ADMS 3.16: • Updated Exporting a Switching Order. • Editorial and formatting updates.
Oct 20, 2023	ADMS 16.1.0: • The product version numbering has changed. The prefix '3' is no longer included in the version numbering for ADMS.
Nov 16, 2023	ADMS 16.2.0: • Updated Switching Orders. • Updated Switching Order Toolbar. • Added Attachments. • Updated Audit Trail. • Added Create a Hot Line Hold from the Line. • Updated Creating a Planned Outage from a Planned Request.
Apr 1, 2024	ADMS 16.7.0: • Updated document format and styling.
Jul 22, 2024	ADMS 16.10.0: • Updated Approving a Switching Order. • Updated Creating a Switching Order.
Dec 10, 2024	ADMS 16.15.0: • Added Creating a Planned Outage from a Switching Order. • Updated Switching Steps for Number of De-Energized Customers and Number of Energized Customers.
Apr 29, 2025	ADMS 16.17.0: • Added a statement in step 7 in Executing and Completing a Switching Order to reflect automatic completion of steps for manually controlled SCADA devices.

Contents

1. Introduction

Switching operations are the formal processes of performing switching changes on the network for the purposes of maintenance, network reconfiguration, restoration of unplanned outages, and augmentation of the distribution network.

The core of switching operations is the "switching order", a structured list of switching instructions that is used to plan, record, and coordinate the execution of the switching procedure.

The switching operations application supports the following types of documents:

- **Switching Order:** A formal document for common switching activities.
- **Daily Log:** The switching log, which is a step-by-step recorder of switching performed by each operator during a shift.
- **Hot Line Hold Document:** A document for hot line hold operations.
- **Planned Request:** A document for planned switching operations.
- **Template:** A pre-defined template that can be used for creating switching order documents.

Once a section of the network has been switched out of service, the jurisdiction for that section is passed from the operations center to the field crew.

The formal document that records the transfer of jurisdiction is the "safety document", also known as a "clearance permit" or "access permit". Before a network section can be switched back into service, the field crew must release the safety document to transfer jurisdiction back to the operation center.

The switching operations application for the manual creation and execution of switching orders is part of the standard ADMS network view user interface. Switching operations are also integrated with the Distribution Network Analysis Functions (DNAF) to automatically generate switching orders from the output of planned outage studies and fault isolation and restoration plans. Safety documents can also be created, issued, and released as part of the standard switching operations functionality.

The functionality provided by the switching operations functions and the appearance of the switching operation client can be modified using the Switching Order Server Configuration Editor and Switching Order Client Configuration Editor applications.

2. Displays Overview

This chapter describes all the switching order application's displays.

2.1. Switching Orders

The Switching Order form is used for switching orders.

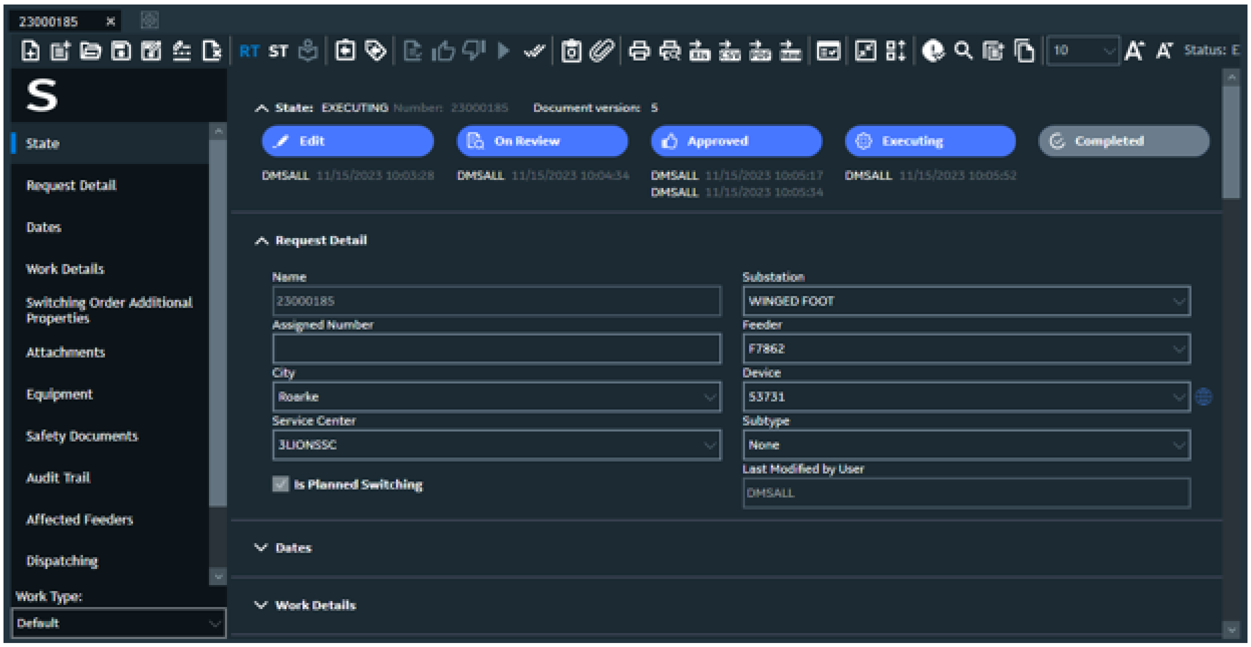


Figure 1. Switching Order Form

The Switching Order form consists of the following sections:

- **State:** Shows the state and the validation and authorization process of the switching order.
- **Request Detail:** Contains information about the switching order.
- **Dates:** Shows the scheduled and actual start and end dates.
- **Work Details:** Shows work request details.
- **Switching Order Additional Properties:** Contains custom fields, which are configurable.
- **Attachments:** Shows the operator's attached documents.
- **Equipment:** Provides details for the associated devices.
- **Safety Documents:** Provides details for the associated safety documents.
- **Audit Trail:** Logs the operator's actions for this switching order.
- **Affected Feeders:** Shows stations and feeders that are associated with switching steps.
- **Dispatching:** Shows information for dispatching a document to field clients.
- **Step Blocks Overview:** Provides details for the switching step blocks.

- **Switching Steps:** Contains switching steps that are associated with the document.

The switching order form has a toolbar to control the required processes.

The switching order form display is tabbed so that multiple switching orders can be handled concurrently. The tabs along the top of the form show the name of each currently loaded switching order.

At the bottom of the form is the Switching Order server's status indicator. The real-time or study server should always be connected. If the current operating mode server is not connected, a warning message appears, the switching order functions are not operating, and the system administrator should be notified.

2.1.1. Switching Order Toolbar

The toolbar at the top of the switching order form contains the buttons for the overall control of the switching order (see the below image).

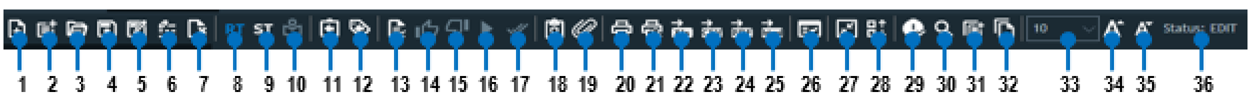


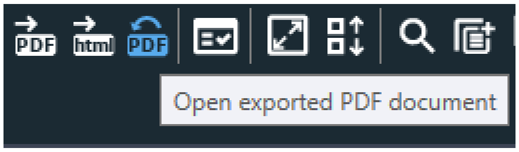
Figure 2. Switching Order Toolbar

Switching Order Toolbar Key for the above image					
1.	Create New Switching Order	2.	Create New Template	3.	Open Switching Order
4.	Save Switching Order	5.	Save Switching Order Under New Name	6.	Rename Switching Order
7.	Delete Switching Order	8.	Switch to RT (Real-Time) Mode	9.	Switch to ST (Study) Mode
10.	Initialize Study Environment	11.	Open New Return to Normal Switching Order	12.	Open New Switching Order with Clearance Steps
13.	Validate Switching Order	14.	Approve Switching Order	15.	Reject Switching Order
16.	Execute Switching Order	17.	Complete Switching Order	18.	Create Safety Document from Switching Order
19.	Attach Document	20.	Print	21.	Print Preview
22.	Export Switching Order to Excel	23.	Export Switching Order to CSV	24.	Export Switching Order to PDF
25.	Export Switching Order to HTML	26.	Online Validation	27.	Expand/Collapse Header Information
28.	Expand/Collapse Steps Grid View	29.	Check SWO Conflict	30.	Search Switching Orders
31.	Create Template From Switching Order	32.	Create a Copy of Switching Order	33.	Change the Font Size
34.	Increase the Font Size	35.	Decrease the Font Size	36.	Document Status

The switching order can be exported using default or custom templates to CSV, PDF, HTML, and

Excel formats. For more information about creating and configuring custom templates, refer to the *Switching Order Client Configuration Editor User Guide*.

When you export a switching order to PDF, the Open Exported PDF Document button appears on the toolbar. Clicking this button opens the recently exported PDF document.



When the interface to the Workforce Management System (WFM) is enabled, the Send to WFM button appears on the toolbar. Clicking this button sends a document to an external URL and receives the new work request ID from the WFM. For more information about configuring the interface to the WFM, refer to the *Switching Order Client Configuration Editor User Guide*.

Buttons for enabling and disabling the Switching Order recording mode also appear on the Study Mode toolbar.

2.1.2. Switching Order Context Menu

To access the switching order context menu, right-click anywhere on the switching order properties. The following options are available:

- **Copy Selected Steps:** Copies all selected steps in the Switching Steps pane.
- **Copy All Steps:** Copies all steps in the Switching Steps pane.
- **Paste Steps:** Paste steps in the Switching Steps pane.
- **Copy/Paste Uncompleted Steps to Daily Log:** Copies uncompleted steps from the Switching Steps pane and pastes them into the Daily Log.

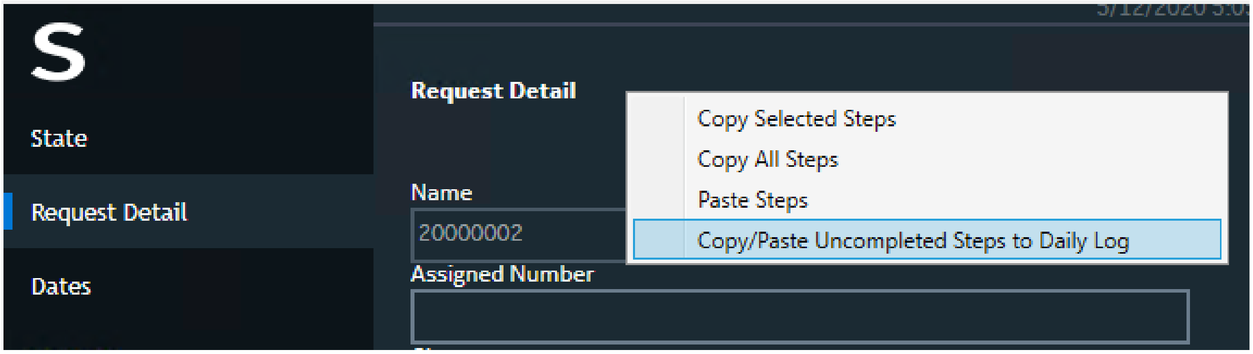


Figure 3. Switching Order Context Menu

2.1.3. Switching Order Sections

2.1.3.1. State

The Switching Order State section displays the status of the switching order as it moves through its life cycle.

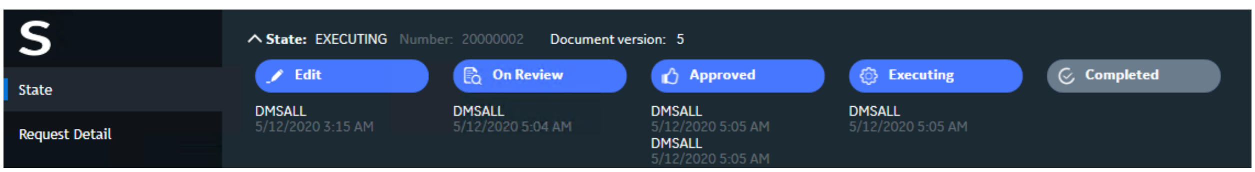


Figure 4. Switching Order State

This section displays the past and current states of the switching order, timestamps, and the names of users who performed an action for each state.

The status field at the right of the Switching Order toolbar displays the current state of the switching order. The states of the switching order are:

CREATED	The new blank switching order is created.
VALIDATED	The switching order has been saved and changes are not permitted.
APPROVED	The switching order has been approved and is awaiting execution.
EXECUTING	The switching order steps are being executed.
COMPLETED	The steps of the switching order have been completed and the action times have all been entered. No further changes to any information in the switching order are permitted.

You can validate, reject, approve, execute, and complete the switching order using the buttons on the Switching Order toolbar. You can also update the document state using the associated button at the bottom of the switching order form.

> **Note**
>
> The switching order workflow can be configured to exclude the review and approval stages,

> that is, skip the Validate and Approve statuses and hide the related fields and controls in the Switching Order form. For more details, refer to the *Switching Order Server Configuration Editor User Guide*.

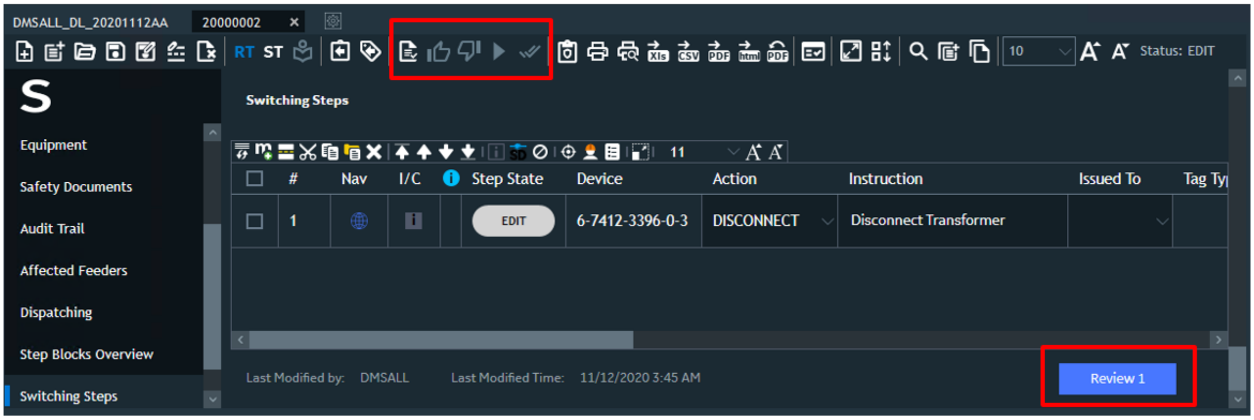


Figure 5. Updating the Switching Order State

2.1.3.2. Request Detail

The Request Detail section describes the switching order details.

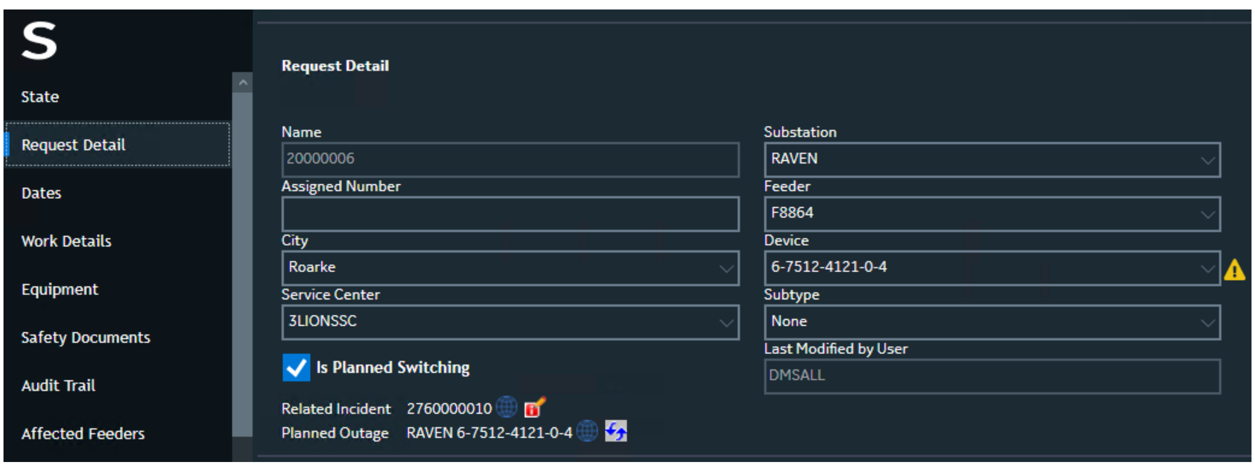


Figure 6. Switching Order Request Detail

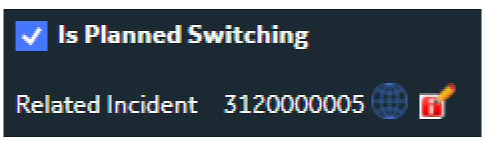
The fields with a gray background are updated automatically and cannot be changed. The fields with a white background must be entered manually.

- **Name:** The unique name of the switching order.

  If the Name field is blank, it is automatically populated with the generated number when the switching order is saved or validated. The name can be changed using the Rename button on the Switching Order toolbar.

- **Assigned Number:** A custom number for the switching order. The operator manually enters this number.

- **City:** The name of the city associated with the device affected by the switching operation.
- **Service Center:** The name of the service center associated with the device affected by the switching operation.
- **Substation:** The name of the substation associated with the device affected by the switching operation.
- **Feeder:** The name of an associated feeder.
- **Device:** The name of a device that is associated with the the switching order.
- **Subtype:** Switching order subtype (for example, MV or LV). Subtypes are defined by the SwitchOrderSubtype coded translation item.
- **Is Planned Switching:** When selected, the switching order is considered to be planned. When not selected, the switching order is considered to be unplanned.
- **Last Modified by User:** Name of the user who made the latest update.
- **Related Incident/Planned Outage:** Shows the associated incident or planned outage information. This section only appears if a switching order is associated with an incident or planned outage.



  - **Name:** The name of the incident/planned outage that is associated with the switching other.
  - **Locate Button:** Locates the device of record on the network view.
  - **Update Incident/Planned Outage Button:** Opens the Update Incident or Update Planned Outage display.
  - **Planned Outage is Cancelled Bitmap:** The Planned Outage is Cancelled bitmap appears for a switching order that is associated with a cancelled planned outage.

> **Note**
>
> A switching order can be associated with only one incident or planned outage.

You can remove association between a switching order and an incident on the Switching tab of the Update Incident display in the OMS Display Plugin. The association is also automatically removed when the switching order is deleted.

The system shows a prompt to remove a switching order when cancelling or removing the associated planned outage (for example, from the Planned Outages tab of the OMS Display Plugin).

- **Related Planned Request:** Shows the associated planned request information. This section only appears if a switching order is created from a planned request.

  - **Name:** The name of the planned request that is associated with the switching order.
  - **Open Button:** Opens the planned request document.

  The association between a switching order and a planned request is automatically removed when the switching order is deleted.

2.1.3.3. Dates

The Dates section shows the document scheduled and actual start and end dates.

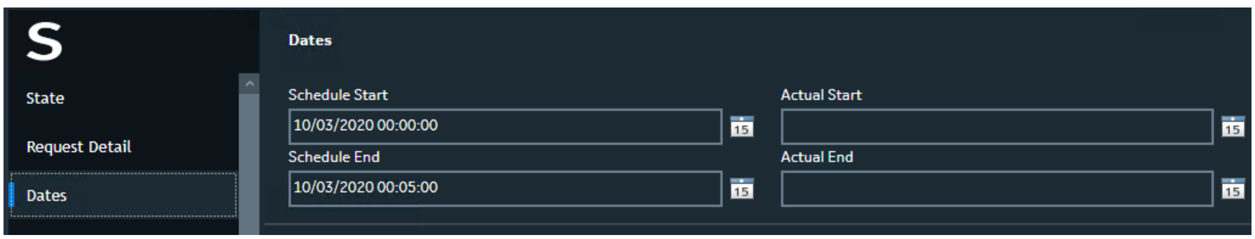
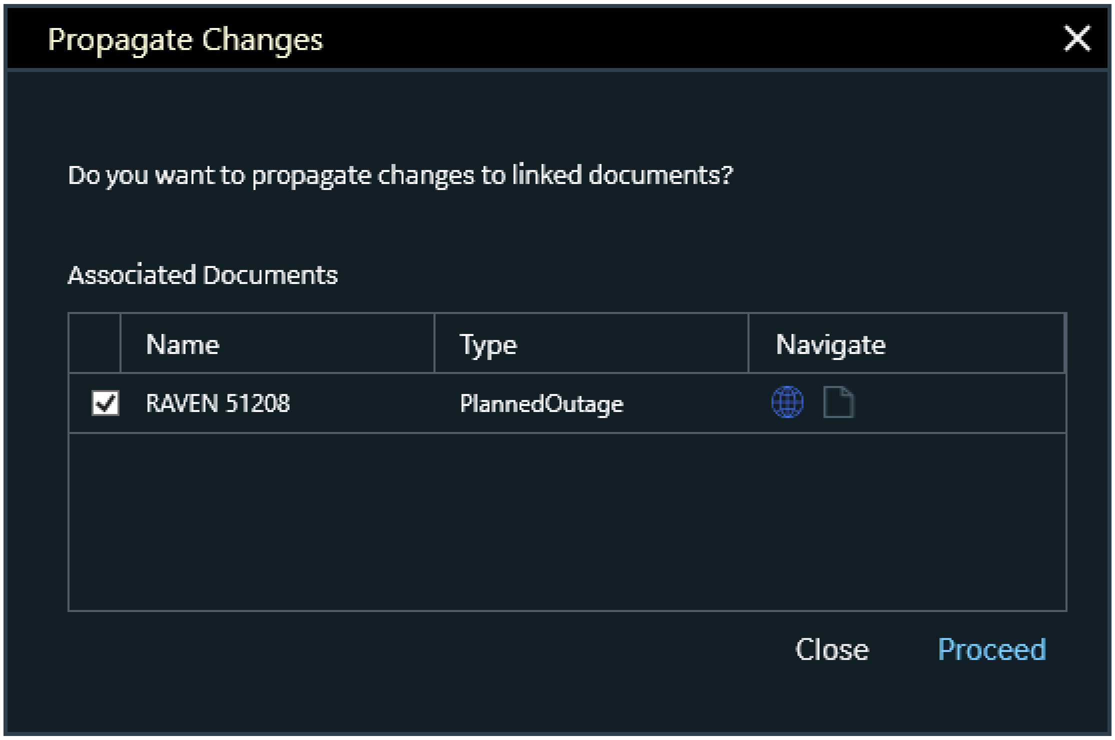


Figure 7. Switching Order – Dates Section

The Dates section includes the following fields:

- **Scheduled Start/End:** Scheduled start and end dates for the planned switching operations.
- **Actual Start/End:** Actual start and end dates for the switching operations.

When you update start and end dates for a switching order that is associated with a planned outage, the system prompts you whether you want to propagate the date changes to the planned outage.



2.1.3.4. Work Details

The Work Details section provides details for the associated work request.

Figure 8. Switching Order - Work Details Section

The Work Details section includes the following fields:

- **Work Request:** The switching order work request ID.

  The work request ID is generated automatically when saving a document. When sending a document to the Workforce Management System (WFM), the work request ID is updated with the ID generated by the WFM.

- **Search Keywords:** Keywords that filter switching orders on the Switching Order Search form.
- **Description:** Comments about the switching operation.

2.1.3.5. Attachments

The Attachments section shows the documents attached by operators. Examples of attachment types include pictures, PDF, HTML, Word, Excel, and other types of standard formats.

Figure 9. Switching Order — Attachments Section

The Attachments section includes the following fields:

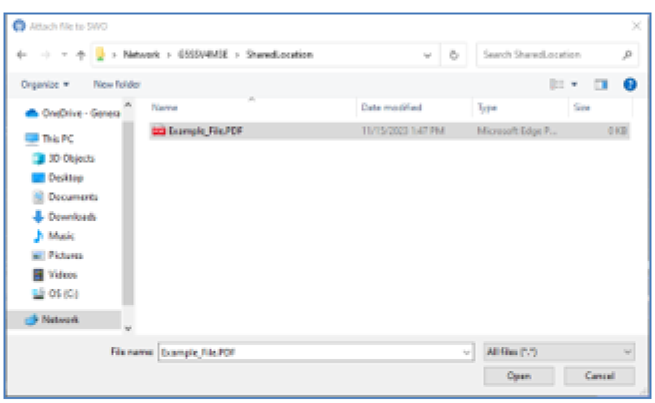
- **Document Name:** The name of the attached document.
- **Actions:**
  - **Open:** The document is opened with the default windows application installed on the system.
  - **Remove:** The document is removed from the Attachments section list.

> **Note**
>
> When removing an attachment, the actual file is not deleted from the source location.

To add documents to the attachments list, do the following:

> **Note**
>
> Operators are responsible for ensuring that the attached file is present in a shared location and accessible to all clients.

- Click on the Attach Document button on the Switching Order Toolbar to open the Attach file to SWO (windows explorer) dialog.

  Navigate to a shared location (previously set on your system). Select the file and click Open to add the file to the Attachments section list.

2.1.3.6. Equipment

The Equipment section provides details for the associated isolating devices and temporary grounds.

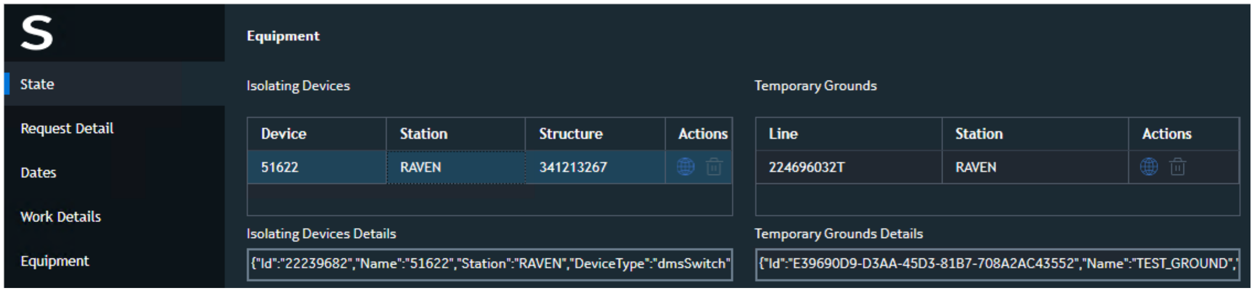


Figure 10. Switching Order – Equipment Section

The Equipment section includes the following fields:

- **Isolating Devices:** The name, station, structure ID, and device details for isolating devices.
- **Temporary Grounds:** The line name, station, and device details for temporary grounds.

To add switching devices to the list of isolating devices, do one of the following:

- Drag and drop the required switches from the network view into the list.
- Add the required switches using the Add Isolation Switch to Switch Order option from the switching device context menu on the network view.
- Add the required switches using the Add Isolation Switch or Add Selected as Isolation Switches options from the Switching Steps section context menu.

To add lines to the list of temporary grounds devices, do the following:

- Drag and drop the required lines from the network view into the list.
- Add the required lines using the Add to Switch Order as Temporary Ground option from the line context menu on the network view.

To add device details to the Isolating Device Details and Temporary Ground Details fields, do one of the following:

- Drag and drop the required objects from the network view into the detail fields.
- Add the required switches using the Add Isolation Switch context menu option from the Switching Steps section.

2.1.3.7. Safety Documents

> **Note**

The Safety Documents functionality requires integration with the Safety Document application.

The Safety Documents section provides details for the associated safety documents.

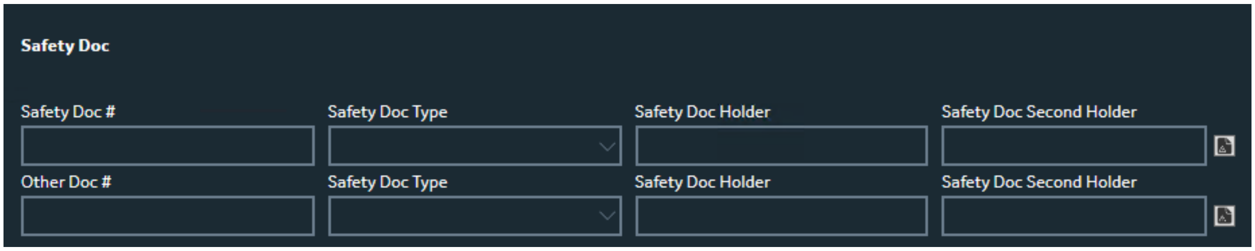
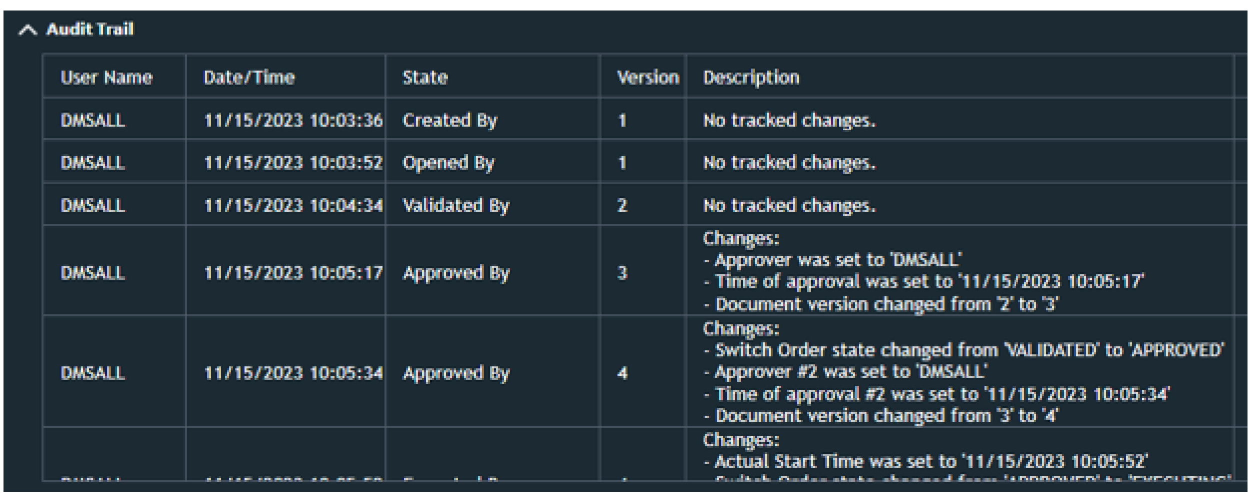


Figure 11. Switching Order - Safety Documents Section

2.1.3.8. Audit Trail

The Audit Trail section logs the operator’s and crew’s actions for this switching order. When the switching order is opened, updated, validated, rejected, approved, executed, or completed, the relevant record is added, including the operator’s name, date and time, action name, document version, and a list of changes.

The document version number is incremented each time the document properties or switching steps are modified or the state is changed (except for when transitioning from Approved to Executing without changing document properties so that crews in-field have the approved document version).


Audit Trail

User Name	Date/Time	State	Version	Description
DMSALL	11/15/2023 10:03:36	Created By	1	No tracked changes.
DMSALL	11/15/2023 10:03:52	Opened By	1	No tracked changes.
DMSALL	11/15/2023 10:04:34	Validated By	2	No tracked changes.
DMSALL	11/15/2023 10:05:17	Approved By	3	Changes: - Approver was set to 'DMSALL' - Time of approval was set to '11/15/2023 10:05:17' - Document version changed from '2' to '3'
DMSALL	11/15/2023 10:05:34	Approved By	4	Changes: - Switch Order state changed from 'VALIDATED' to 'APPROVED' - Approver #2 was set to 'DMSALL' - Time of approval #2 was set to '11/15/2023 10:05:34' - Document version changed from '3' to '4'
				Changes: - Actual Start Time was set to '11/15/2023 10:05:52'

Figure 12. Switching Order Audit Trail

2.1.3.9. Affected Feeders

The Affected Feeders section shows substations and feeders that are associated with devices in the Switching Steps section.

Note

For station internal devices above the feeder head breakers, the Feeder field shows All.

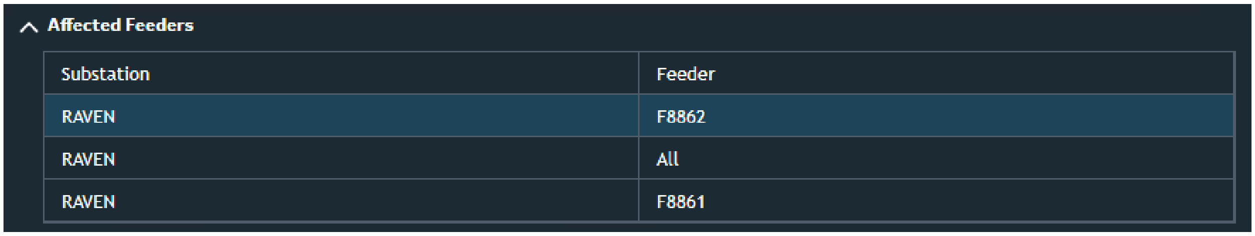


Figure 13. Switching Order Affected Feeders

2.1.3.10. Step Blocks Overview

The Step Blocks Overview section provides details for the switching step blocks. You can group associated switching steps into blocks and specify block dependencies.

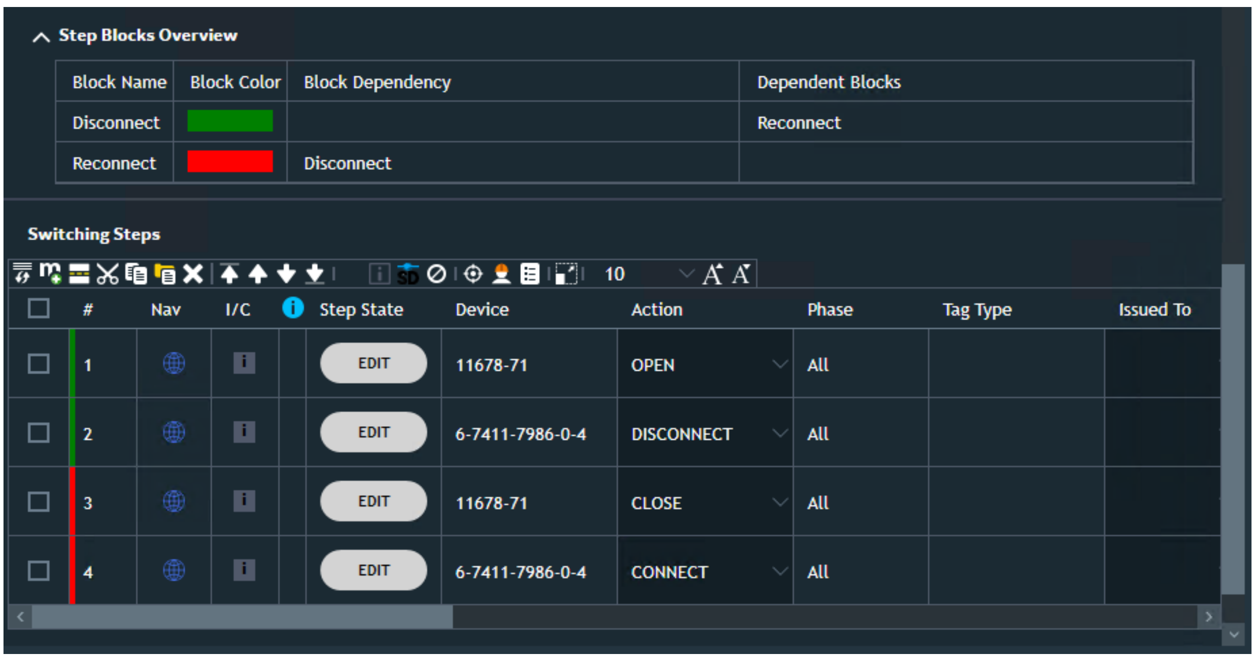


Figure 14. Switching Order Step Blocks Overview

The Step Blocks Overview section includes the following fields:

- **Block Name:** Step block name.
- **Block Color:** Step block color.
- **Block Dependency:** Step block dependency.
- **Dependent Blocks:** Dependent blocks.

2.1.3.11. Switching Steps

The Switching Steps section contains the switching steps of the current switching order.

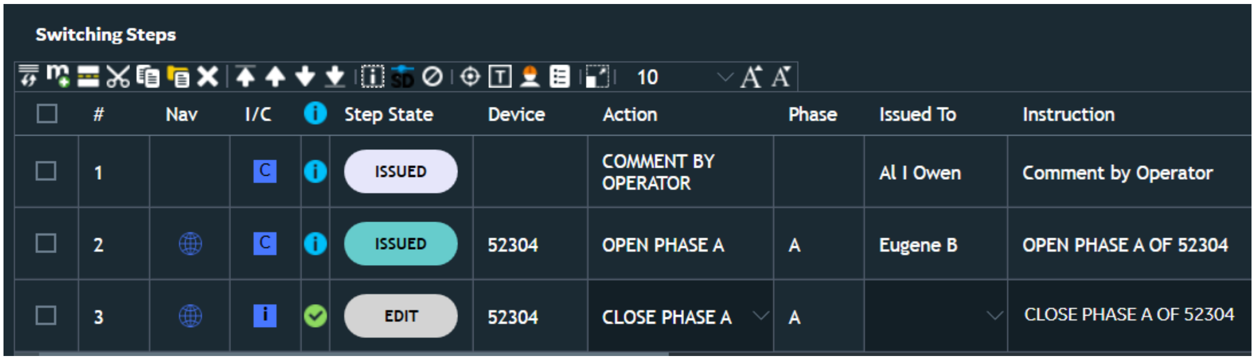


Figure 15. Switching Steps

Switching steps can be added to the switching order at any time until the switching order is Completed.

> **Note**
>
> Switching steps that are added to a switching order in the Validated, Approved, or Executing state are numbered as sub-numbers. To disable the sub-numbering, set the UseOldStyleStepNumbering parameter to true in `SwitchOrderClient.Config.xml`.

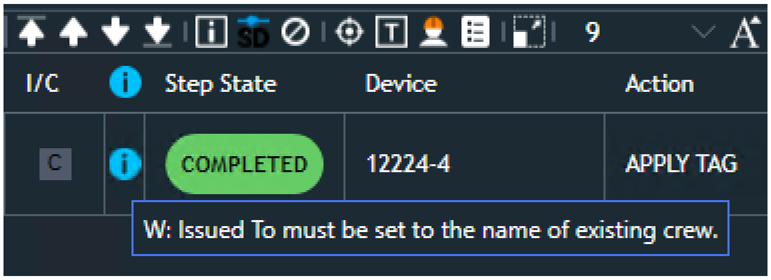
The Step State field in the switching steps grid shows the current state of the switching step. The switching step states are:

EDIT	The new switching step is created.
ISSUED	The switching step has been issued and is waiting for execution. Issued steps cannot be modified or deleted.
COMPLETED	The switching step has been implemented and completed. Completed steps cannot be modified, deleted, or cancelled.
CANCELLED	The switching step has been cancelled. Cancelled steps cannot be modified or deleted.
HISTORICAL	The switching step had been implemented in the past. Historical steps are added when creating a switching order from an incident. Historical steps cannot be modified, deleted, or cancelled.

The switching steps grid contains the following fields:

- **Select:** Allows you to select the switching step. You can select more than one switching step.
- **Step:** The step's number.
- **Nav:** Navigates to the device's location on the geographic view, or on the SCADA automated view, or the URD loop schematic view.
- **Issue:** Issues the step to the employee specified in the Issued To field. When a step is issued, the Issue icon tooltip shows warnings associated with the step issuance, if any.

- **Complete:** Marks the switching step as completed if the device's state and action match. The button tooltip also indicates if the switching step is dispatched to a mobile crew for completion.
- **Validation:** Step validation status (green is successfully validated, blue is validated with warnings, red is validated with errors). Hover over the validation bitmap to see validation messages.



- **Step State:** The current step state (for example, Edit or Issued). The Step State background color is different for different step states. You can configure the Step State field background and text colors using the Switching Orders Client Configuration Editor.
- **Device:** The device's name.
- **Action:** The switching step's action. The available actions depend on the device type.
- **Phase:** The phase to be worked on in each switching step. This phase depends on the selected switching step's action.
- **Tag Type:** The tag type to be applied by the switching step. Available only for tagging steps.
- **Issued To:** Name of the employee assigned to the switching step.
- **Issued Time:** Date and time when the step was issued.

> **Note**
>
> The Issued Time of a step that was issued during the "extra hour" (an hour before the clock is adjusted backwards when Daylight Saving Time is changed to standard time) is marked with an asterisk symbol ("*"). In this case, the Completed Time may be "before" the Issued Time.

- **Completed Time:** Date and time when the step was marked as completed.

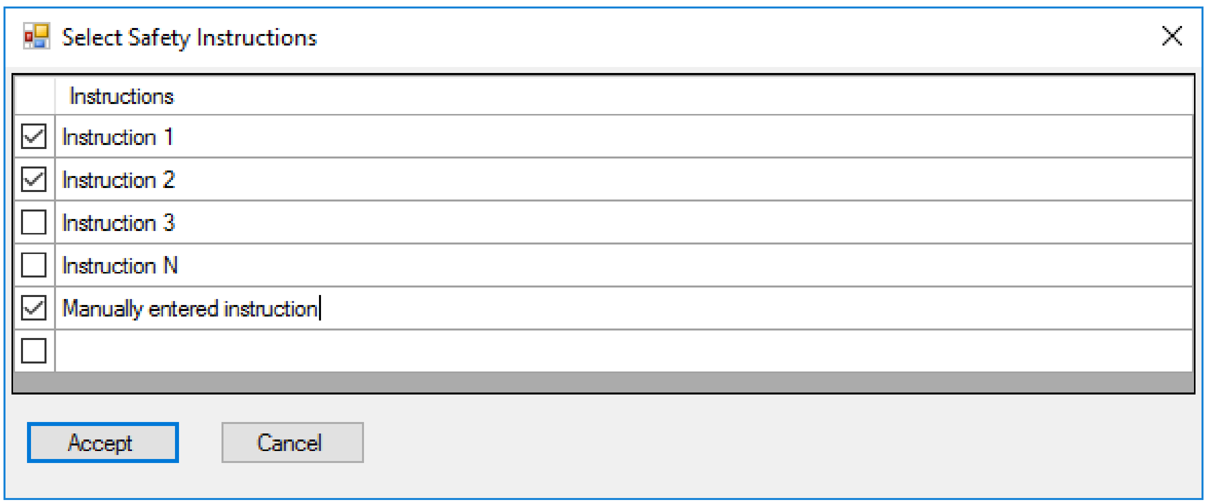
> **Note**
>
> When the Completed Time of a step is within the "extra hour", it is marked with an asterisk symbol ("*").

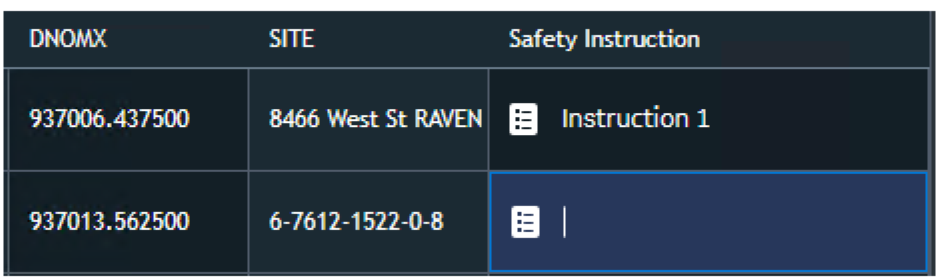
- **Instruction:** The instruction to the switching step. The instruction text is populated automatically according to the action selected, but it can be edited if required. To enter multi-line text, use the "Shift + Enter" key combination.

> **Tip**

To prevent an instruction text from being overridden, enclose the text in square brackets (for example, "[Text to keep]". This text will be kept in the instruction when you select different actions.

- **Comments:** Text field containing entered comments.
- **Issued By:** Name of the user who issued the switching step.
- **DNOM X:** X coordinate of the associated device.
- **Site:** Alternate ID or address of the associated device.
- **Safety Instruction:** Text field containing safety instructions. To enter multi-line text, use the "Shift + Enter" key combination. You can also select instructions from a pre-defined list. To open the Select Safety Instructions display, click the Set Switching Instructions button on the Switching Steps toolbar.





- **Scheduled Time:** The planned date and time for executing a specific switching step. This is used for planned switching when the execution time is set in advance.
- **Number of De-Energized Customers:** Displays the number of customers de-energized when a specific action is performed in the switching steps. In study mode, the step determines the count of de-energized customers. The column shows the count, and the customers keys are saved to the switching order.
- **Number of Energized Customers:** Displays the number of customers energized when a specific action is performed in the switching steps. In study mode, the step determines the count of energized customers. The column shows the count, and the customers keys are saved

to the switching order.

> **Note**
>
> The Number of De-Energized Customers and Number of Energized Customers columns are visible only when Planned Outage creation is enabled from the switch order.

Custom columns can be defined in the Switching Order Client Configuration Editor.

For more details about creating a switching order and performing operations with switching steps, see General Usage.

2.1.3.11.1. Switching Steps Toolbar

The toolbar at the top of the switching steps section contains the buttons to control the switching steps (see the below image).

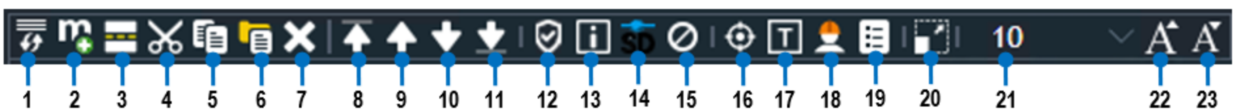


Figure 16. Switching Steps Toolbar

Switching Steps Toolbar Key for the above image			
1.	Add Return to Normal Steps	2.	Add Manual Step
3.	Add/Remove Step Separator	4.	Cut Selected Steps
5.	Copy Selected Steps	6.	Paste Copied Steps
7.	Delete Step	8.	Move Current Step to Top
9.	Move Current Step Up	10.	Move Current Step Down
11.	Move Current Step to Bottom	12.	Check Before Operate
13.	Issue Selected Steps (enabled in the Executing state only)	14.	Add Selected Steps to Safety Document
15.	Cancel Steps in Execute Mode	16.	Mark Selected Devices on Network View
17.	Track/Untrack Devices on Network View (appears in the Executing state only)	18.	Employee Info Pop-Up
19.	Set Safety Instructions for Selected Steps	20.	Expand/Collapse Steps Grid
21.	Change the Font Size	22.	Increase the Font Size
23.	Decrease the Font Size		

In study mode, the Switching Steps toolbar differs from the one in the real-time environment.

Figure 17. Switching Steps Toolbar in Study Mode

Switching Steps Toolbar in Study Mode Key for the above image			
1.	Add Return to Normal Steps	2.	Add Manual Step
3.	Add/Remove Step Separator	4.	Cut Selected Steps
5.	Copy Selected Steps	6.	Paste Copied Steps
7.	Delete Step	8.	Move Current Step to Top
9.	Move Current Step Up	10.	Move Current Step Down
11.	Move Current Step to the Bottom	12.	Check Before Operate
13.	Implement Selected Steps	14.	Clear Implementing Steps in ST Mode
15.	Add Selected Steps to Safety Document	16.	Cancel Steps in Execute Mode
17.	Track/Untrack Devices on Network View (appears in the Executing state only)	18.	Employee Info Pop-Up
19.	Set Switching Instructions	20.	Expand/Collapse Steps Grid
21.	Change the Font Size	22.	Decrease the Font Size
23.	Increase the Font Size		

2.1.3.11.2. Switching Steps Context Menu

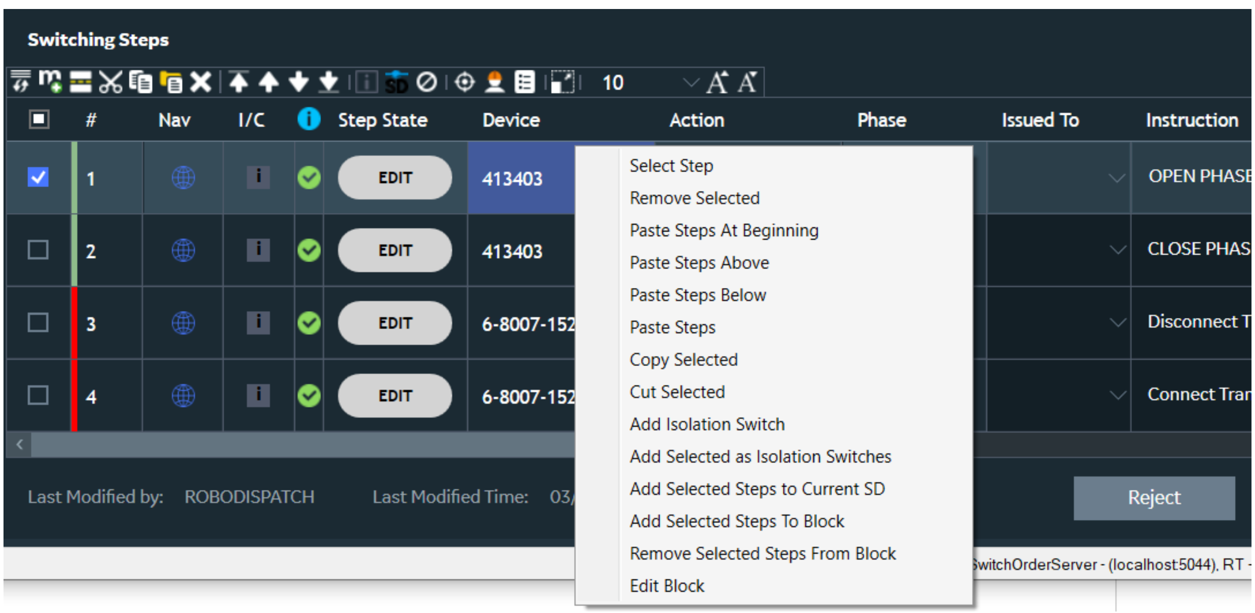


Figure 18. Switching Steps Context Menu

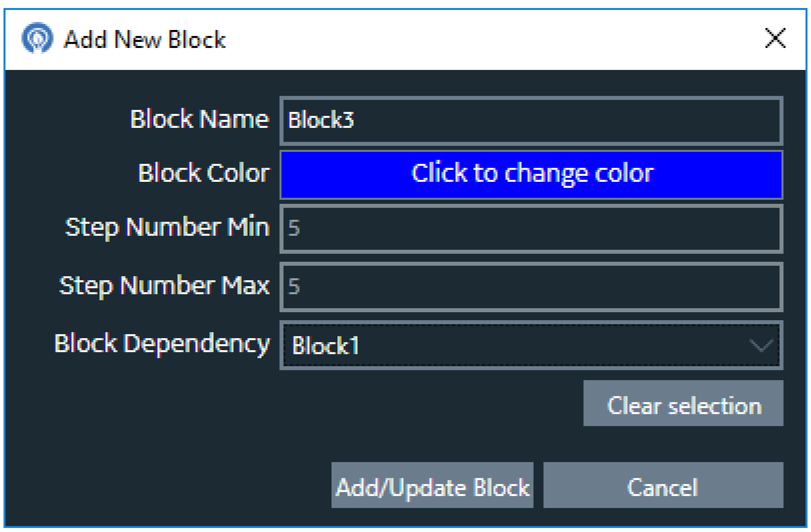
> **Note**
>
> Certain context menu options appear only when steps are selected in the list.

To access the switching step context menu, right-click a switching step. The following options are available:

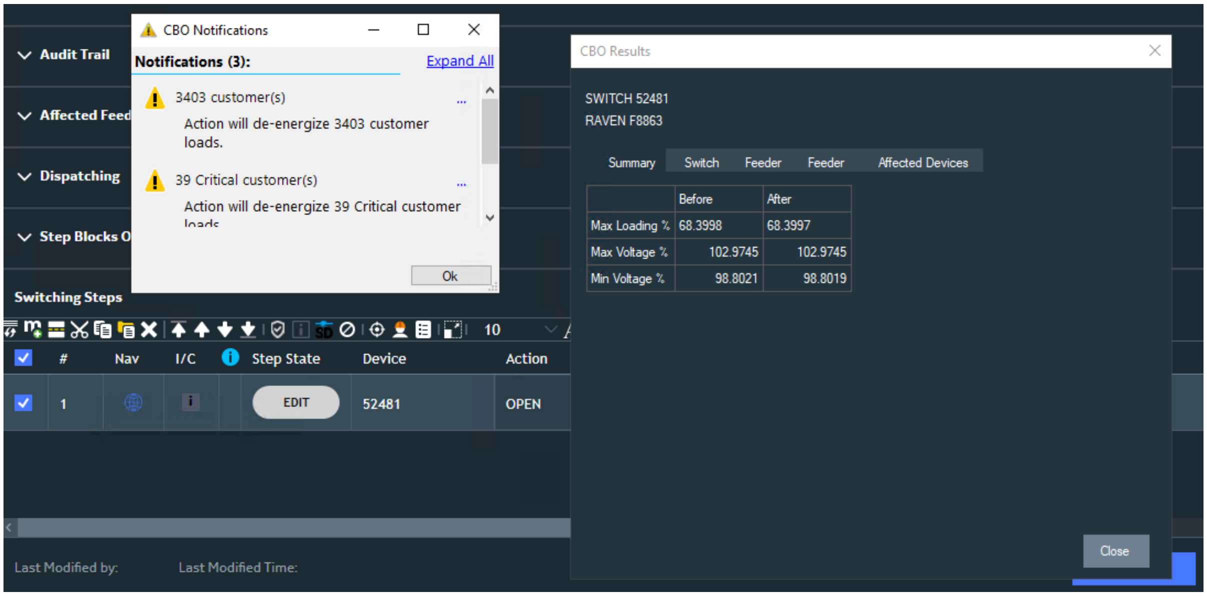
-

Select Step: Selects or deselects the step.

- **Remove Selected:** Removes the selected steps from the switching order.
- **Paste Steps at Beginning:** Pastes the copied steps at the top of switching steps list.
- **Paste Steps Above:** Pastes the copied steps above the selected step.
- **Paste Steps Above:** Pastes the copied steps below the selected step.
- **Paste Steps:** Pastes the copied steps.
- **Copy Selected:** Copies the selected steps.
- **Cut Selected:** Cuts the selected steps.
- **Add Isolation Switch:** Add the selected switch to the list of isolation switches in the Equipment section.
- **Add Selected as Isolation Switches:** Add the selected switches to the list of isolation switches in the Equipment section.
- **Add Selected Steps to Current SD:** Adds the selected steps to the safety document.
- **Add Selected Steps to Block:** Opens the Add New Block dialog box to create a block of switching steps. The switching step blocks appear in the Step Block Overview section. Step blocks in the switching steps list are marked with the selected color.



- **Remove Selected Steps from Block:** Removes the selected steps from the associated step block.
- **Edit Block:** Opens the Edit Block dialog box to edit the associated step block.
- **Check Before Operate:** Performs the Check Before Operate (switching validation) for the switching step. For more information, refer to the *Distribution Management System User Guide*.



2.1.3.12. Work Type

Use the Work Type field at the bottom of the Switching Order control panel to select a work type for the switching order document. This applies the configured workflow for the selected combination of a switching order document type and work type. For example, you may have different name pattern, reject behavior, and number of reviewers for switching order documents of the Clearance/Isolation and Hold work types. For more details, refer to the *Switching Order Server Configuration Editor User Guide*.

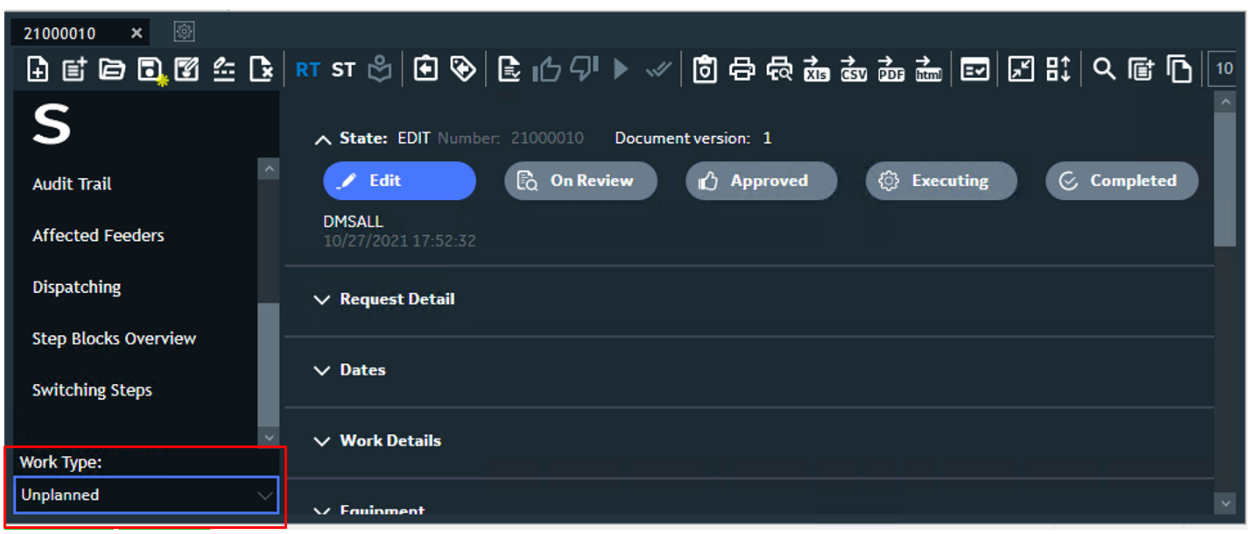


Figure 19. Switching Order Document Work Type

2.1.3.13. Switching Order Additional Properties

You can configure the Switching Order Additional Properties section to show custom text fields, dropdown menus, and checkboxes. If configured, this section appears under the Work Details section. For more information about configuring the Switching Order Additional Properties section, refer to the *Switching Order Server Configuration Editor User Guide*.

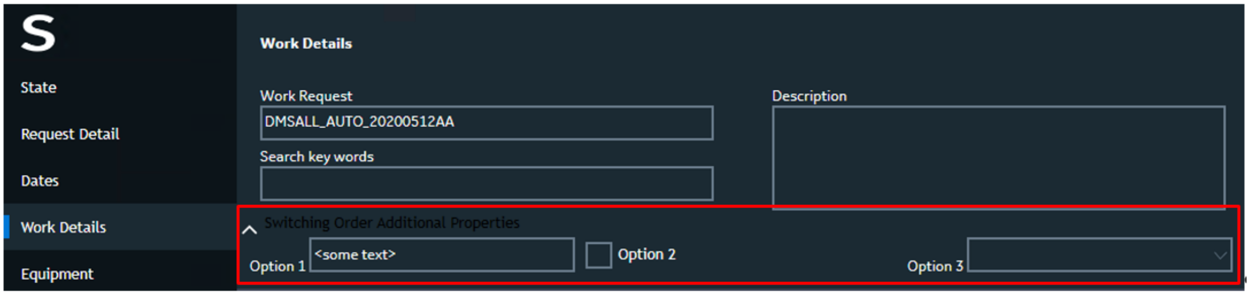


Figure 20. Switching Order Additional Properties

2.1.3.14. Dispatching

The Dispatching section allows you to dispatch switching orders to field crews that are equipped with the Mobile application. For more information, see Dispatching a Switching Order.

This section visibility is configurable. For more information, refer to the *Switching Order Client Configuration Editor User Guide*.

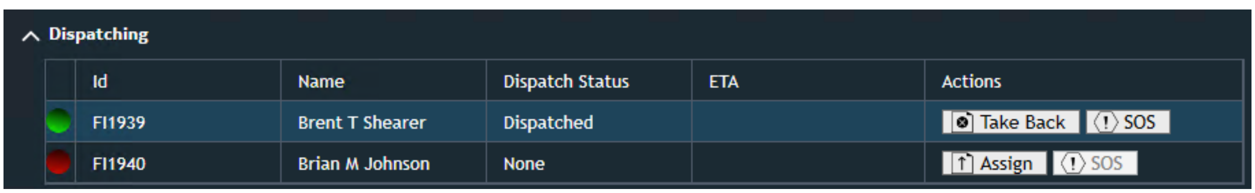


Figure 21. Switching Order Dispatching

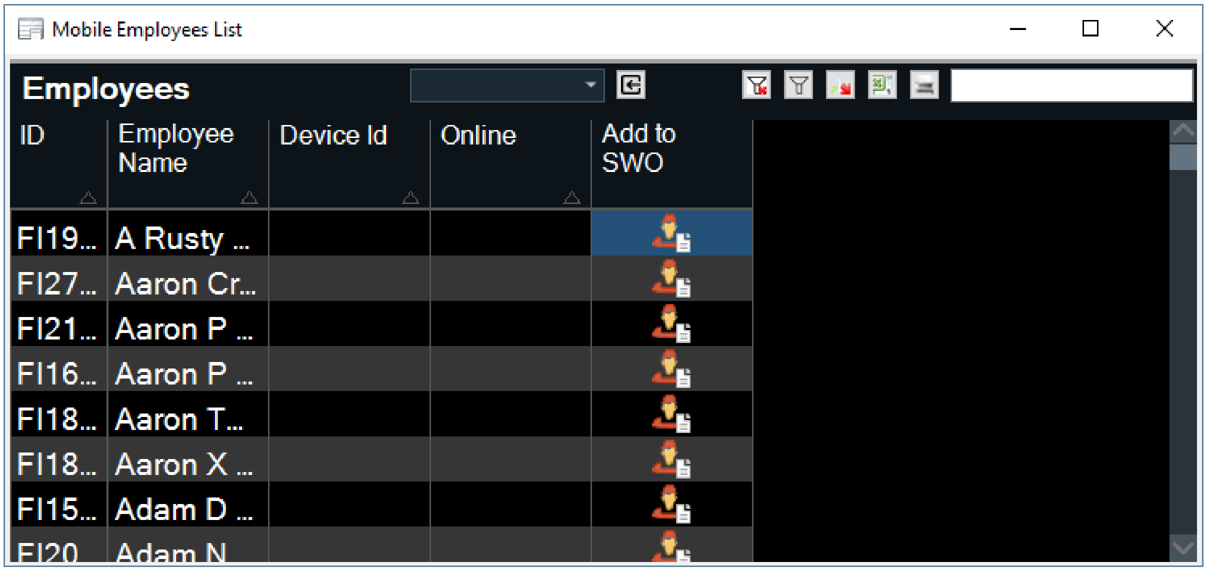
The Dispatching section includes the following fields:

- **Logon Indicator:** The crew logon indicator (for example, Green - Logged On and Red - Logged Off).
- **ID:** ID of the crew.
- **Name:** Name of the crew.
- **Dispatch Status:** Crew dispatching and completing status.
- **ETA:** Crew's estimated time of arrival (ETA).
- **Actions:** The dispatcher actions:
  - **Dispatch:** Assigns a switching order to a crew.
  - **Takeback:** De-assigns a switching order from a crew.
  - **SOS:** Issues the SOS signal for a crew to immediately stop working on the switching order.

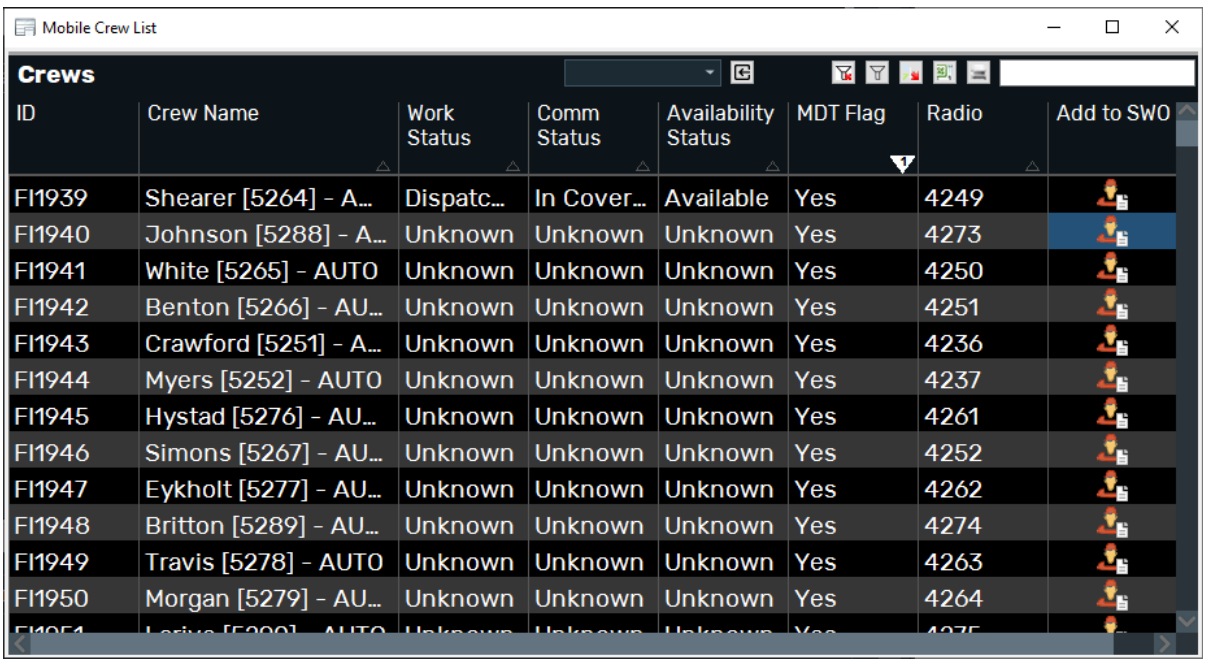
The Dispatching section context menu includes the following options:

-

Add New Employee: Opens the Mobile Employees List display to add an employee to the Dispatch Employee list.



- **Add New Crew:** Opens the Mobile Crew List display to add a crew to the Dispatch Employee list.



- **Remove Selected:** Removes the selected crew or employee from the dispatching list.

2.1.3.15. Email Notifications

The Email Notifications section allows you to specify the recipients for notifications about switching order validation, approval, execution, rejection, completion, and deletion events.

The visibility of this section is configurable. For more information, refer to the *Switching Order Server Configuration Editor User Guide*.

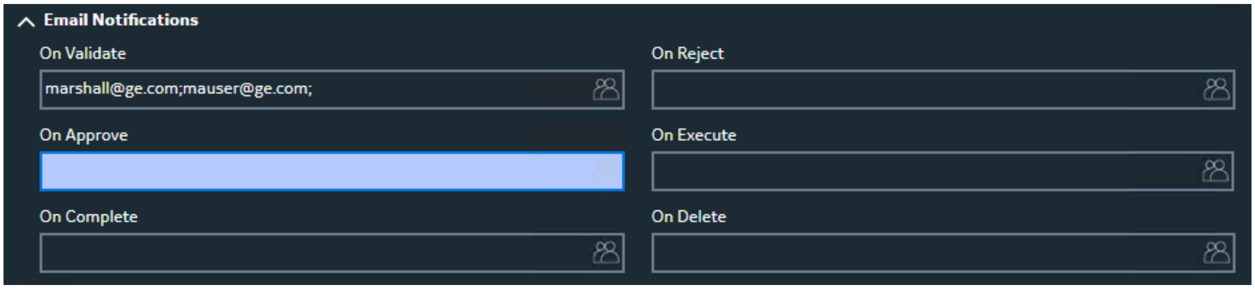
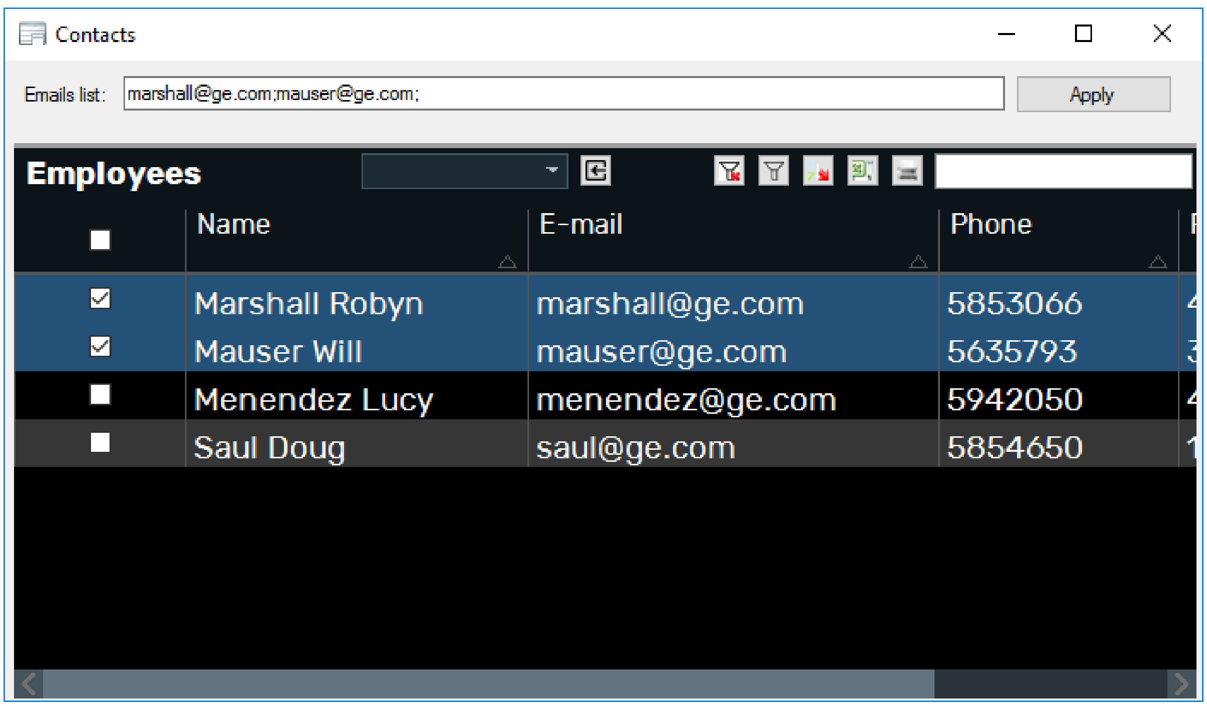


Figure 22. Email Notifications

You can specify the recipients' email addresses as follows:

- Manually enter email addresses or contact names separated by semicolons in the recipient field.
- Click the Contacts button for the required event and then select the recipients using the contacts list.



2.2. Unplanned Switching Operations - Daily Log

The daily log (below image) executes and records unplanned switching operations during an operator’s shift. It allows the operator to maintain a log of all switching operations over the period of a shift. Switching steps can be entered into the log and executed on an adhoc basis.

Note

> Daily log can be edited only by the operator who created it.

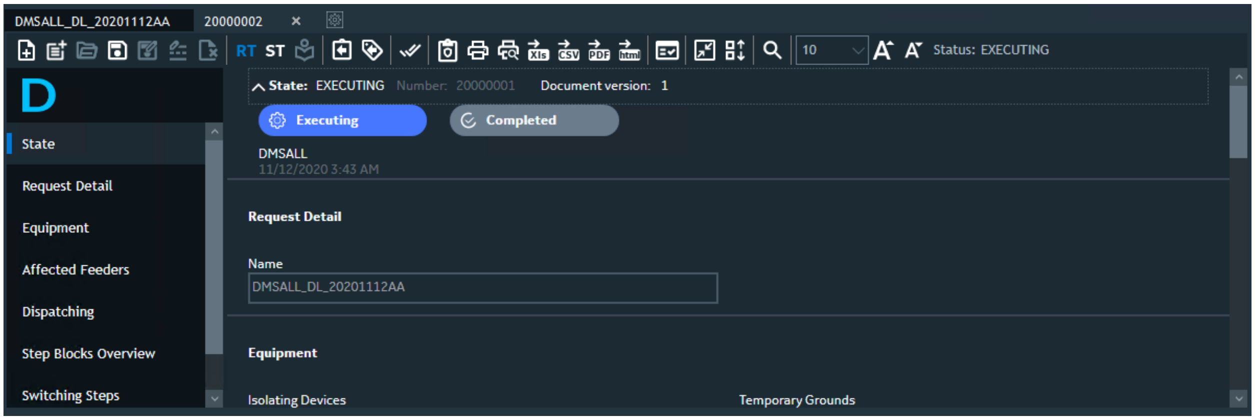


Figure 23. Daily Log

The daily log is accessed from a tab on the switching order form, allowing both methods of switching operations to be used concurrently.

The daily log is automatically created with the name of the user when the first switching step is entered. All subsequent adhoc switching steps are added to the switching log until the end of the shift, when the user completes the daily log and logs off. The below image shows the Daily Log toolbar.

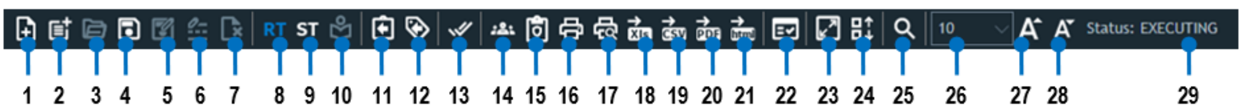


Figure 24. Daily Log Toolbar

Daily Log Toolbar Key for the above image					
1.	Create New Switching Order	2.	Create New Template	3.	Open Switching Order
4.	Save Switching Order	5.	Save Switching Order Under New Name	6.	Rename Switching Order
7.	Delete Switching Order	8.	Switch to Real Time Mode	9.	Switch to Study Mode
10.	Initialize Study Environment	11.	Open New Return to Normal Switching Order	12.	Open New Switching Order with Clearance Steps
13.	Switching Order Completed	14.	Send to WFM	15.	Create Safety Document from Switching Order
16.	Print	17.	Print Preview	18.	Export Switching Order to Excel
19.	Export Switching Order to CSV	20.	Export Switching Order to PDF	21.	Export Switching Order to HTML
22.	Online Validation	23.	Show/Hide Step Grid View Information	24.	Show/Hide Header Information
25.	Search Switching Orders	26.	Change the Font Size	27.	Increase the Font Size
28.	Decrease the Font Size	29.	Document Status		

The states of the Daily Log are:

CREATED	The Daily Log is created.
EXECUTING	The switching order steps are being executed.
COMPLETED	The steps of the switching order are completed and the action times are entered. No further changes to any information in the switching order are permitted.

After you click the Complete button, the daily log becomes a read-only form. The Completed Time field is populated with the date and time of completion.

The operator completes the daily log at the end of a shift or when it is overpopulated with many completed steps. The daily log can be completed even with no switching steps added.

To start a new daily log, you must complete and close the currently opened daily log. You can create a new Daily Log at any time from the network view. To create a new log, select the device and click the Add to Daily Log button either on the toolbar or from the right-click menu.

To test the unplanned switching steps before the real-time implementation, you can also initialize the study environment and switch between the real-time and study modes.

2.3. Hot Line Hold Operations

Hot line hold documents are used when disabling auto-reclosing or automatic control for a telemetered device and placing a Hot Line Hold tag on the device while a crew is working on an energized line section protected by the tagged device.

The Hot Line Hold form is used for hot line hold documents.

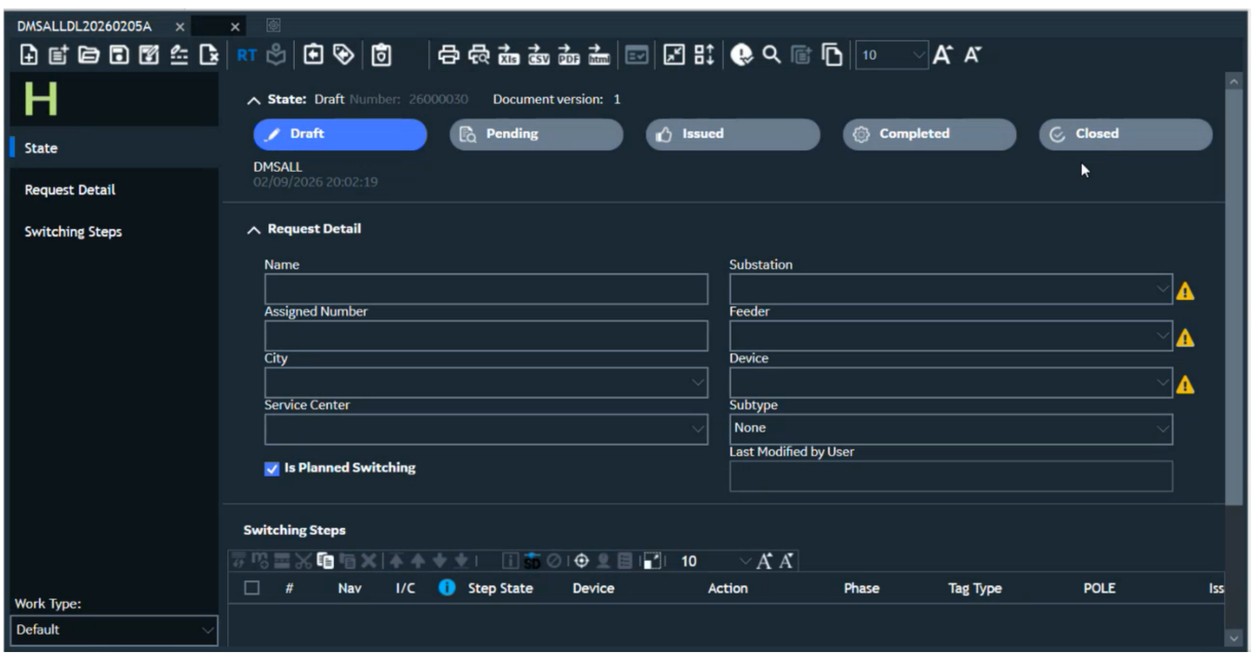


Figure 25. Hot Line Hold Form

2.3.1. Hot Line Hold State

The Hot Line Hold State section displays the status of the hot line hold document as it moves through its life cycle.

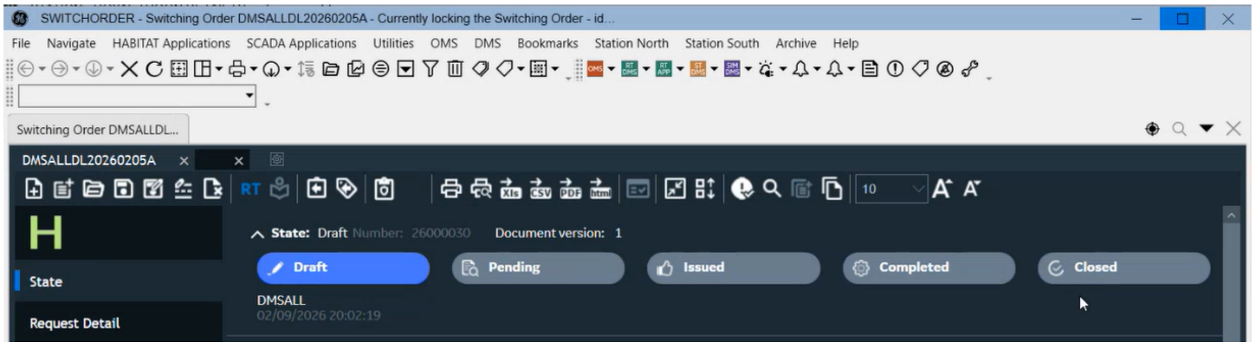


Figure 26. Hot Line Hold States

Hot Line Hold States:

1. **Draft:** Initial stage of the Hot Line Hold document. The document can be edited in this state.
2. **Pending:** The document undergoes review and validation.
3. **Cancelled:** The Hot Line Hold document is cancelled.
4. **Issued:** The Hot Line Hold document is issued.
5. **Completed:** The Hot Line Hold document is marked as completed.
6. **Closed:** The Hot Line Hold document is marked as closed.

> **Note**
>
> Hot Line Hold documents have two execution stages. The first stage typically includes steps to apply tags, and the second stage includes steps to remove the applied tags.

2.3.2. Switching Steps

The Switching Steps section contains the switching steps of the hot line hold document.

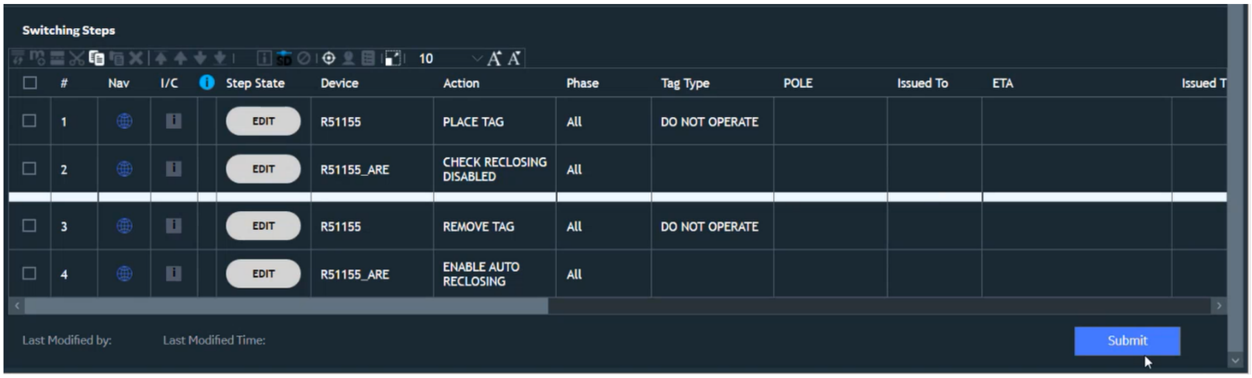


Figure 27. Hot Line Hold Switching Steps

2.3.3. Pending Hot Line Hold Documents

The Current Pending Hot Line Hold display allows you to search for active hot line hold documents. You can access this display from the Options page.

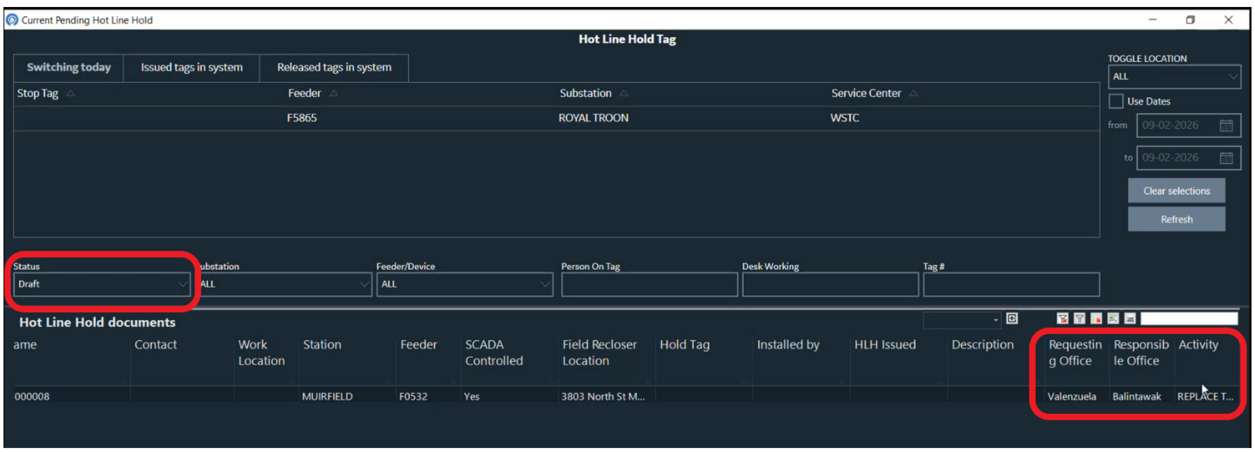


Figure 28. Pending Hot Line Hold Documents

The Status dropdown lists Hot Line Hold states that can be used for filtering. The section includes the following fields:

- **Requesting Office:** Office initiating the request.
- **Responsible Office:** Office accountable for the task.
- **Activity:** Action being performed.

2.4. Planned Requests

Planned request documents are used when planning switching operations. You can create a switching order from a planned request.

The Planned Request form is used for planned request documents.

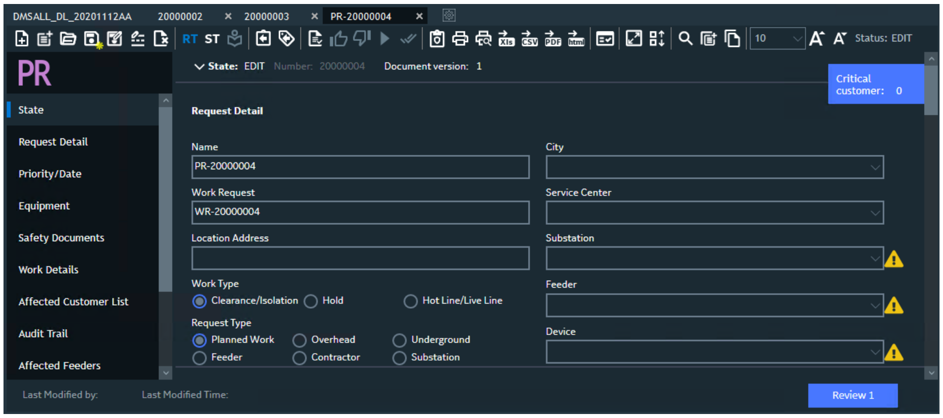


Figure 29. Planned Request Document

Most of hot line hold document sections are similar to switching order sections. For more detail of Equipment, Safety Documents, Audit Trail, Affected Feeders, Switching Order Dispatching, Step Block Overview sections, refer to Switching Order Sections.

2.4.1. Request Detail

The Request Detail section describes the planned request document.

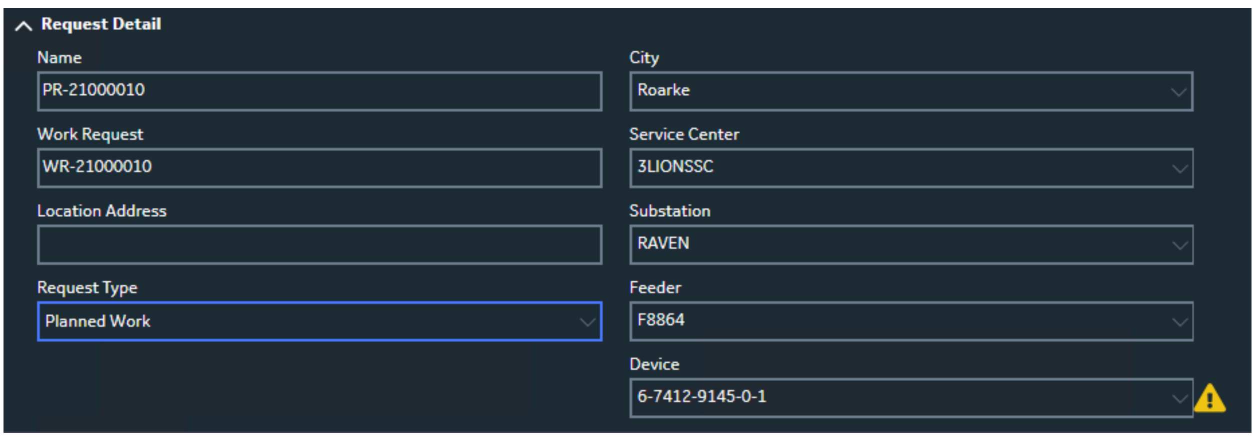


Figure 30. Planned Request Detail

The Request Detail section includes the following fields:

- **Name:** The unique name of the planned request document. If the Name field is blank, it is automatically populated with the generated number when the document is saved or validated.
- **Work Request:** The name of the work request that is associated with the planned request document.
- **Location Address:** Address of the location associated with the planned request.
- **Request Type:** The planned request type (Planned Work, Overhead, Underground, Substation, Feeder, or Contractor).
- **City:** The name of the associated city.

- **Service Center:** The name of the associated service center.
- **Substation:** The name of the associated substation.
- **Feeder:** The name of an associated feeder.
- **Device:** The name of the device that is the subject of the document.

2.4.2. Priority and Date

The Priority and Date section shows the document priority and scheduling data.

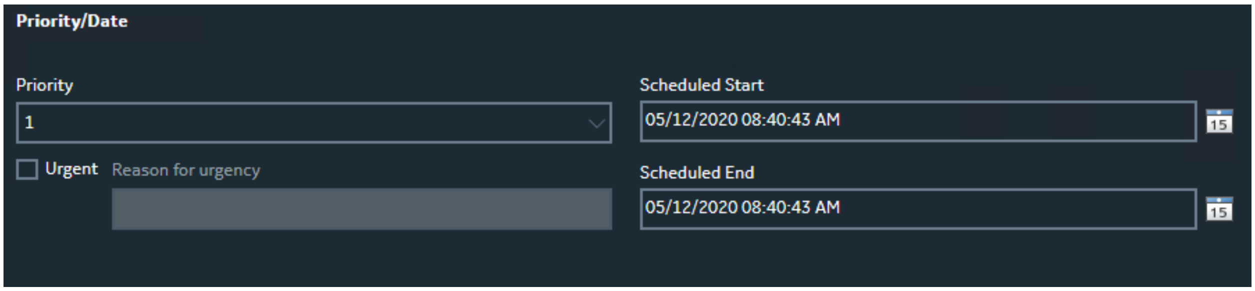
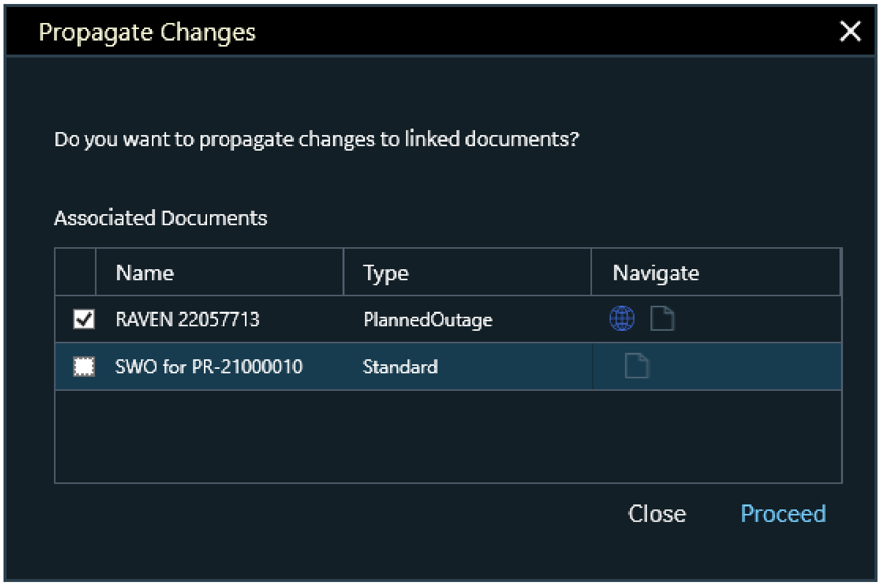


Figure 31. Planned Request Priority and Date

The Priority and Date section includes the following fields:

- **Priority:** The planned request priority.
- **Urgent:** Specifies whether the planned request is urgent.
- **Reason for Urgency:** Free-form text to describe the urgency reason.
- **Scheduled Start Date:** The date when the planned request is scheduled to start.
- **Scheduled End Date:** The date when the planned request is scheduled to end.

When you update start and end dates for a planned request that is associated with a switching order or planned outage, the system prompts you whether you want to propagate the date changes to the associated switching order or planned outage.



2.4.3. Work Details

The Work Details section provides details for the associated work.

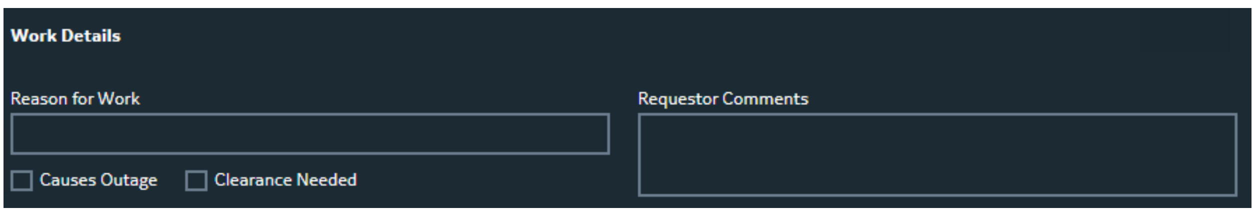


Figure 32. Planned Request Work Details

The Work Details section includes the following fields:

- **Reason for Work:** Free-form text to describe the reason for work.
- **Causes Outage:** Specifies whether the work results in customer de-energization. If selected, the Create Planned Outage button appears under the Switching Steps section, near the Create Switching Order button.
- **Clearance Needed:** Specifies whether clearance is needed for the work.
- **Requestor Comments:** Comments from the work requestor.

2.4.4. Affected Customer List

The Affected Customer List section provides details for the affected customers.

Affected Customer List Count: 7

Customer Name	Address	Phone Number	Service Point	Account Number	Actions	Customer Key	IsCritic
Bree Chris W	8115 STONE NE	5635790	SP_13359118	555207456		555207456555207	False
Benton Wendy V	9374 WALNUT ST	5884229	SP_13359118	555346552		555346552555346	False
Christian Cherri D	9374 CEDAR SW	5776367	SP_13359118	555282266		555282266555282	False
Simons Steve S	8117 STONE N	5661558	SP_13359118	555357672		555357672555357	False
Shaw Gayle Y	9375 CHERRY ST	5867540	SP_13359118	555211525		555211525555211	False
Larson Tory A	8117 RIVER RD	5581467	SP_13359118	555246716		555246716555246	False
Bree Virginia U	8117 SAND RD	5868893	SP_13359118	555295298		555295298555295	False

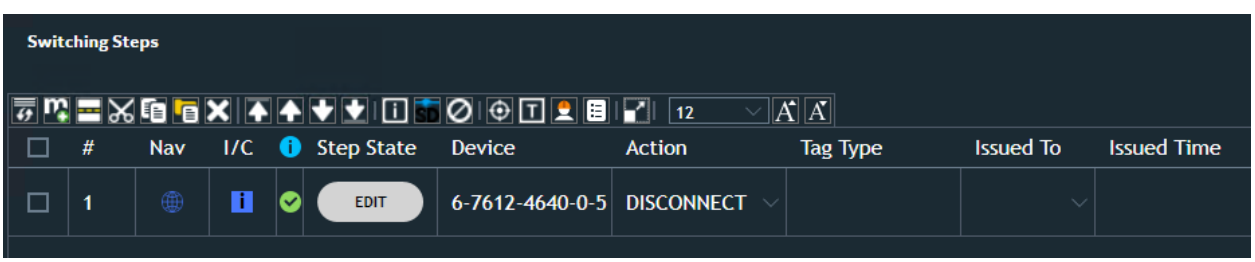
Figure 33. Planned Request Affected Customer List

The Affected Customer List section includes the following fields and controls:

- **Customer Name:** Customer name.
- **Address:** Customer address.
- **Phone Number:** Customer phone number.
- **Service Point:** Service point that the customer is associated with.
- **Account Number:** Customer account number.
- **Customer Key:** Unique customer key.
- **Is Critical:** Specifies whether this is a critical customer.
- **Actions:**
    - **Locate:** Navigates to the customer on the network view.
    - **Remove:** Removes the customer from the affected customers list.
- **Add from RT:** Adds traced customers from the real-time network view to the customer list.
- **Add from ST:** Adds traced customers from the study mode network view to the customer list.

2.4.5. Switching Steps

The Switching Steps section contains the switching steps of the planned request document.



Switching Steps

	#	Nav	I/C		Step State	Device	Action	Tag Type	Issued To	Issued Time
	1				EDIT	6-7612-4640-0-5	DISCONNECT			

Figure 34. Planned Request Switching Steps

The Planned Request Switching Steps section is similar to the Switching Order Switching Steps section. For more information, see Switching Steps.

2.5. Switching Order Options Page

On the Switching Order form, click the Options button to view more switching order options.

The Options page shows options for creating a new document (switching order, hot line hold, template, or planned request) and searching for a document. It also displays shortcuts to the 20 most recent documents. The Clear History option clears all shortcuts from this view.

> **Note**
>
> The latest 20 switching order names are saved in the RecentList_SwitchOrder.RecentSwitchOrder.<User_Name>.xml file, which is located at C:\Users\<User>\AppData\Roaming\General Electric Technology GmbH\Advanced Presentation Framework\<Version>.

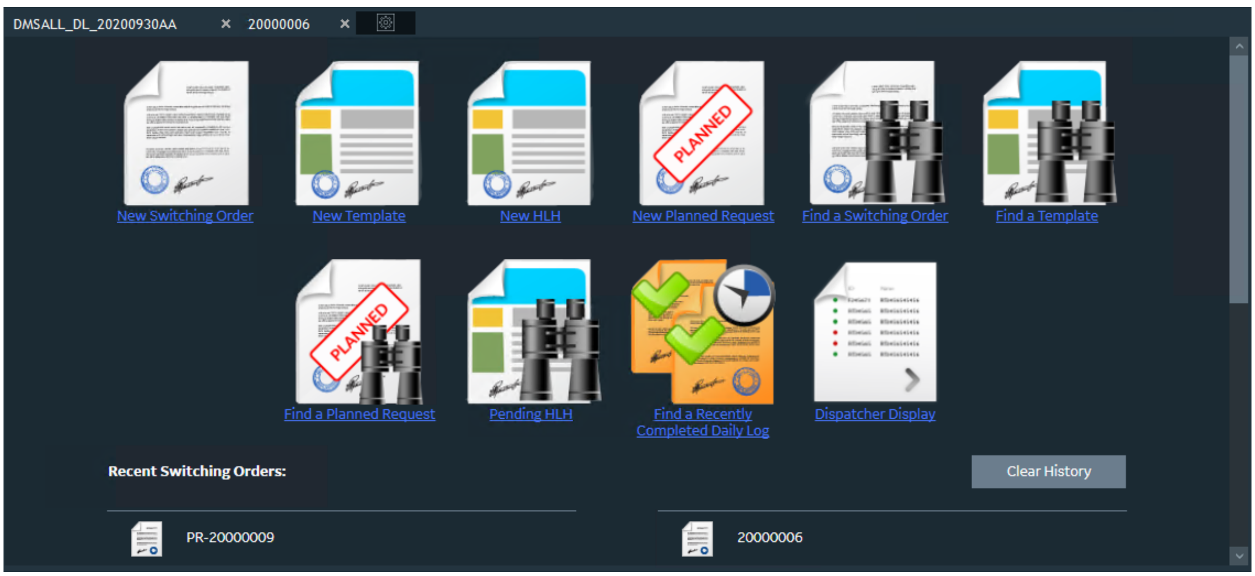


Figure 35. Options Page

2.6. Switching Order Search

The Switching Order Search pop-up allows you to filter and open switching order documents for edit or review.

To open the Switching Order Search window, do one of the following:

- On the Switching Order Form toolbar, click Open File.
- On the Switching Order Form toolbar, click Search.

- On the geographic (common) toolbar, click Switching Order Search.

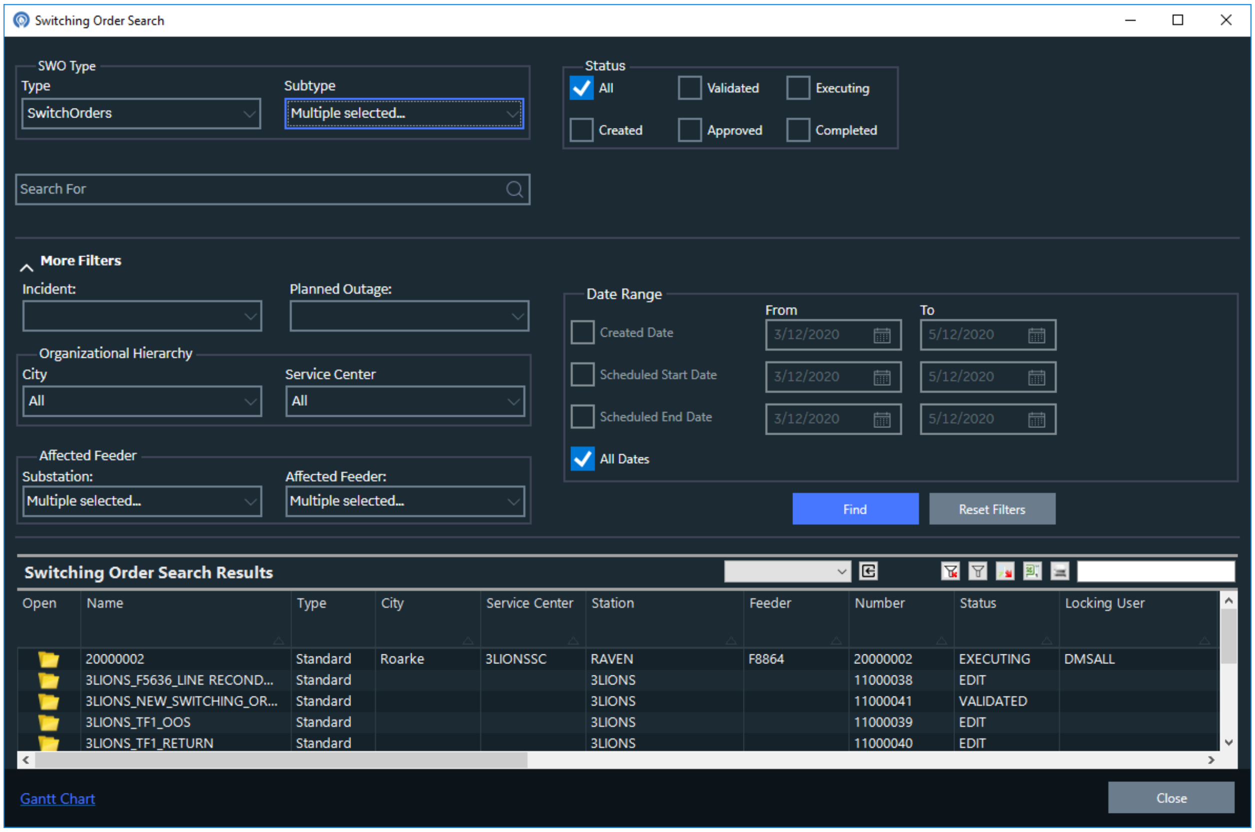


Figure 36. Switching Order Search Pop-Up Window

The Switching Order Search pop-up display includes the following fields and controls:

- **Type:** Filters switching order documents by type (All, Switch Order, Daily Log, Template, or Hot Line Hold).
- **Subtype:** Filters switching order documents by one or more subtypes (All, None, MV, or LV).
- **Status:** Filters documents by one or more statuses (All, Created, Validated, Approved, Executing, and Completed).
- **Search For:** Searches by name or search tags.
- **Incident:** Filters documents by the associated incident name.
- **Planned Outage:** Filters documents by the associated planned outage ID.
- **City:** Filters documents by a city.
- **Service Center:** Filters documents by a service center.
- **Substation:** Filters documents by one or more substations.
- **Affected Feeder:** Filters documents by one or more substation feeders.
- **All Dates:** If selected, searches for documents regardless of the date range.
- **Created Date From/To:** The date range for when the documents were created.

- **Scheduled Start Date From/To:** The date range for when the documents are scheduled to start.
- **Scheduled End Date From/To:** The date range for when the documents are scheduled to end.
- **Find:** Click to find switching order documents that correspond to the specified search criteria.
- **Reset Filters:** Click to reset all filters to the default values.

The Switching Order Search Results grid shows a switching order name, type, city, service center, station, feeder, number, status, locking user, created by, assigned number, created date, planned start date, planned end date, description, incident name, and planned outage ID. You can also configure the grid to show the second and third approvers. For more details, refer to the *Switching Order Client Configuration Editor User Guide*.

Use the Open button to open and edit the unlocked switch orders.

Click the Gantt Chart button below the Switching Order Search Results grid shows to show a Gantt chart with timeline for the searched documents.

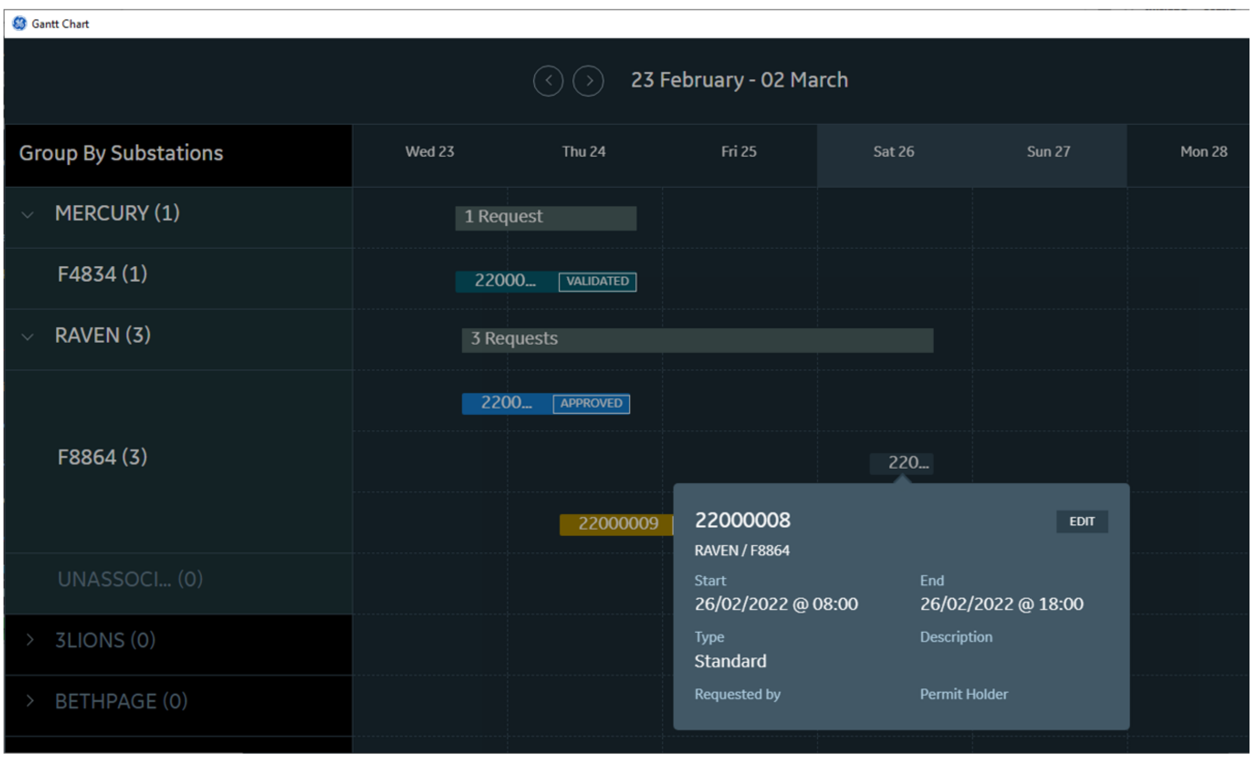


Figure 37. Switching Order Gantt Chart

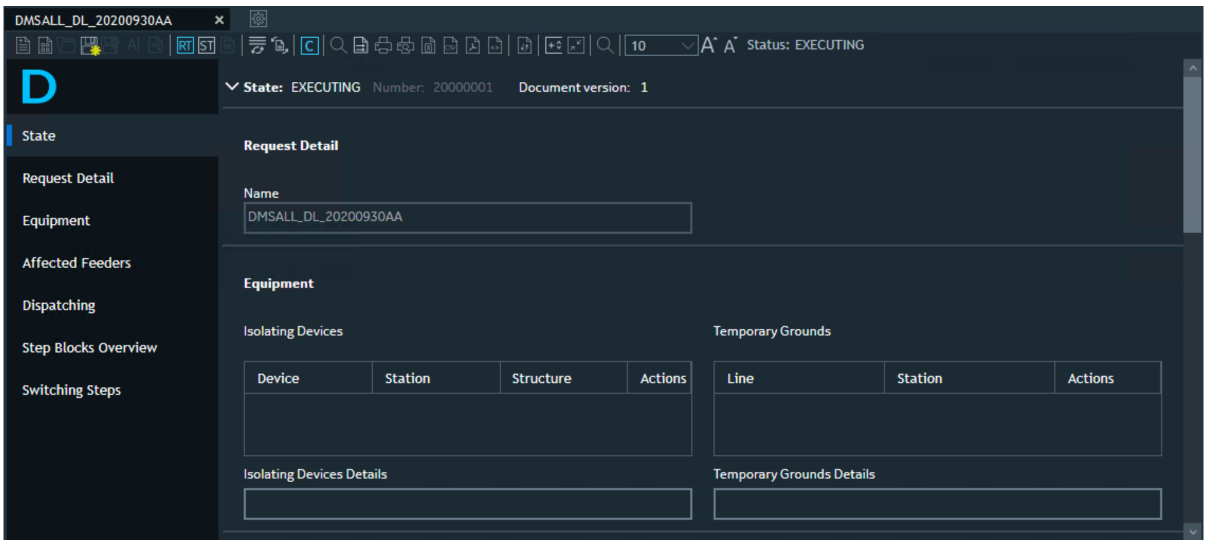
3. General Usage

This chapter describes general functions of the Switching Order application.

3.1. Creating a Switching Order

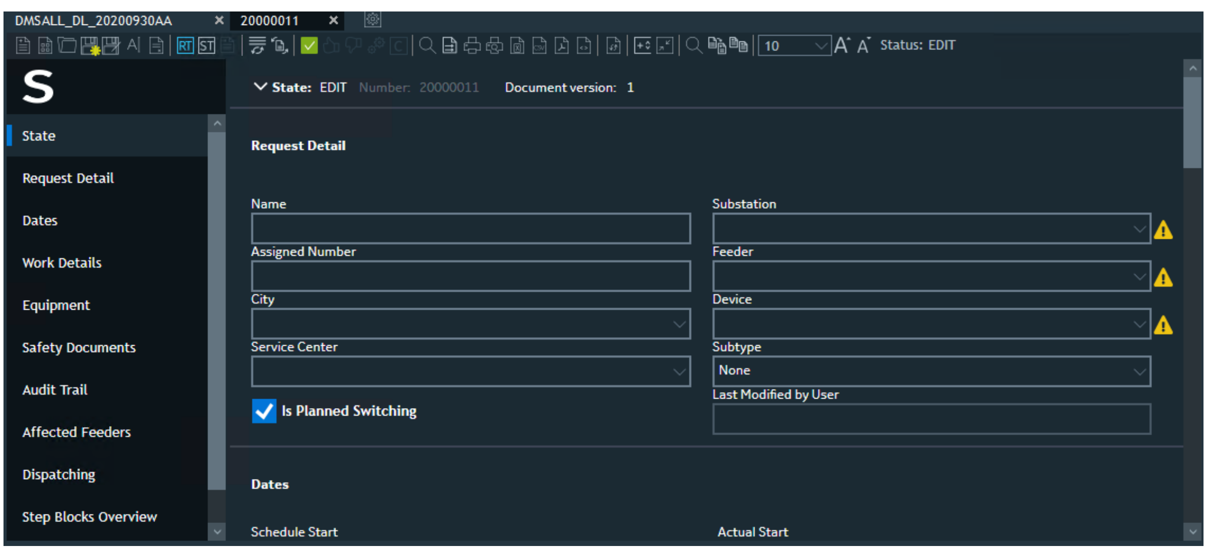
To create a new switching order:

1. Click the Switching Order button on the common toolbar from the network view. The Switching Order display appears showing the Daily Log tab.



2. Click the New Switching Order button on the Switching Order toolbar.

   This opens a new, blank Switch Order.



Note

If the SwitchOrder window is closed abruptly, the operation cannot be aborted.

Note

Switching orders must be cleaned up periodically to ensure the Switching Order Server performs at an optimum level. The frequency of clean-up depends on factors such as server capacity, usage patterns, and business rules.

3.2. Creating a Hot Line Hold Document

You can create a hot line hold document from the Switching Order application and from the network view.

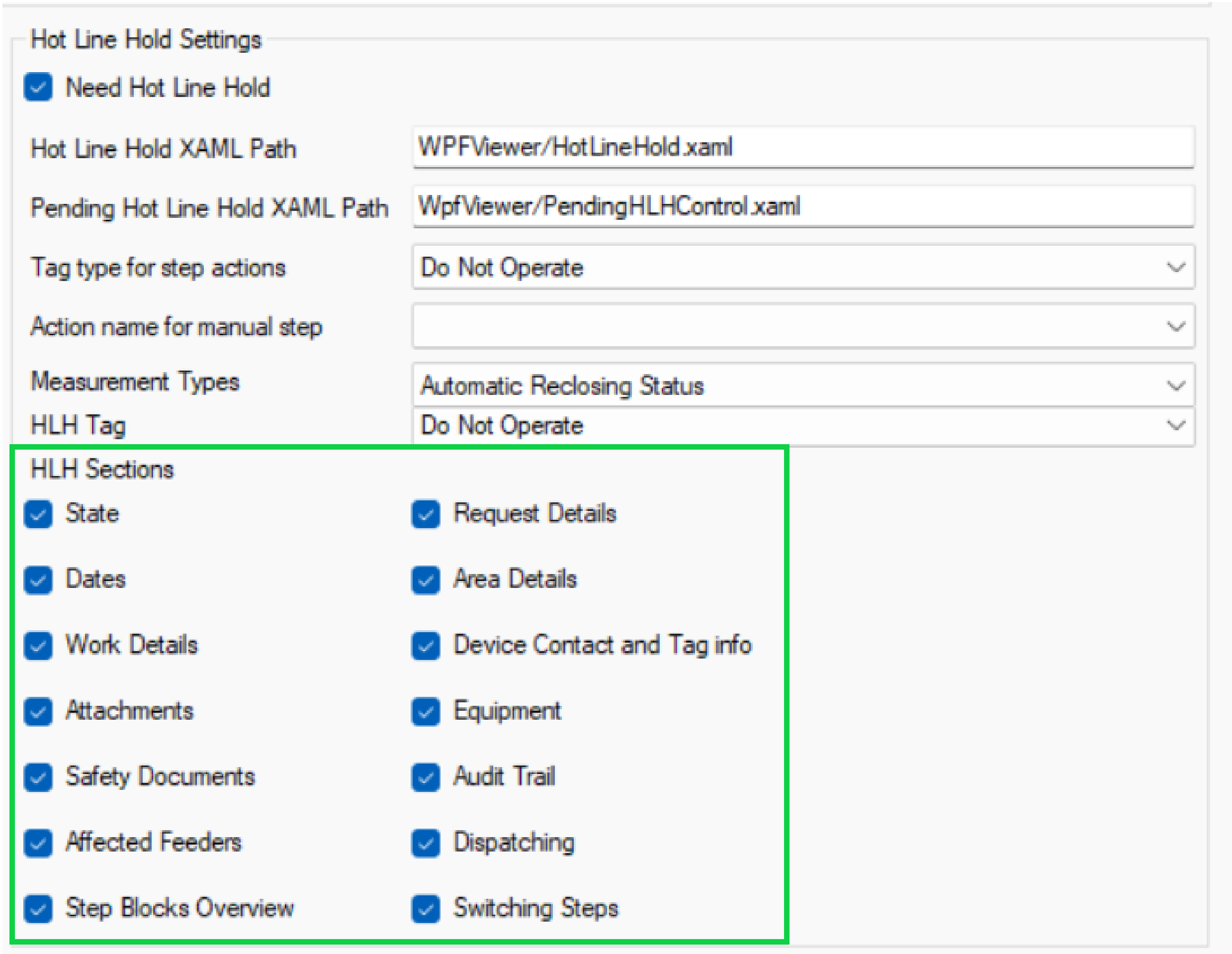
3.2.1. Creating a Hot Line Hold Document from the Switching Order Application

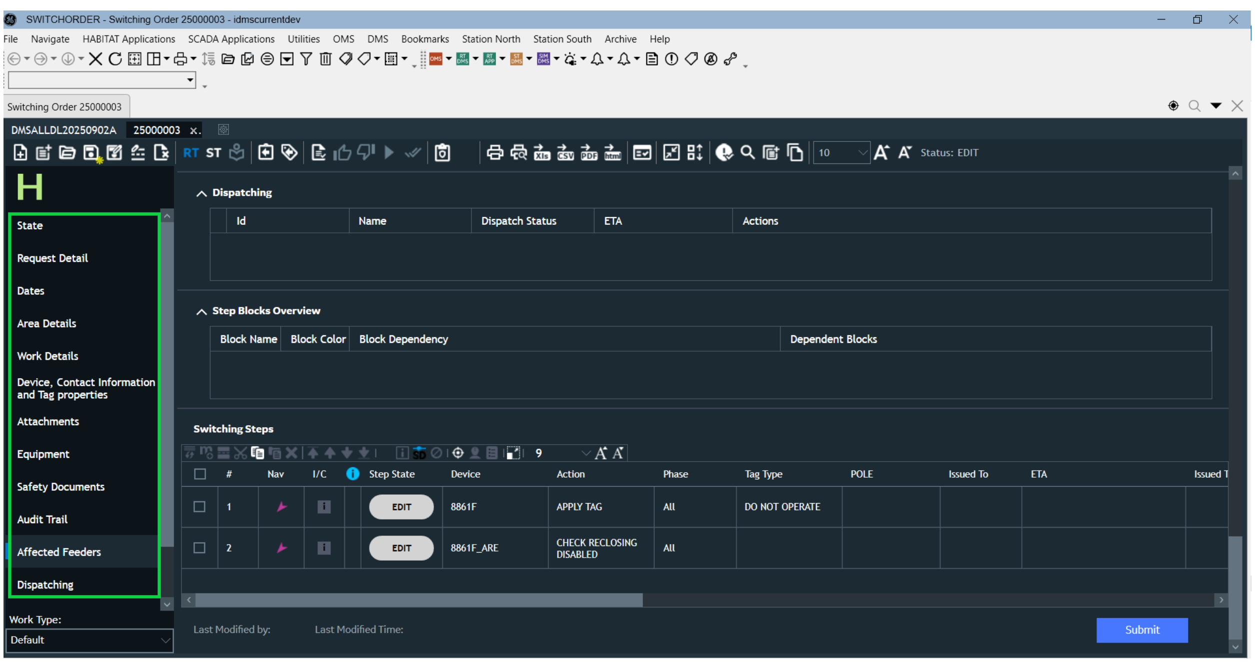
Hot line hold switch order template configuration:

The system automates the execution of Hot Line Hold (HLH) switch orders, reducing the need for manual operator input and improving operational efficiency.

Configure the `SwitchOrderClientConfig` with a master template by defining specific keys for each section and linking the configuration file to the XAML file. This enables dynamic section control, allowing sections such as HLH to be shown or hidden based on configuration settings.

Navigate to the General tab in the Switching Order Client Configuration Editor and select the desired sections using checkboxes to display them under Hot Line Hold Settings in the HLH document.

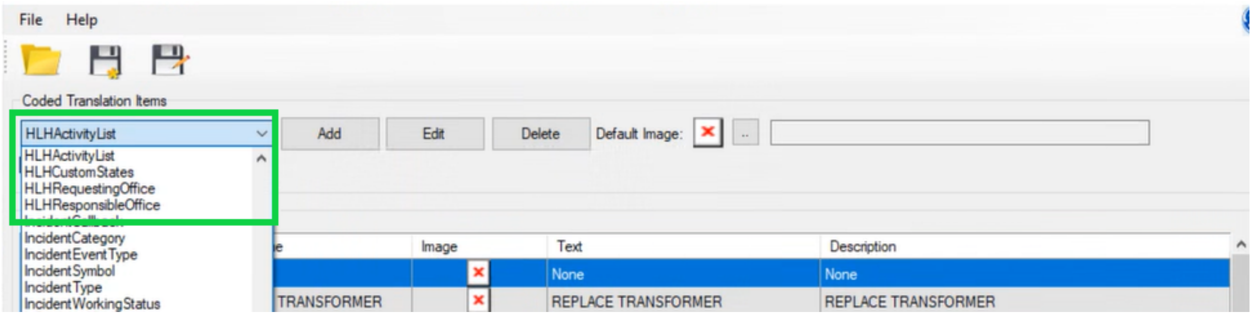




Hot line hold document update:

The **Area Details** section provides fields for entering mandatory information such as Location, Crew Name, Crew Phone Number, and Estimated Completion Time. The Selected Device ID and Selected Device Name fields are auto-populated and disabled.

Users can choose State, Requesting Office, Responsible Office, and Activity from drop-down menus, and modify these values within the Coded Translation Editor configuration settings.

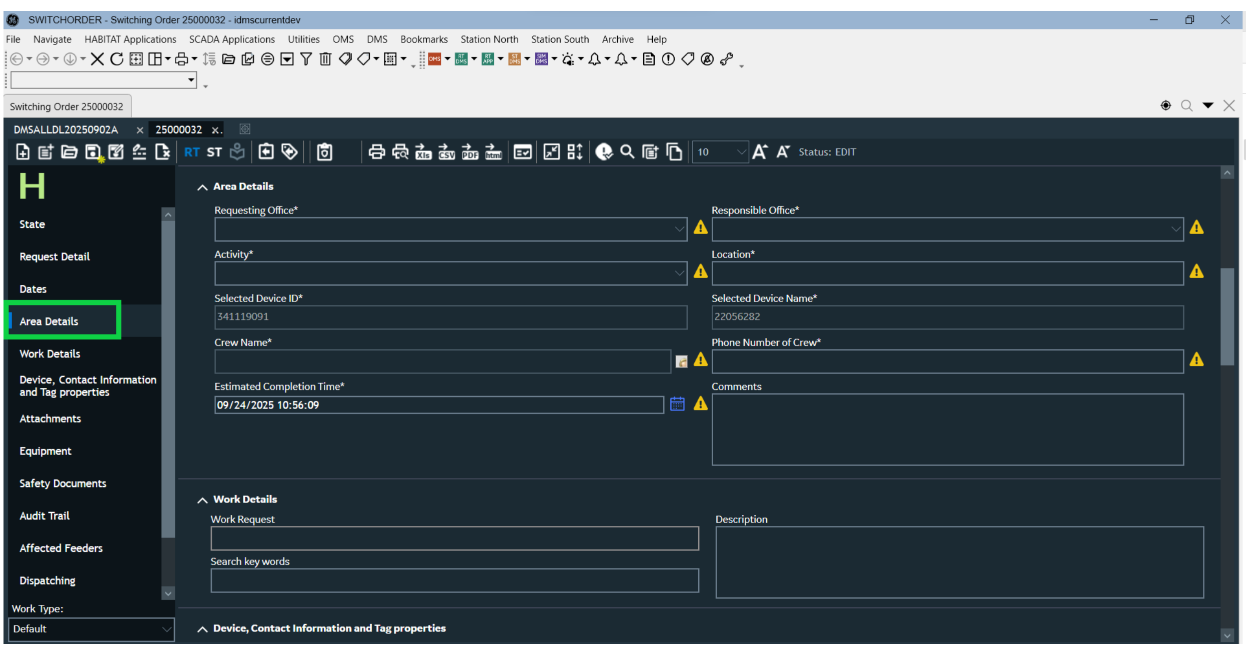


Note

Duplicate the State and Switching Steps sections and dedicate them exclusively to HLH, without affecting other templates.

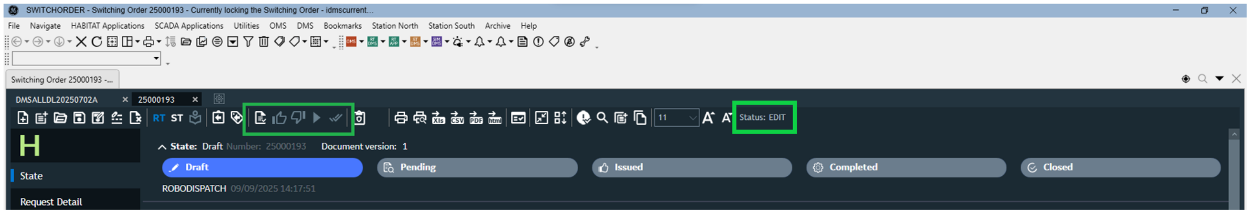
User can enter comments in the comment box even after submitting the HLH document.

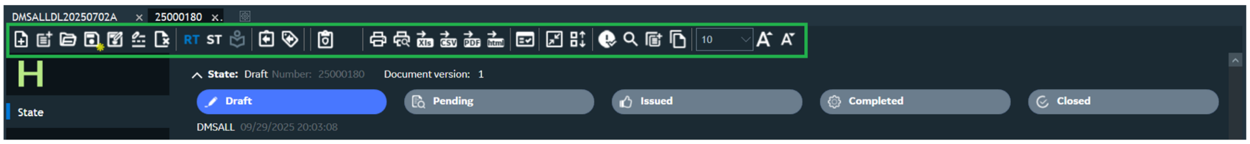
Mandatory fields are marked with an asterisk (*). If a field is left blank, a message appears. A tooltip appears when hovering over the ⚠ notification icon.



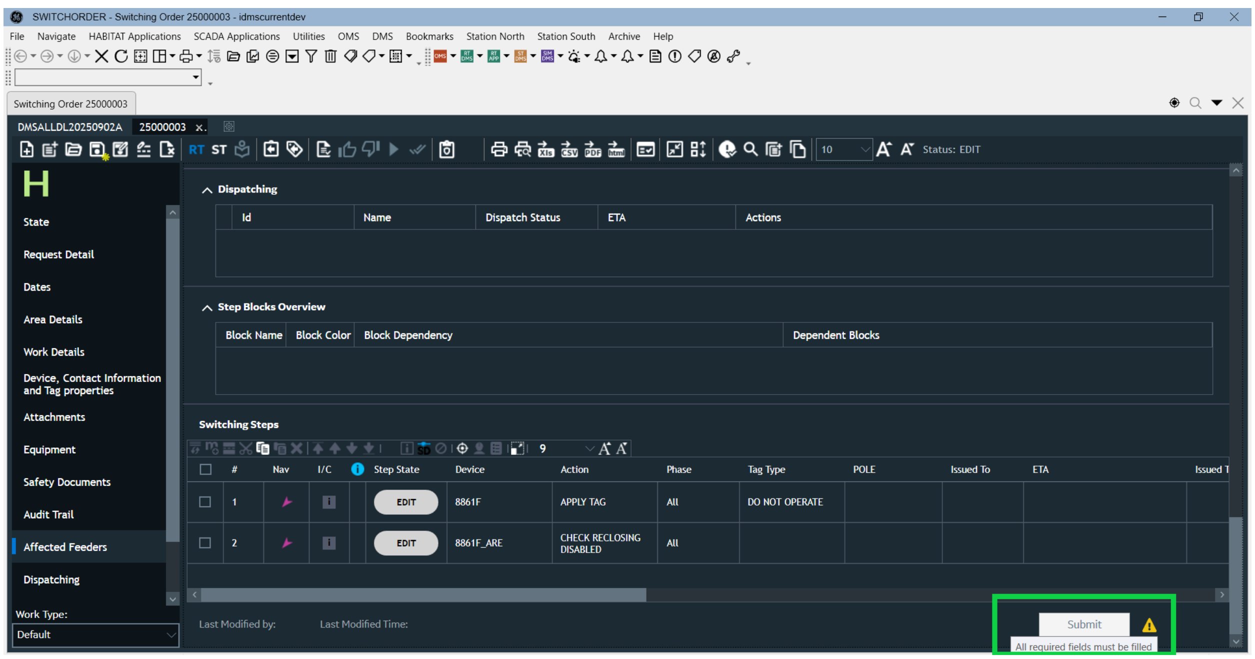
Note

Hide the Status, Validate, Execute, Approve, Reject, and Complete icons in the common tool bar exclusively for HLH, without affecting the switch order.





The **Switching Steps** section is read-only for hot line hold. The **Paste Switching Steps** option in the context menu is disabled. The Submit button remains inactive until all mandatory fields are completed. If mandatory fields are not filled, the following message appears: **All required fields must be filled**.



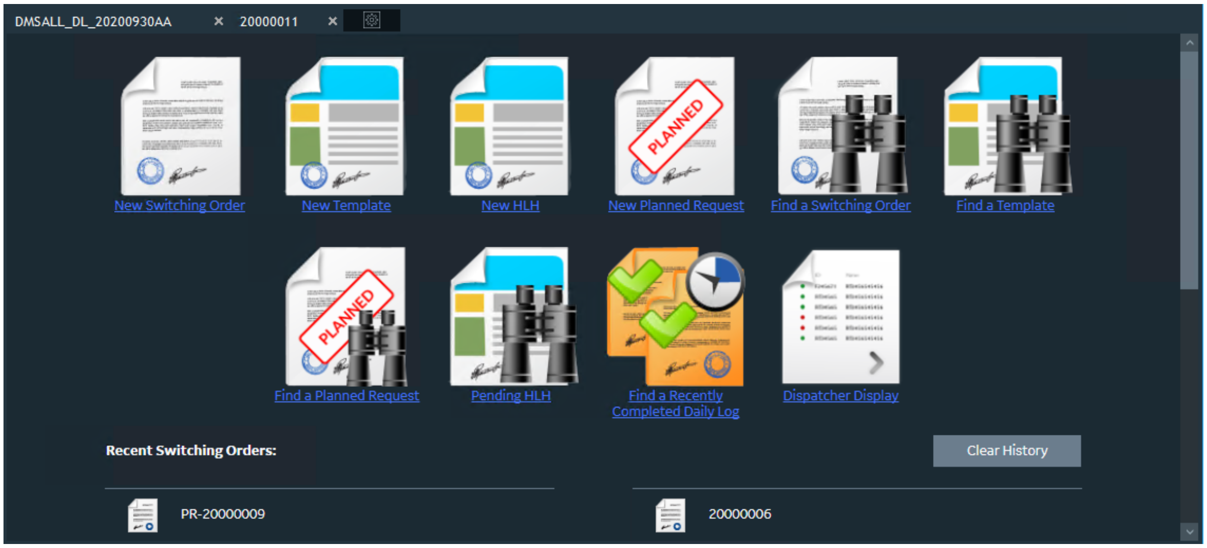
To create a hot line hold document from the Switching Order application:

1. Click the Switching Order button on the common toolbar.

   The Switching Order display appears, showing the Daily Log tab.

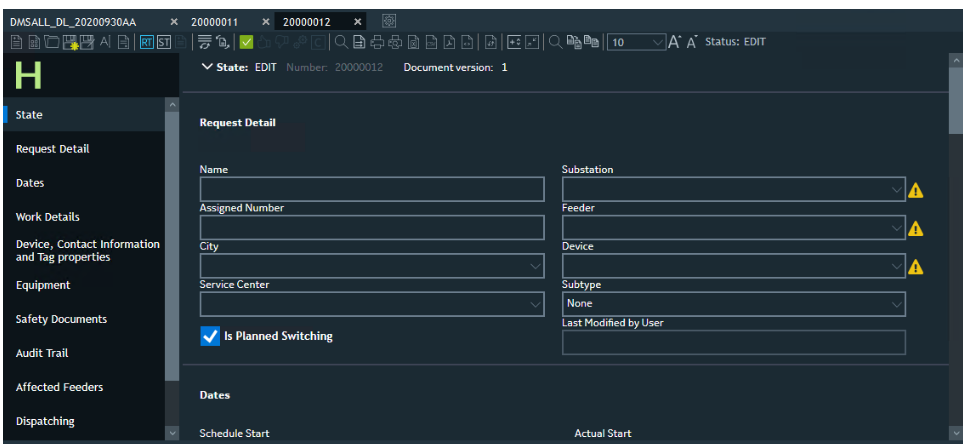
2. On the Switching Order form, click the Options button.

   The Options page appears.



3. Click the New HLH button.

   A new, empty hot line hold document appears.

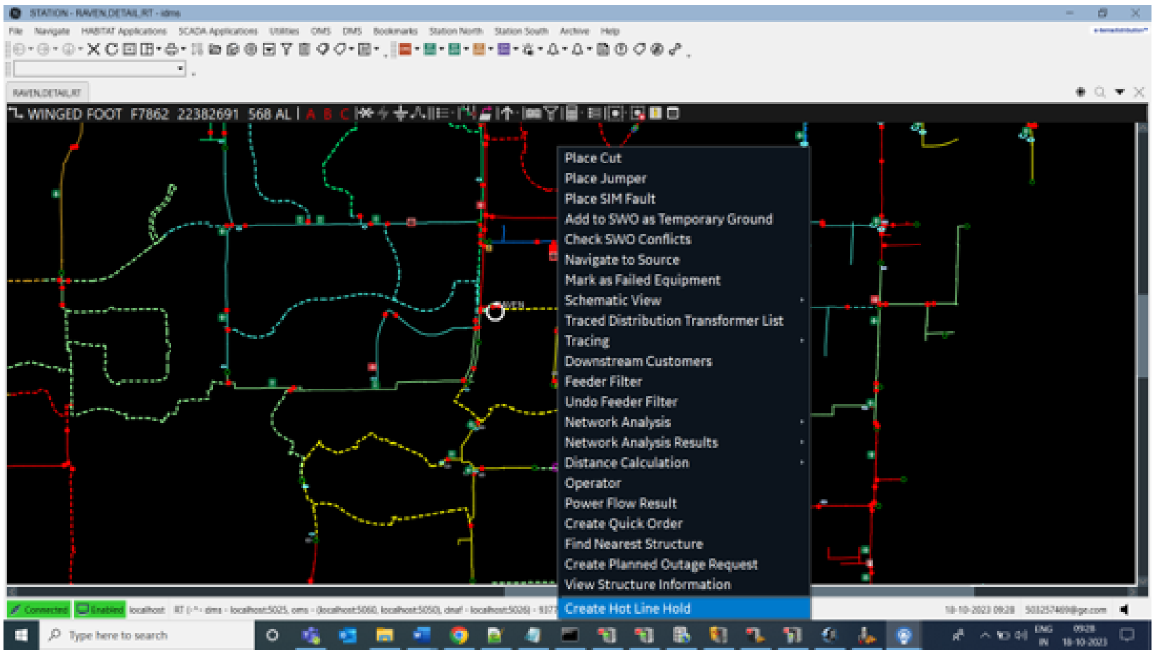


3.2.2. Creating a Hot Line Hold Document from the Network View

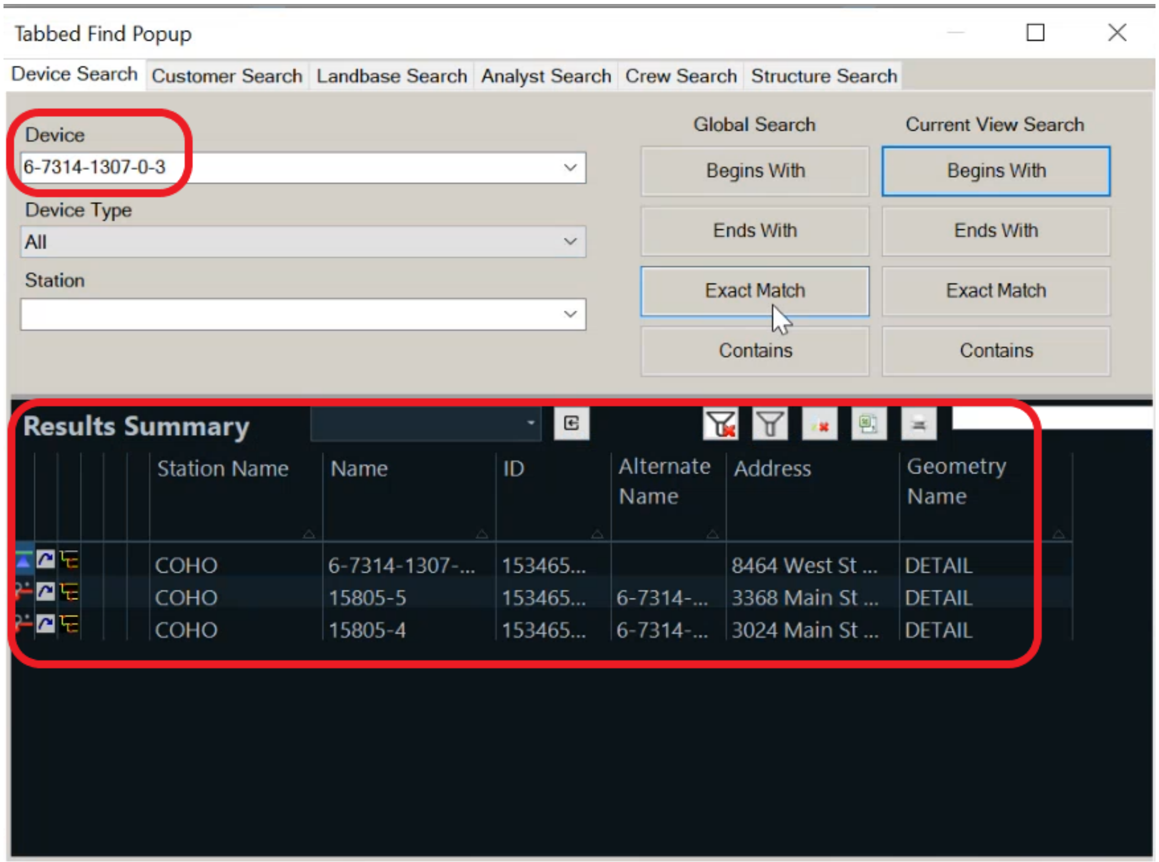
> **Note**
>
> The following steps can be used to create a Hot Line Hold document for any device such as breaker, recloser, line, transformer, switch or capacitor, as required. If the Create Hot Line Hold option is not available in the right-click context menu for a device, enable it using the Viewer Configuration Editor by following the steps in Right Click Menu. The device being selected for creating Hot Line Hold will be the **Selected Device** and the first upstream auto-reclosing device will be the **Protective Device**. Only one device can be selected at a time. Selecting a new asset clears previous selection.

To create a hot line hold document for a device from the network view:

1. Selecting a device from network view by using the following methods for creating hot line hold:
   a. Select device in network view by clicking on the device. The device gets highlighted and zoomed. Right-click on the device from network view. Select **Create Hot Line Hold** from the context menu.

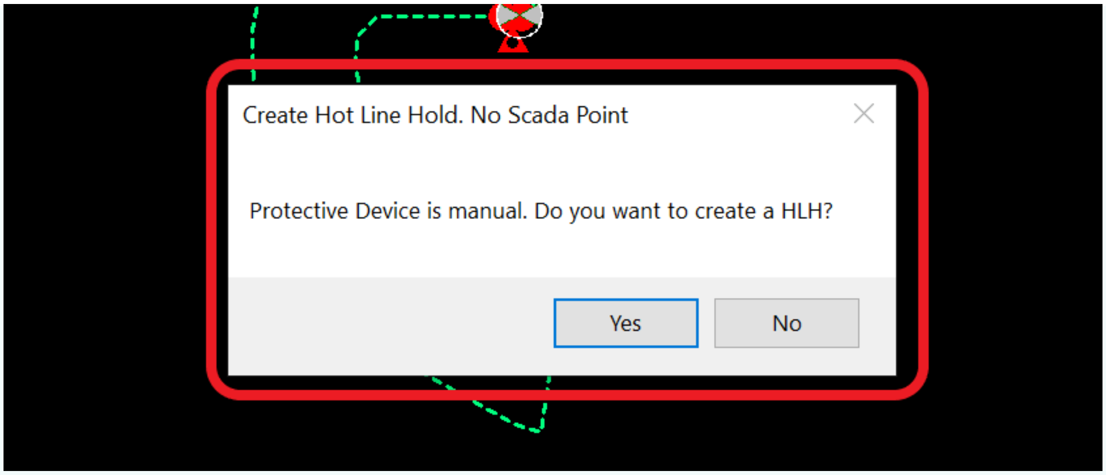


   b. Click search icon in network view. **Tabbed Find Popup** will appear. Search device using Device ID or Device Name in **Device Search**. If no matching asset is found, a user-friendly message (No Results found) will display. Select device from summary to go to network view. The device gets highlighted and zoomed. Right-click on the device in network view. Select **Create Hot Line Hold** from the context menu.

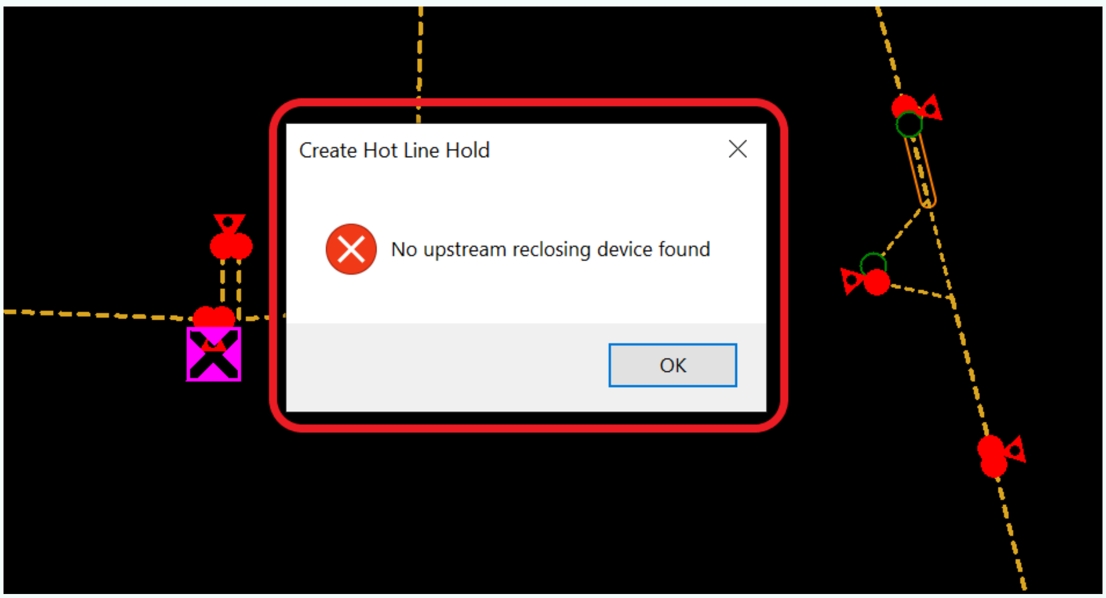


2. The check for upstream protective device and SCADA endpoint is performed systematically. Even if the selected device is a breaker or recloser, system still traces up to the first upstream protective device.

> **Note**
>
> The scenarios for tracing protective device and SCADA endpoint check have been explained below:
>
> a. If upstream protective device is found with protection capability and SCADA endpoint is identified, then it would get redirected to HLH document.
>
> b. If upstream protective device is found with protection capability and SCADA endpoint is not identified. In such case a popup would display: **Protective device is Manual. Do you want to create a HLH?** with Yes/No option. If **Yes** is selected, it will redirect to HLH document. If **No** is selected, then no action will be performed and will stay in the network view.

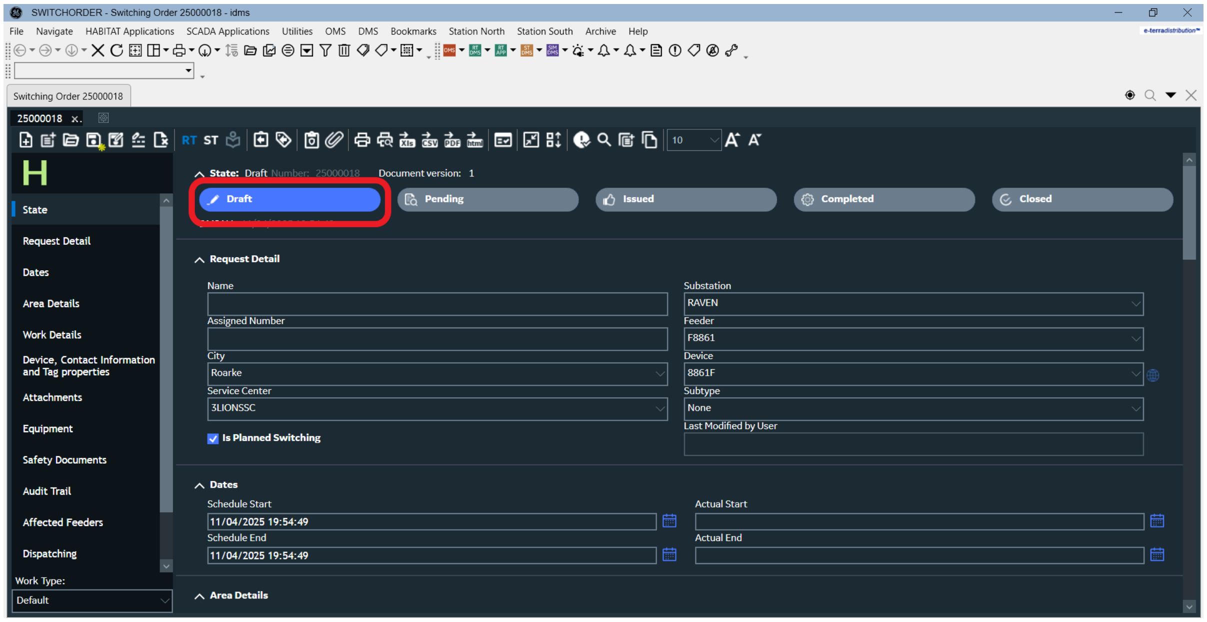


c. If upstream protective is not found with protection capability. In such case an error message: **No Upstream reclosing device found** will popup.



d. If the protection capability of first protective device is disabled, system will trace up to find next upstream protection device.

3. The request gets redirected to HLH document Draft State and state button turns blue.

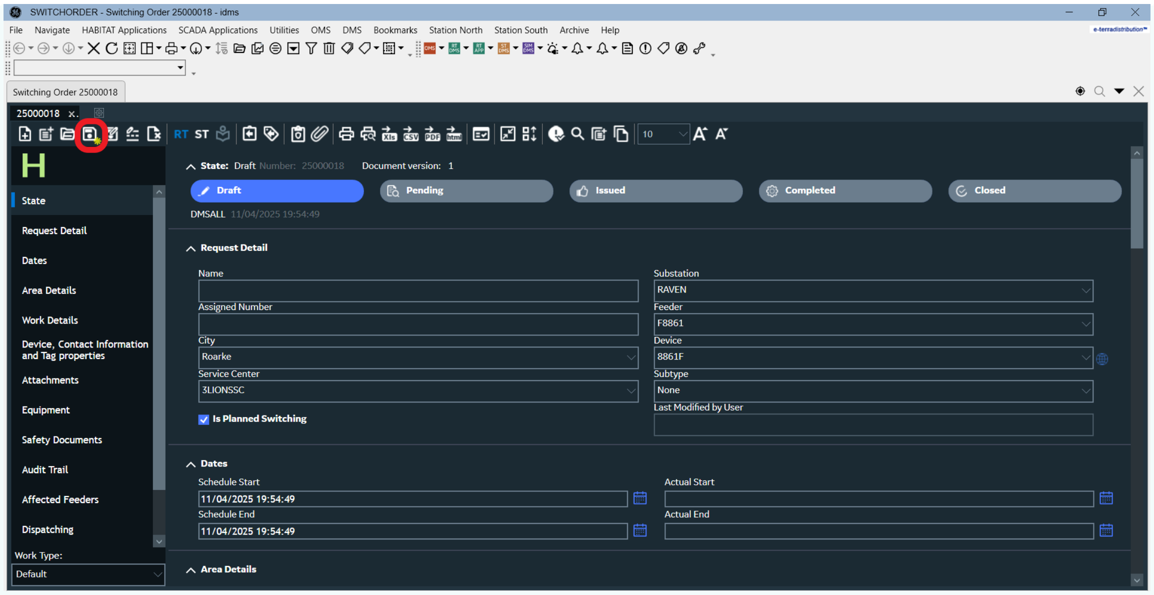


> **Note**
>
> Sections in the HLH draft can be hidden or shown from the Switching Order Client Configuration Editor > General tab. See Hot Line Hold Settings.

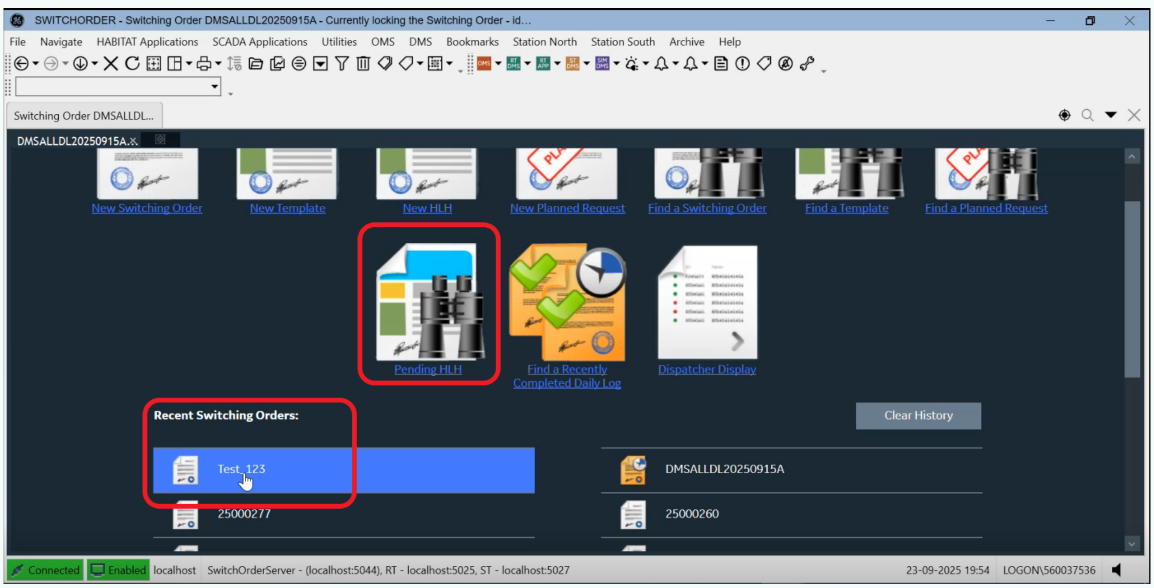
4. Fill in the HLH form which includes both mandatory and optional fields. Mandatory fields are marked with an asterisk (*).

> **Note**
>
> a. The Area Details section provides fields for entering mandatory information such as Location, Crew Name, Crew Phone Number, and Estimated Completion Time. Users can select State, Requesting Office, Responsible Office, and Activity from drop-down lists populated with values configured from the Coded Translation Editor.
>
> b. If mandatory fields are not filled, the following message appears on the tooltip of the alert icon beside the Submit button: All required fields must be filled.
>
> c. The Upstream Protective Device, Selected Device ID and Selected Device Name fields are auto-populated and disabled.
>
> d. In Draft state, the Submit button remains read-only until all mandatory fields are filled.

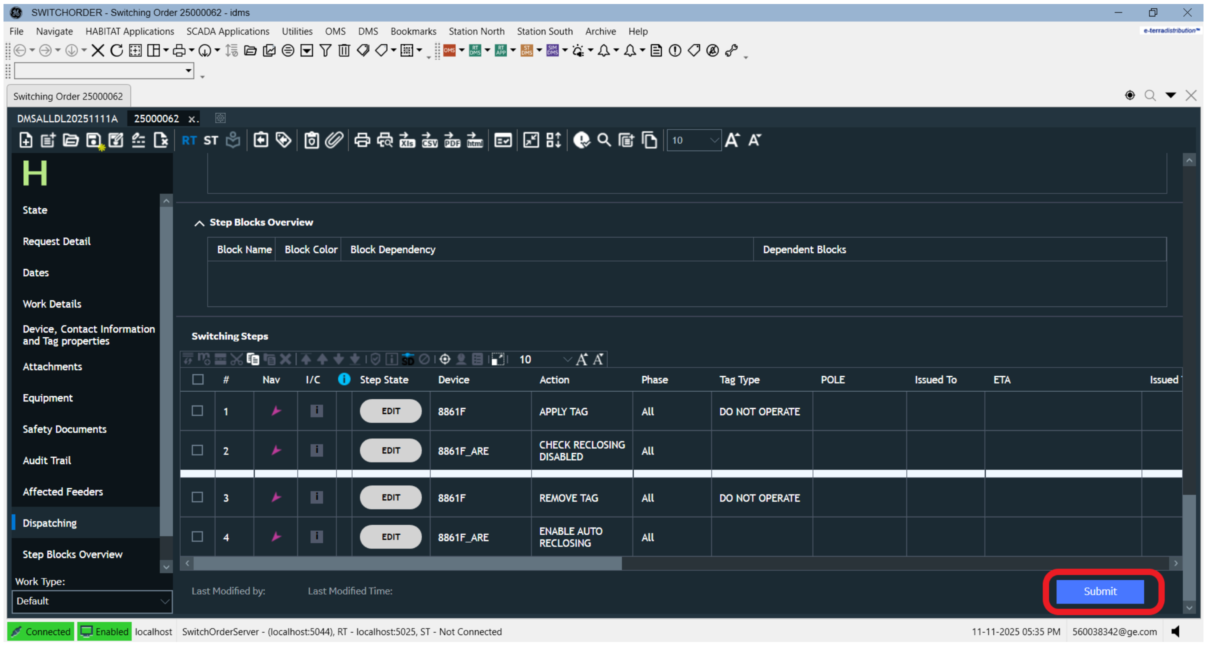
> **Note**
>
> Save the draft using the Save icon in the Quick Access Toolbar.

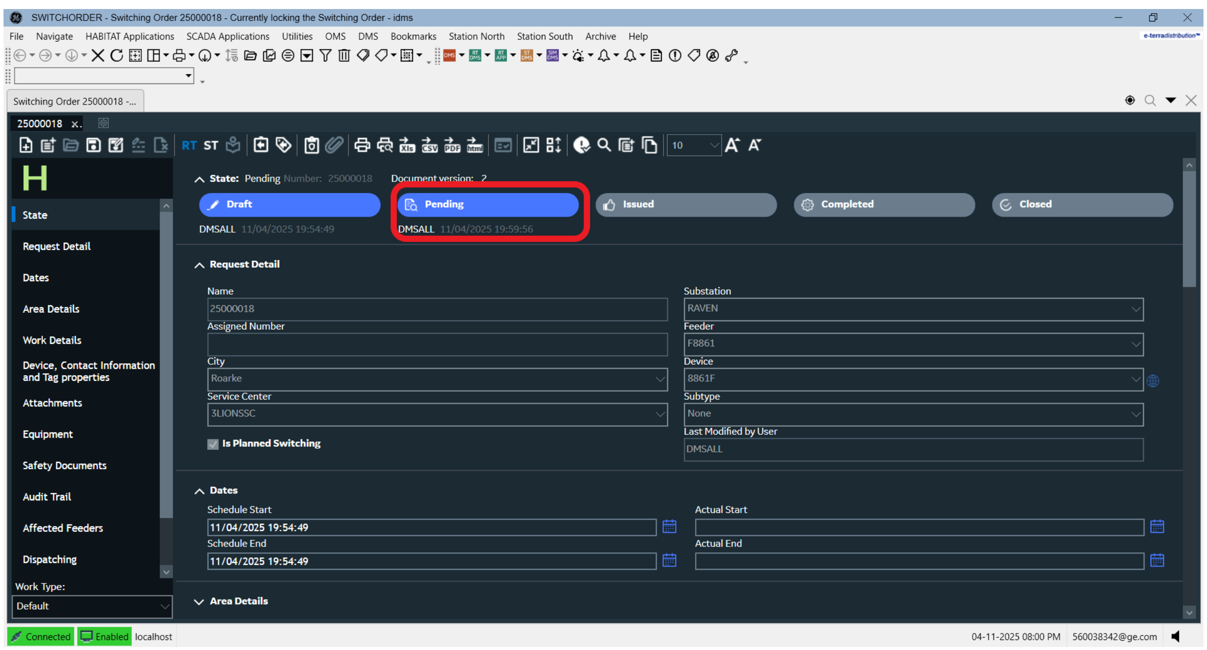


After clicking Save, the document is stored as a draft on the server at a specific file location. To search and open earlier drafted HLH document, click the Options button. Options page would appear. Click on **Pending HLH**.

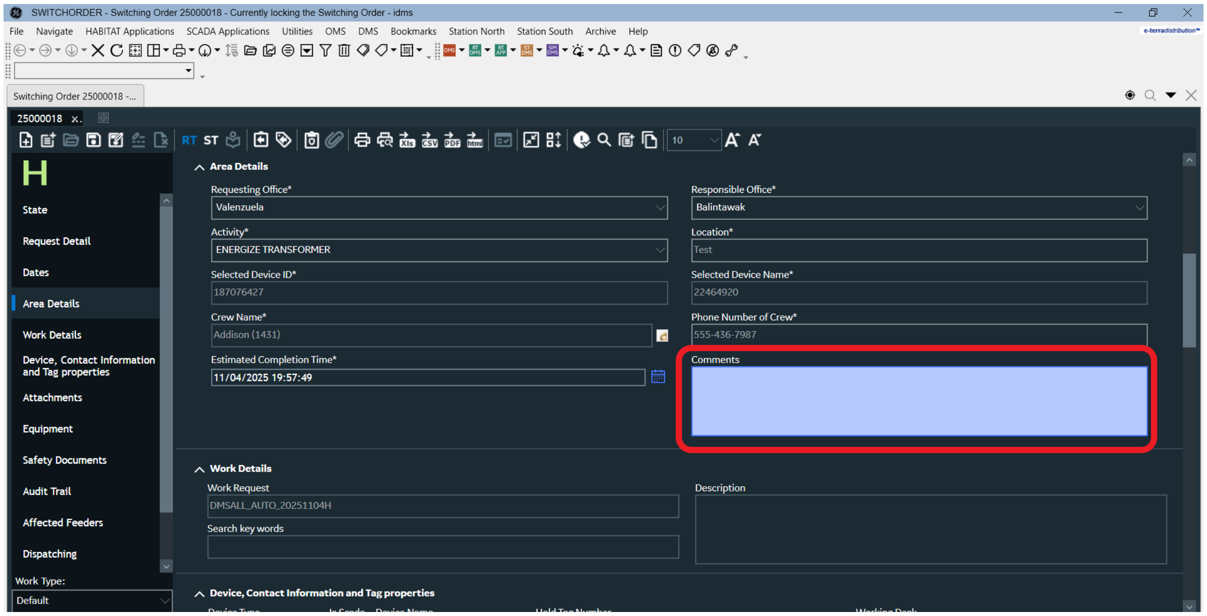


5. Click Submit. On clicking **Submit** the system performs HLH document check for mandatory fields. If the validation check is successful, the HLH document moves to Pending state and the state button turns blue.

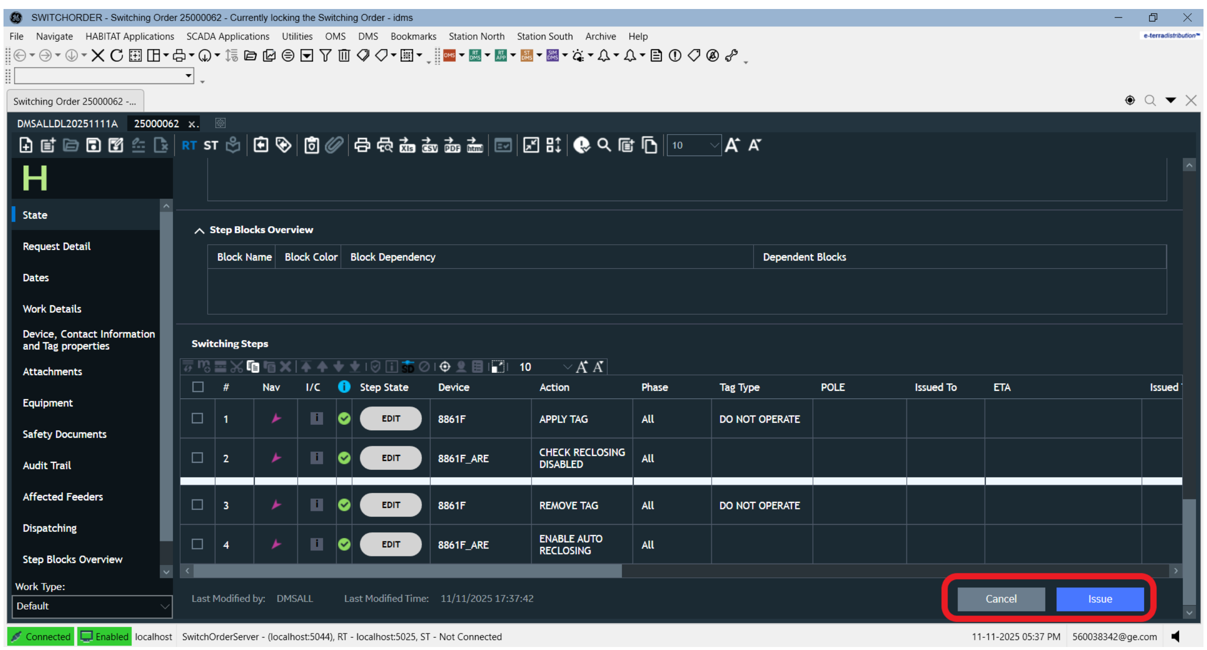




6. Review the HLH document in read only access mode and provide comments.



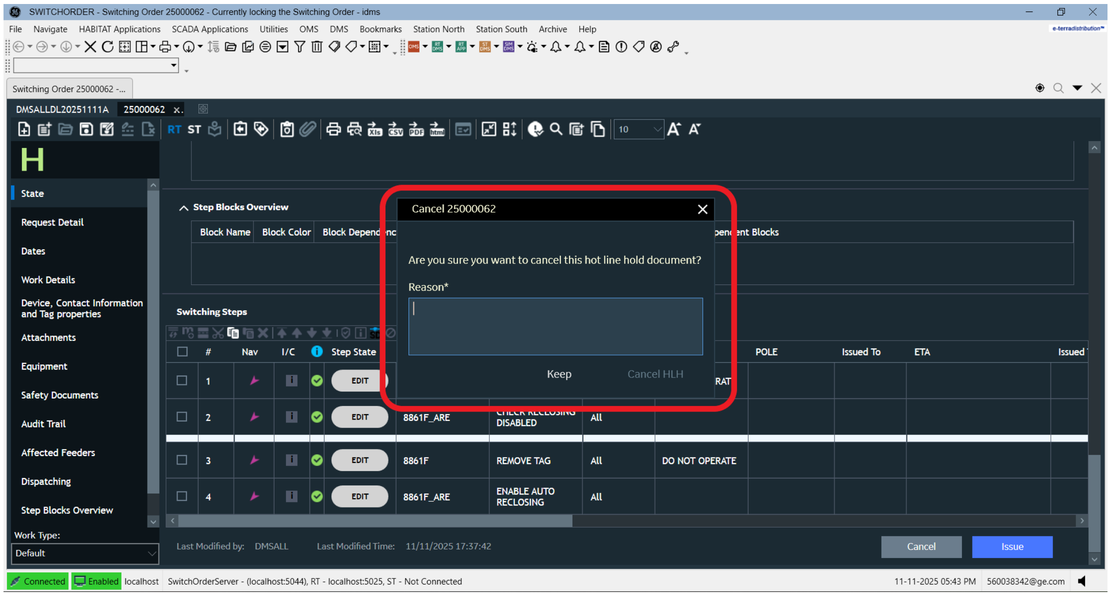
7. Select Cancel or Issue based on the review.



a. If Cancel is selected, a dialog box prompts for a cancellation description. The following conditions must be met to confirm the cancellation:

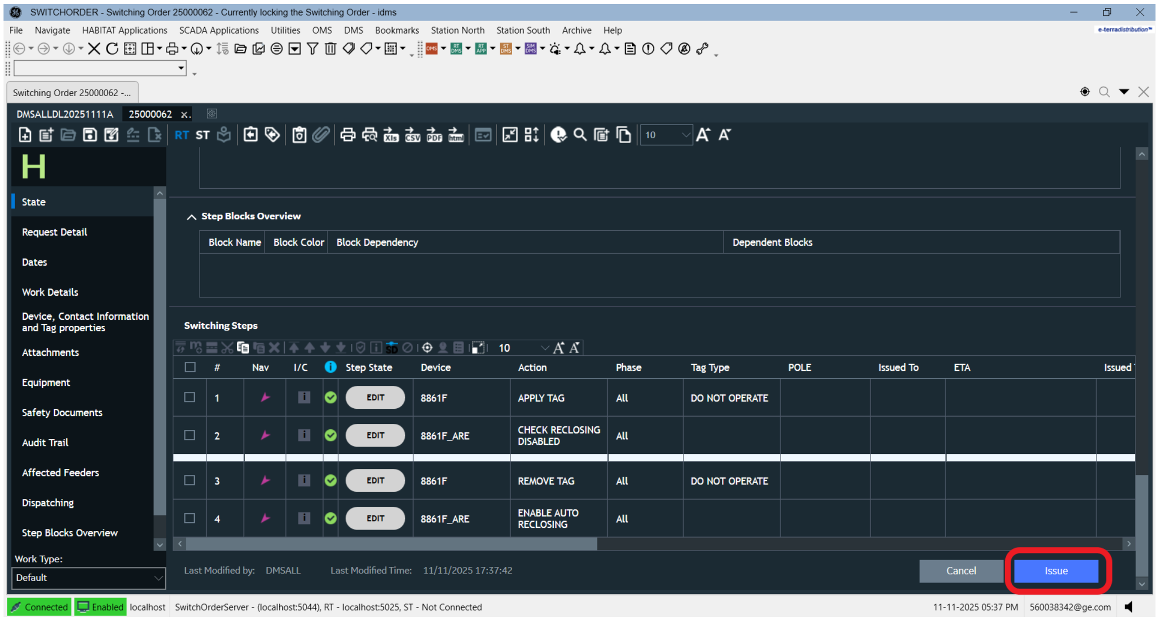
   i. A cancellation description must be entered.

   ii. Cancel HLH must be selected.



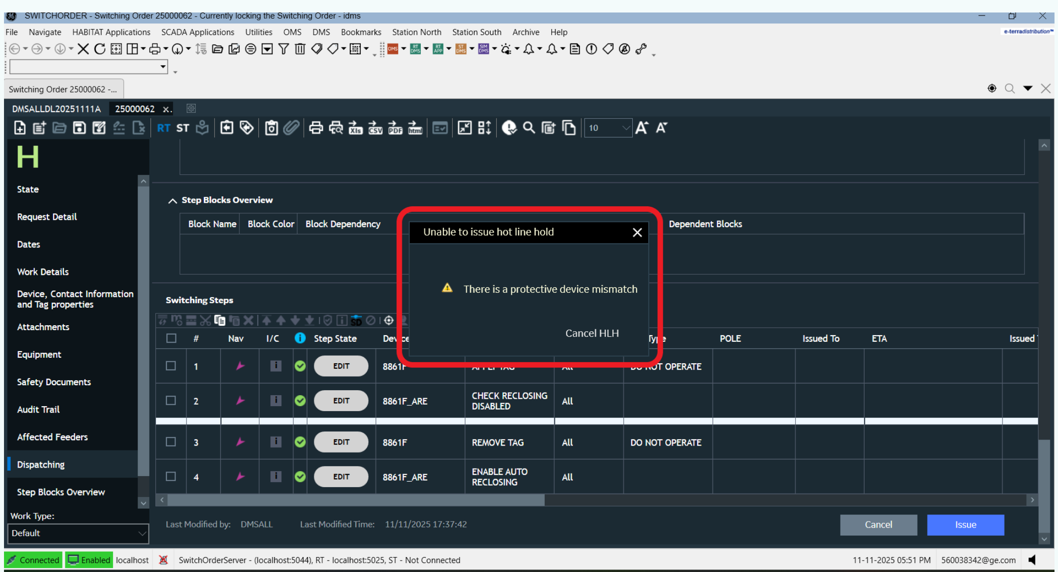
> **Note**
>
> 1. The Cancel HLH button remains inactive until a reason for cancellation is provided.
> 2. If the User clicks on Keep button, popup will get closed and the document will remain in Pending State.
> 3. Upon clicking Cancel HLH button, the document state changes to Cancelled, and the all state buttons grey out. The document remains in read only mode.

b. If the user clicks the Issue button, the system validates the protective device to check if the protective device is same or not. After confirmation of protective device, system executes switching steps. The HLH document changes to Issued state and the state button turns blue. The document version increases by one, and the HLH document becomes read-only.



Note

1. If there is protective device mismatch, a message will appear stating **There is a protective device mismatch** with Cancel HLH button. The Cancel HLH button redirects to the cancellation process.

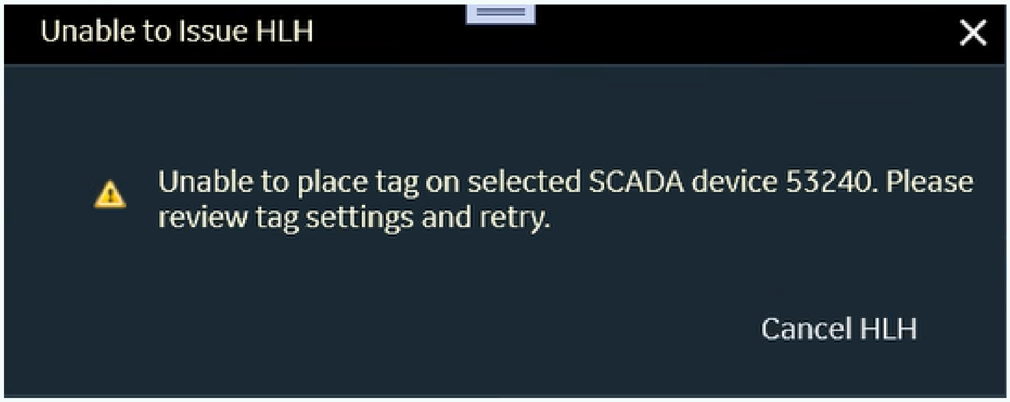


2. If protective device matches, the following switching steps will be executed:
   a. Place the Do-not-operate tag.
   b. Check if auto-recloser is disabled.

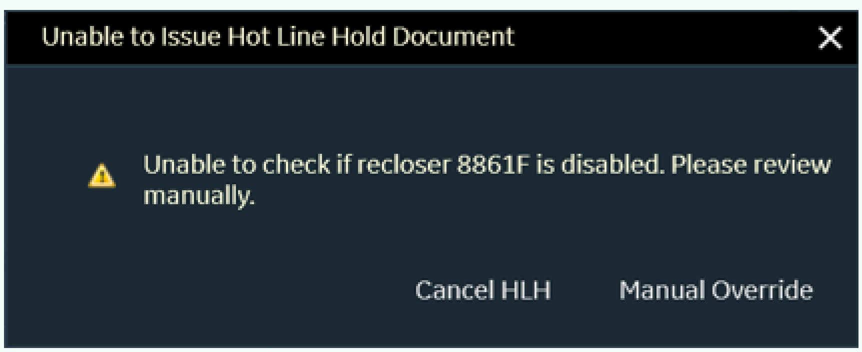
Note

Scenarios for Forward Switching Steps:

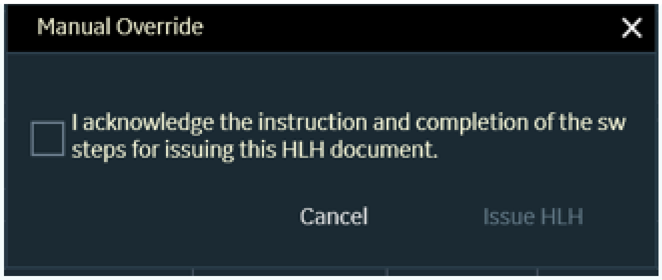
1. SCADA Device - Unable to add Do Not Operate Tag:
    a. If the system is unable to place a tag on a SCADA device, the following popup is displayed

    b. Select the Close (X) button to review the tag settings and retry the operation. After a successful retry, the two forward switching steps are completed.
    c. Select Cancel HLH to initiate the cancellation process.
2. SCADA Device - Unable to disable Auto-Recloser:
    a. When the Issue button is selected for a SCADA device, the following popup is displayed.

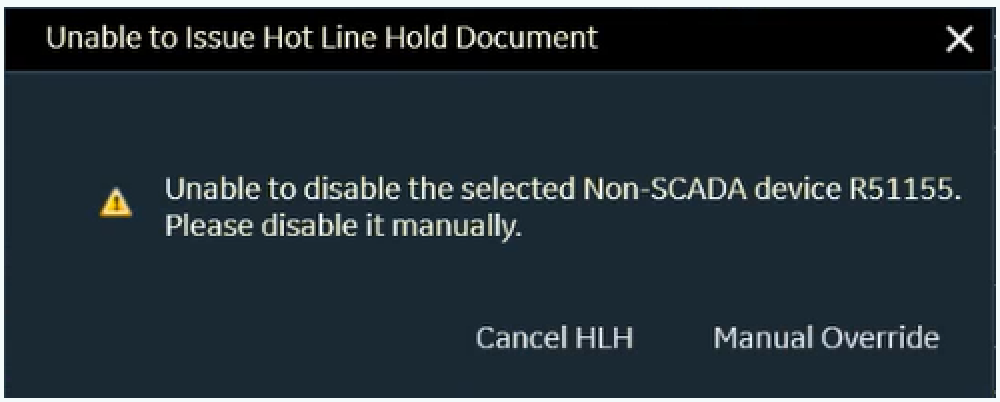
       

    b. Select Cancel HLH to initiate the cancellation process.
    c. Select Manual Override to open an acknowledgement popup displaying the message: **"I acknowledge the instruction and completion of the new sw steps for issuing this HLH document."** The popup provides Cancel and Issue HLH buttons. The Issue HLH button remains disabled until the acknowledgement checkbox is selected.

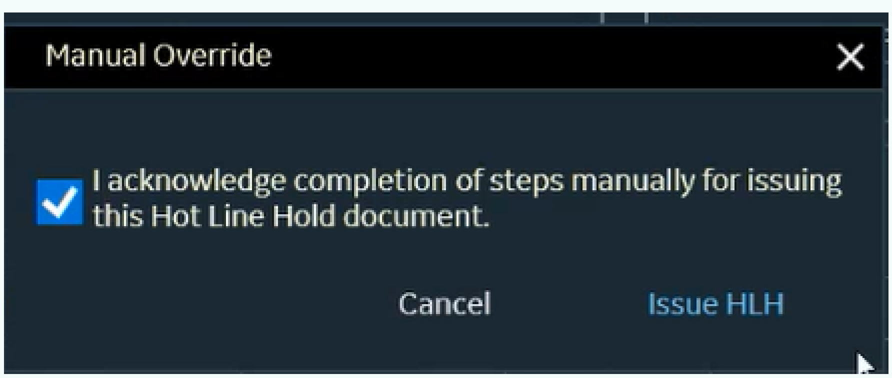


d. Select Cancel to close the popup.
e. Select Issue HLH to complete the forward switching steps.

3. Non-SCADA Device - Unable to disable Auto-Recloser:
   a. When the Issue button is selected for a Non-SCADA device, the following popup is displayed.

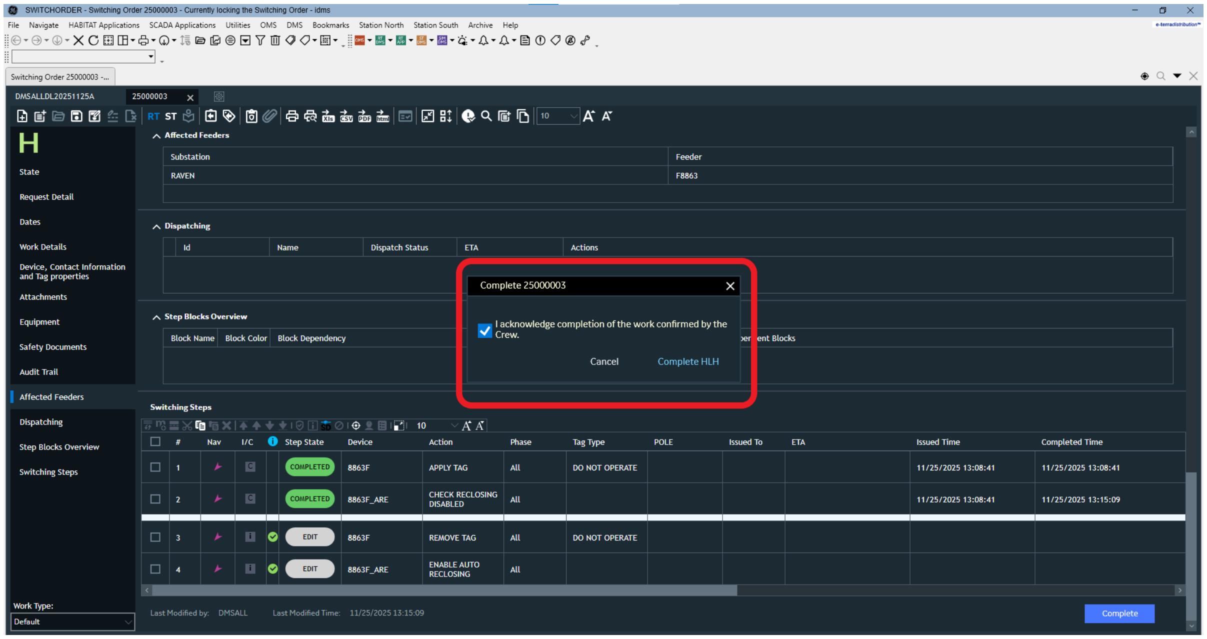


   b. Select Cancel HLH to initiate the cancellation process.
   c. Select Manual Override to open an acknowledgement popup with the message **"I acknowledge completion of steps manually for issuing this Hot Line Hold document."** The popup provides Cancel and Issue HLH buttons. The Issue HLH button remains disabled until the acknowledgement checkbox is selected.

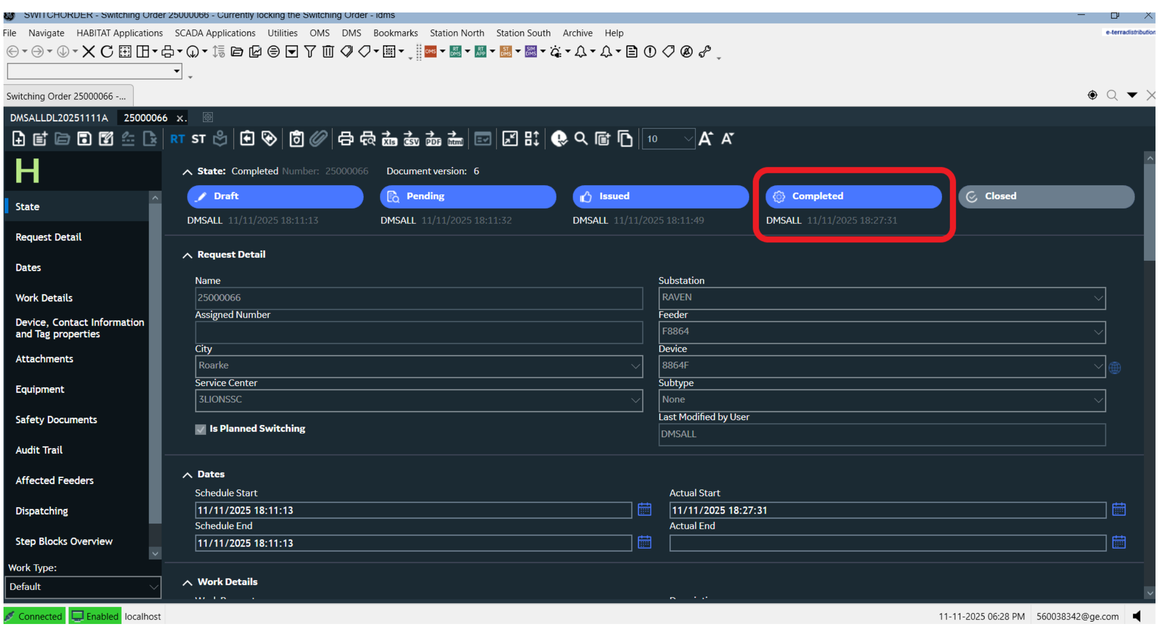


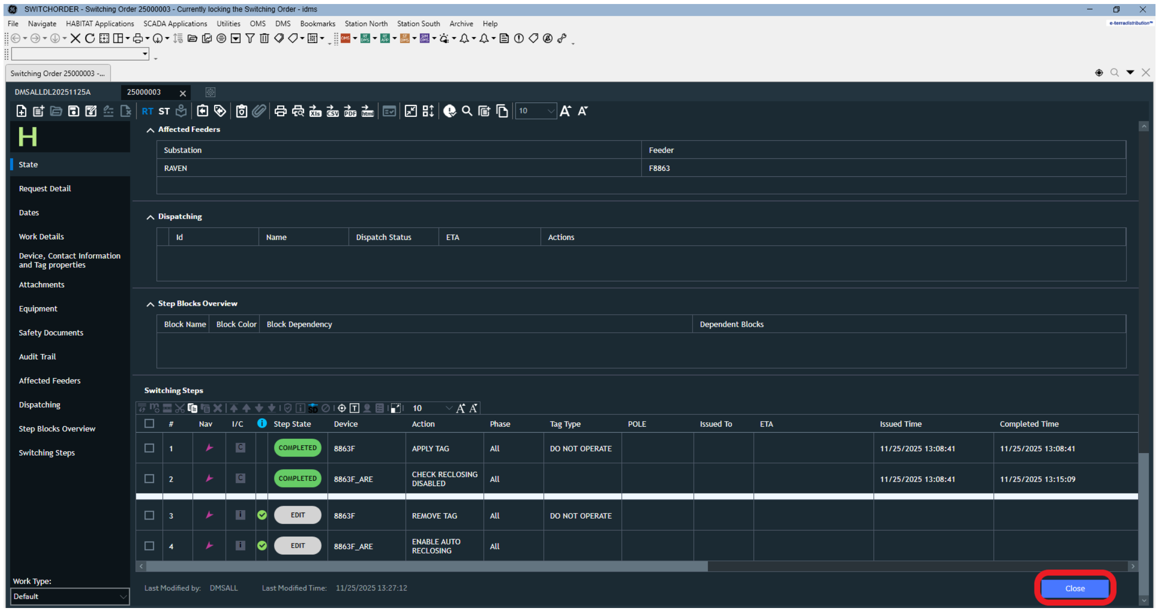
   d. Select Cancel to close the popup.
   e. Select Issue HLH to complete the forward switching steps.

8. Click on Complete button, and an acknowledgement popup appears with Cancel and Complete HLH buttons.

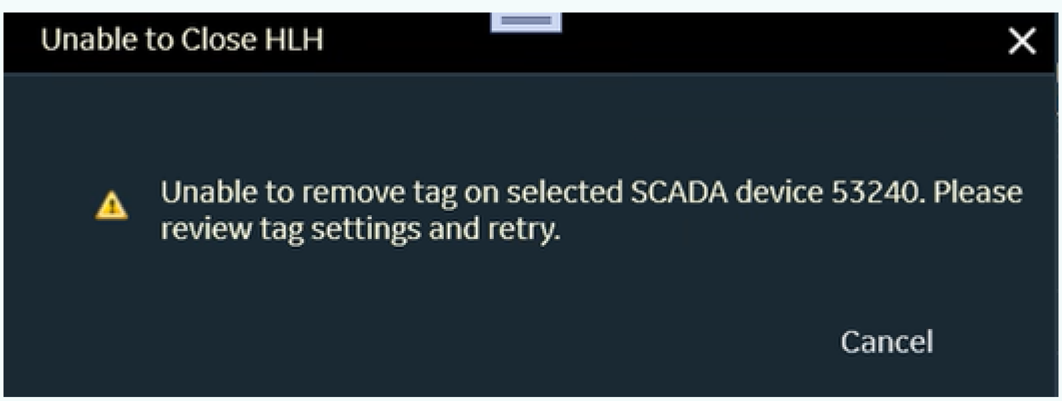


   a. Clicking Cancel closes the popup.
   b. Clicking Complete HLH changes the document state from Issued to Completed turning state button blue, and the document version increases by one, and displays a Close button at the bottom of the HLH document.

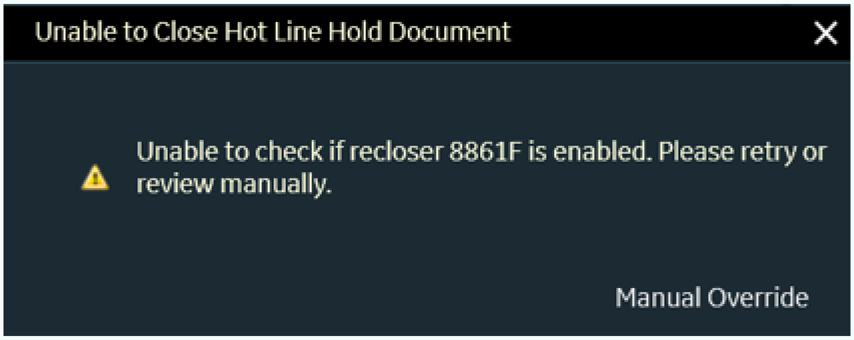




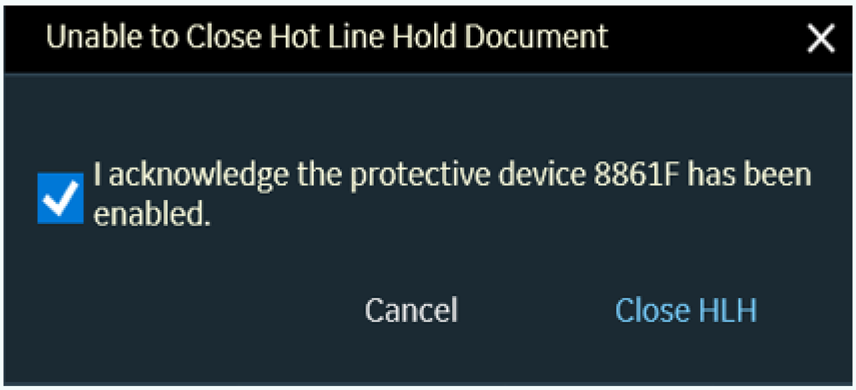
9. Click on Close button. The following switching steps are executed:
    a. Remove the Do-not-operate tag.
    b. Check if auto-recloser is enabled.
10. Upon successful execution, the state changes from Completed to Closed turning state button blue, and the document version increases by one, and the HLH document is read-only.

> **Note**
>
> Scenarios for Backward Switching Steps:
>
> 1. SCADA Device - Unable to remove Do Not Operate Tag:
>   a. If the system is unable to remove the tag from a SCADA device, the following popup is displayed.
>
> 
>
>
>   b. Review the tag settings and retry the operation. After a successful retry, the two backward switching steps are completed.
> 2. SCADA Device - Unable to enable Auto-Recloser:

a. When the Close button is selected for a SCADA device, the following popup is displayed.



b. Select Manual Override to open the acknowledgement popup displaying the message: "I acknowledge the protective device has been enabled."

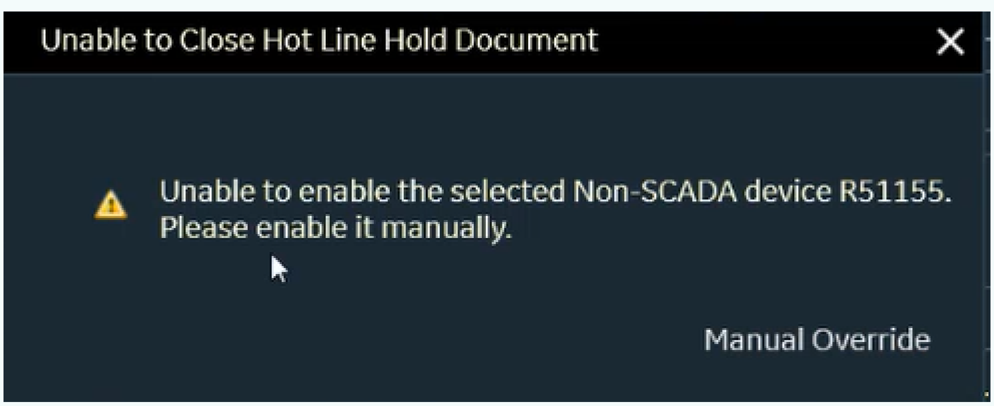


c. Select Cancel to close the popup.

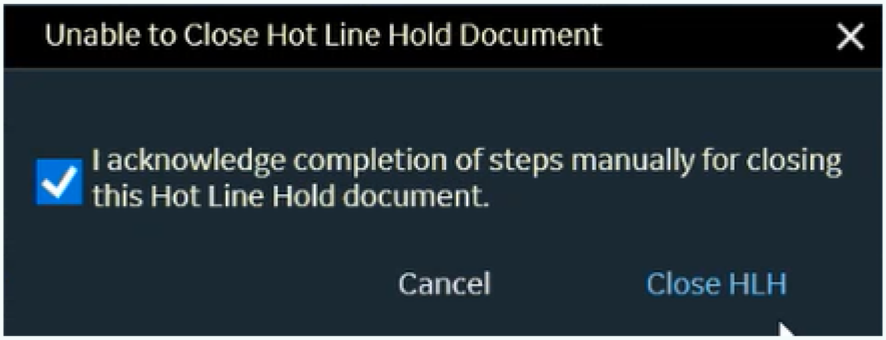
d. Select Close HLH to complete the backward switching steps. The state changes to Closed.

3. Non-SCADA Device - Unable to enable Auto-Recloser:

a. When the Close button is selected for a Non-SCADA device, the following popup is displayed.



b. Select Manual Override to open the acknowledgement popup displaying the message: "I acknowledge completion of the steps manually for closing this Hot Line Hold document".



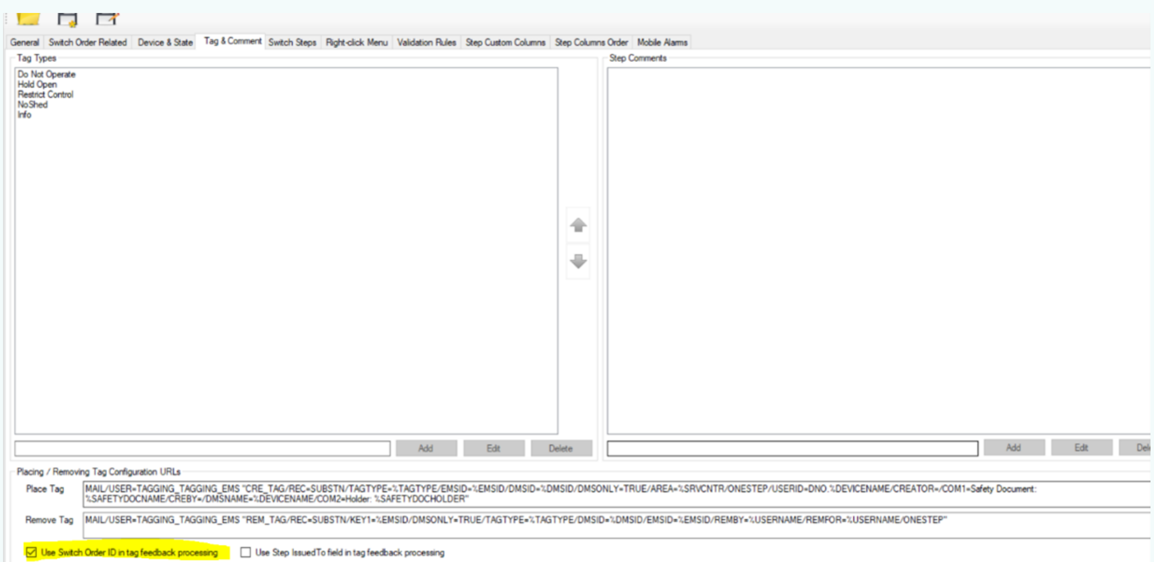
c. Select Cancel to closes the popup.

d. Select Close HLH to complete the backward switching steps. The state changes to Closed.

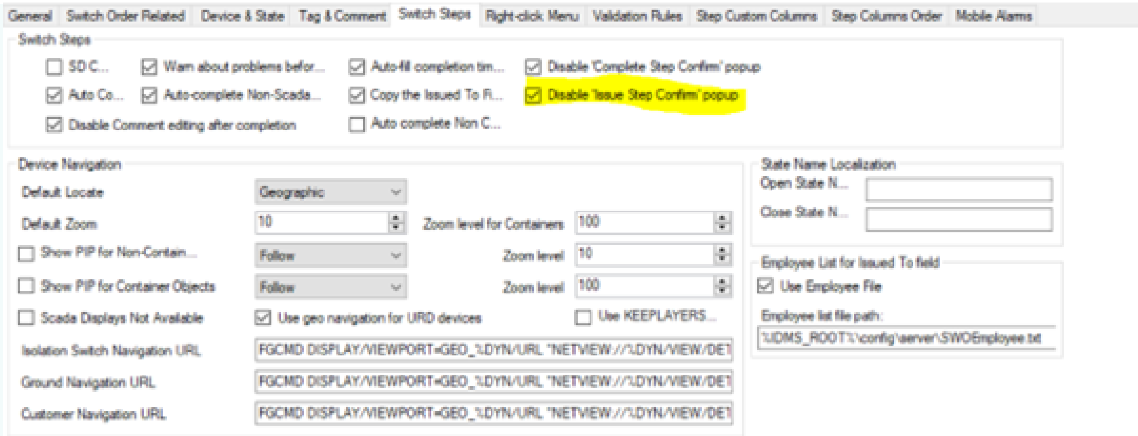
Note

Multi-Tagging and Tag Dependencies:

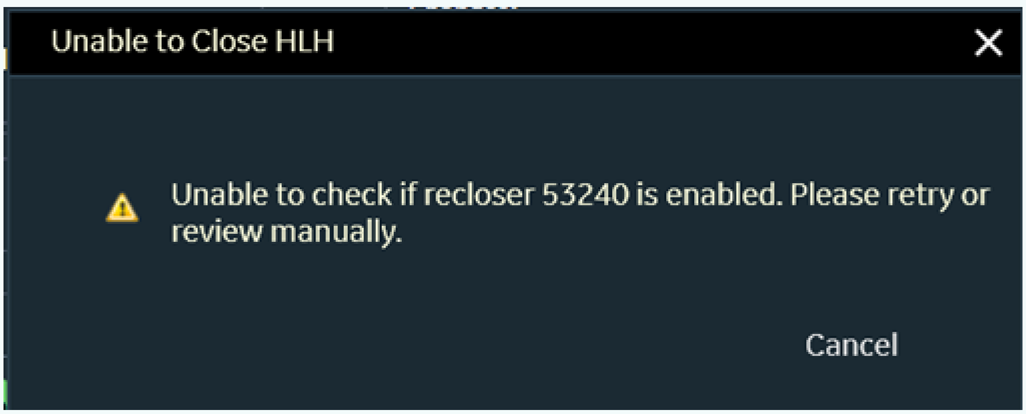
1. To enable multi-tagging in HLH, select the following checkbox in the Switching Order Configuration Editor under the **Tag & Comment** tab.



2. To suppress confirmation message popups during switching step execution, select the following checkbox in the **Switch Steps** tab of the Switching Order Configuration Editor.



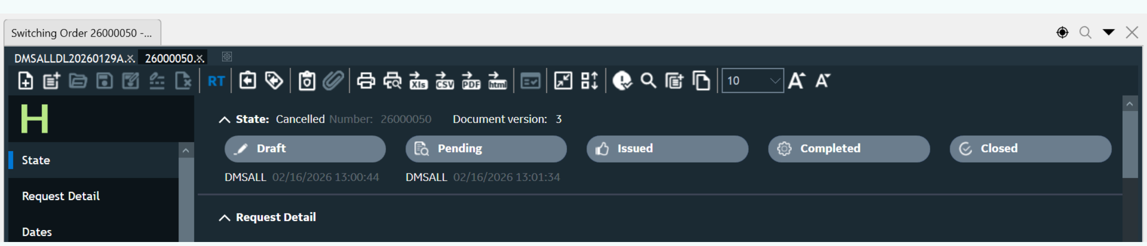
3. When multiple tags are applied to a recloser, removing one tag removes only the selected tag. Due to tag dependency, the recloser remains in the disabled state and the following popup is displayed. The document remains in the Completed state.

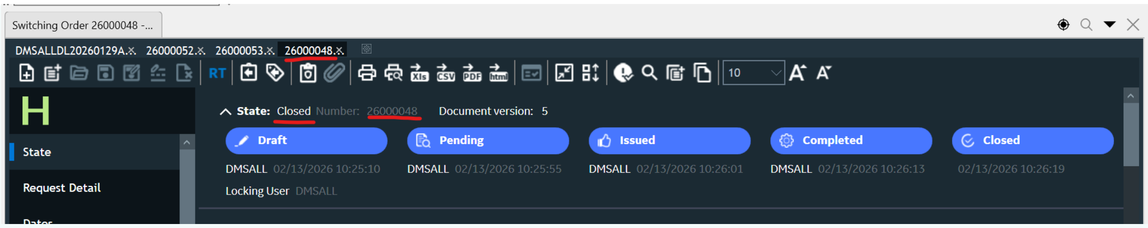


4. The recloser is not enabled until all tags applied to it are removed.
5. When the final tag is removed, the recloser is enabled and any other tag-dependency documents that are open in active tabs are automatically moved to the Closed state.

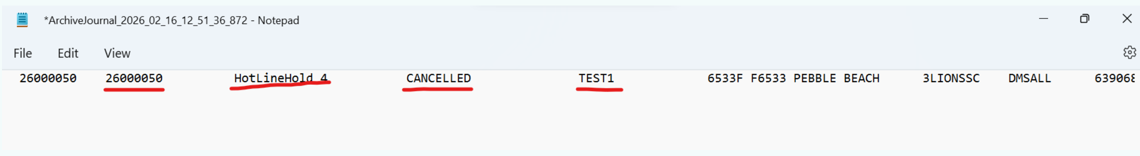
Note

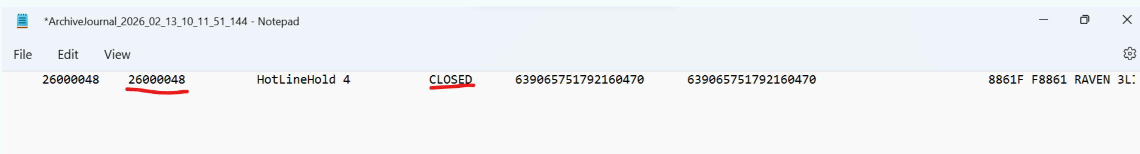
1. Data archiving is performed for two states: Cancel and Close.



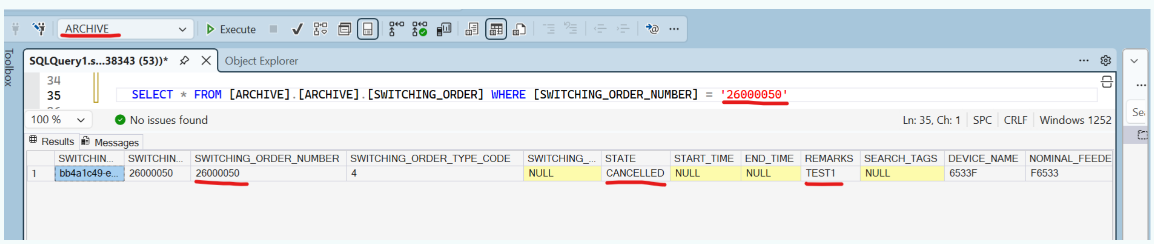


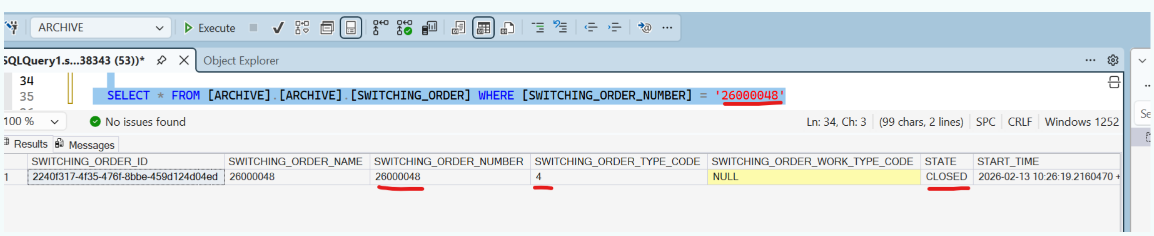
2. All important information like **Requesting Details**, **Area Details**, **Order Number**, **State** and so on, for Hot Line Hold documents is stored in a journal file. For the Cancel state, an additional field, **Remark**, stores the value entered in the comments field.





3. Data from the journal files is captured in a relational SQL database.



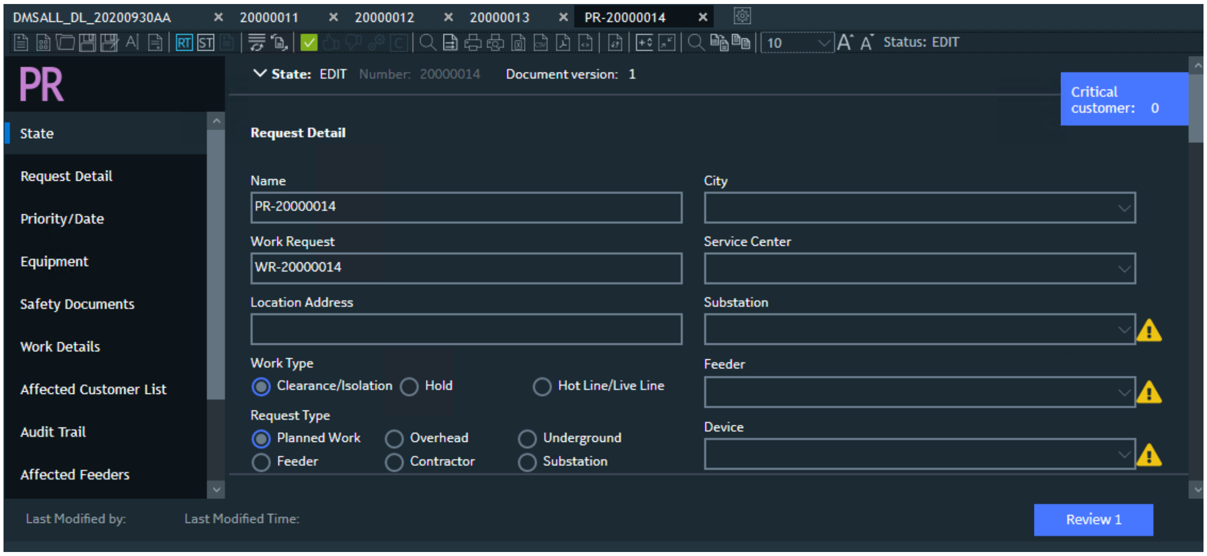


3.3. Creating a Planned Request Document

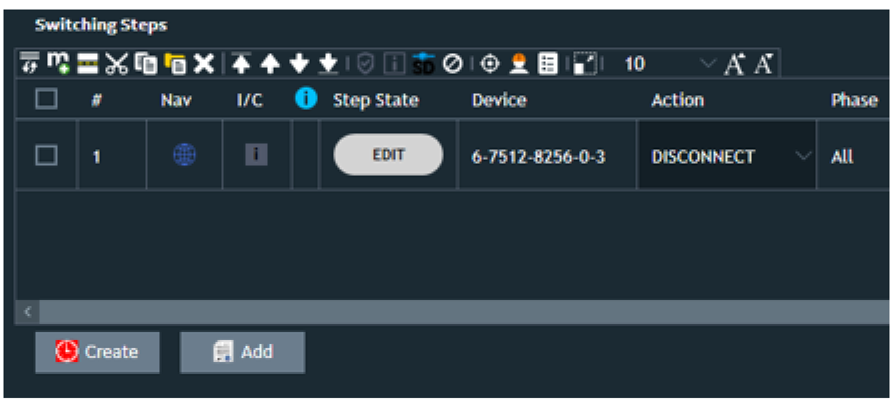
To create a planned request document:

1. Click the New Planned Request button on the common toolbar or on the Switching Order Options page.

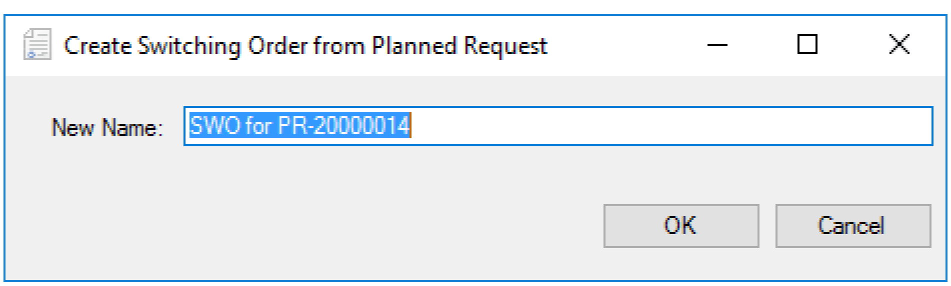
   A new, empty planned request document appears.



2. Update the planned request document as needed. For more details, see Planned Requests.
3. Create a planned outage from the planned request using the Create Planned Outage button under the Switching Steps section. For more information, see Creating a Planned Outage from a Planned Request.
4. Add a switching order document from the planned request using the Add button under the Switching Steps section.

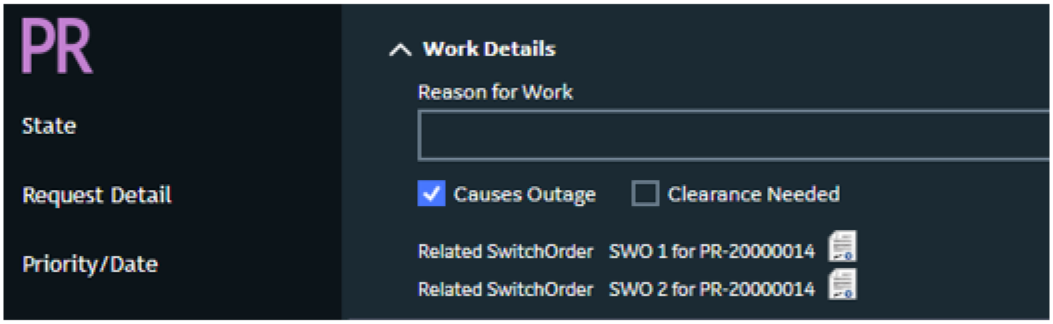


Clicking this button opens a dialog to indicate the name of the new Switching Order being added to the planned request.



Multiple switching orders can be added to the planned request and these are displayed in the

Work Details section of the planned request.



3.3.1. Creating a Planned Outage from a Planned Request

To create a planned outage from a planned request document:

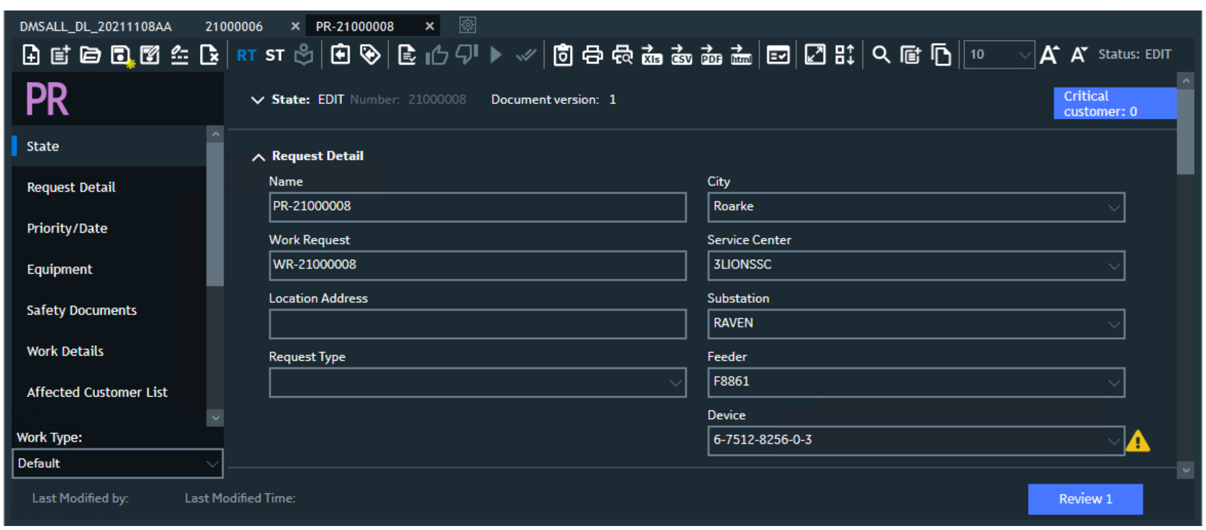
1. Click the New Planned Request button on the common toolbar or on the Switching Order Options page.

   A new, empty planned request document appears.

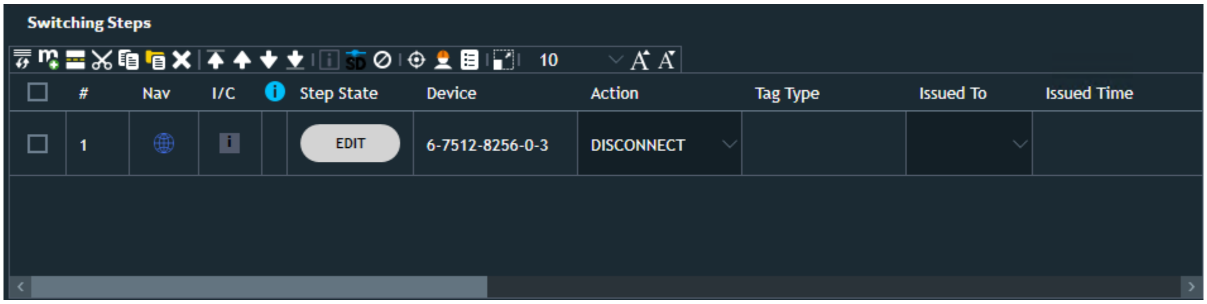
2. On the network view, navigate to the device of record (device to be associated with the planned request document).

3. Select the Add to Switching Order option from the device toolbar or right-click menu.

   The Request Details section is populated with device details.

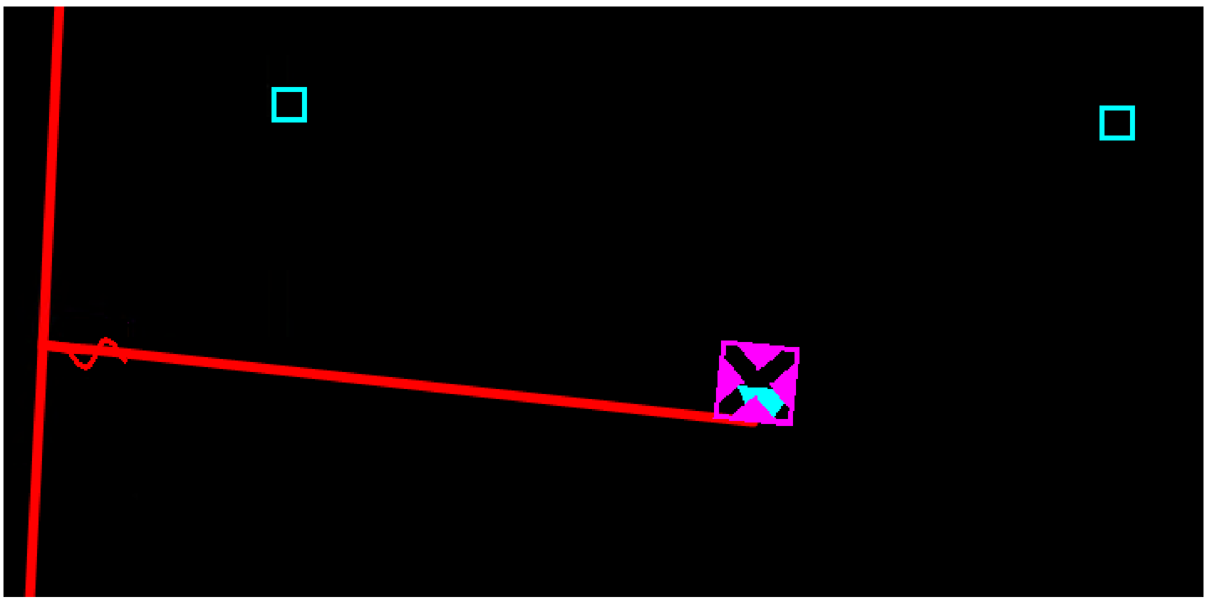


The Switching Steps section includes a step to change the state of the device of record.



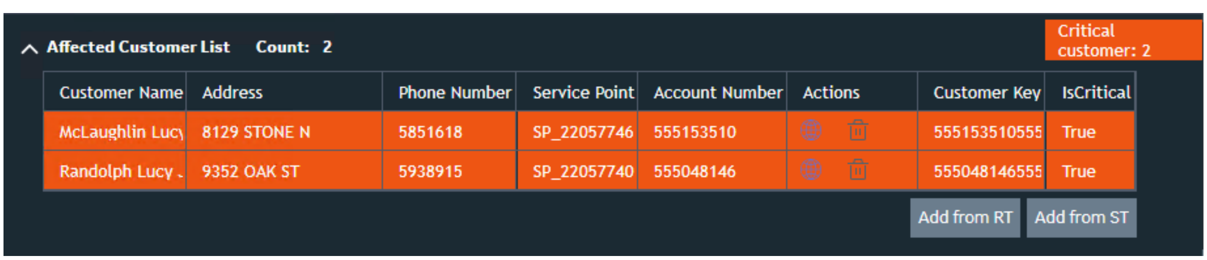
4. On the network view, select the Trace Down option for the device of record.

   The customers downstream of the device of record are traced.

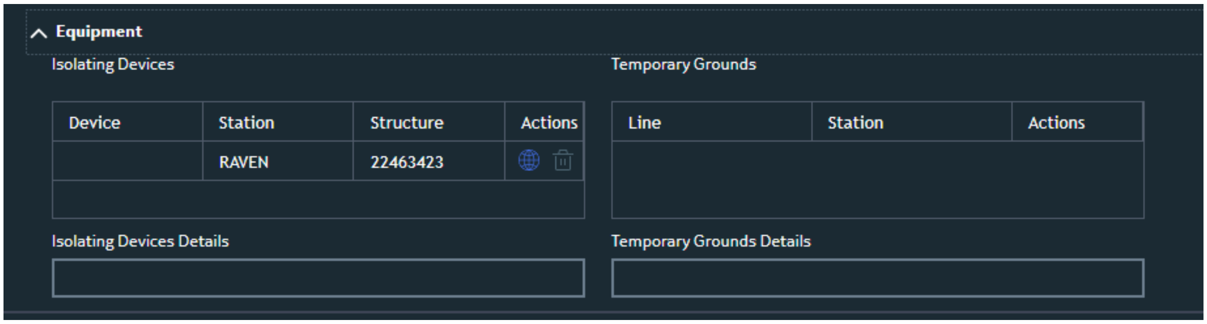


5. In the switching order form, click the Add from RT button in the Affected Customer List section.

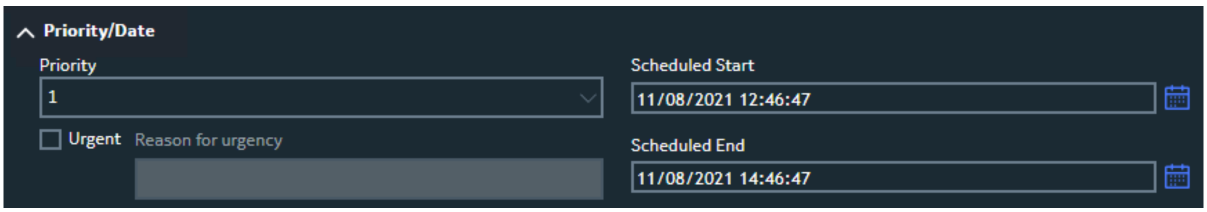
   The traced customers are added the list of affected customers.



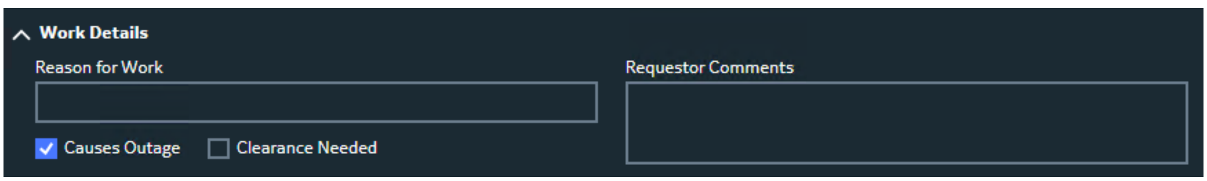
6. On the network view, navigate to an isolation device upstream of the device of record.
7. In the switching order form, drag and drop the isolation device into the Isolating Devices group in the Equipment section.



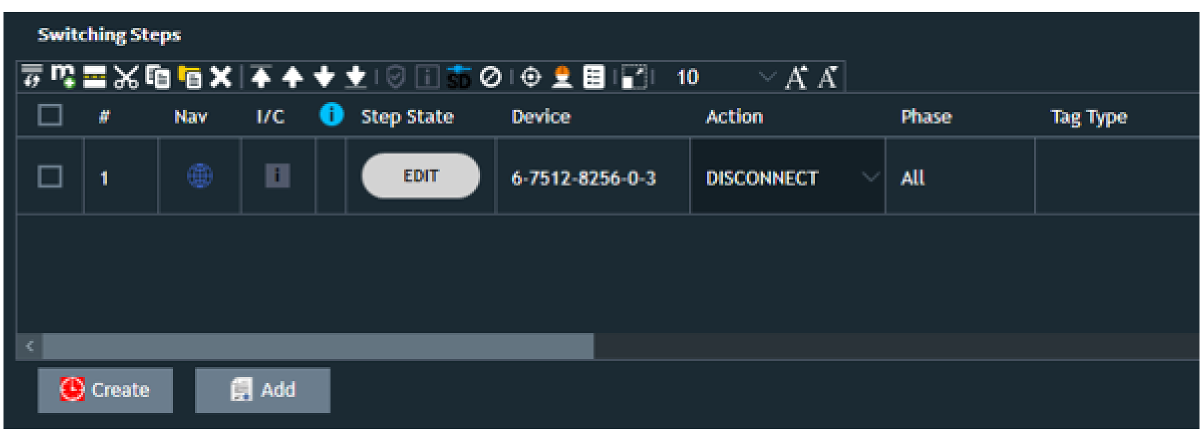
8. In the switching order form, set the planned outage start and end dates in the Priority/Date section.



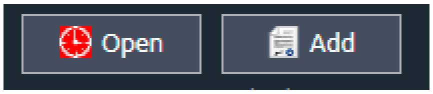
9. In the switching order form, select the Causes Outage option in the Work Details section.



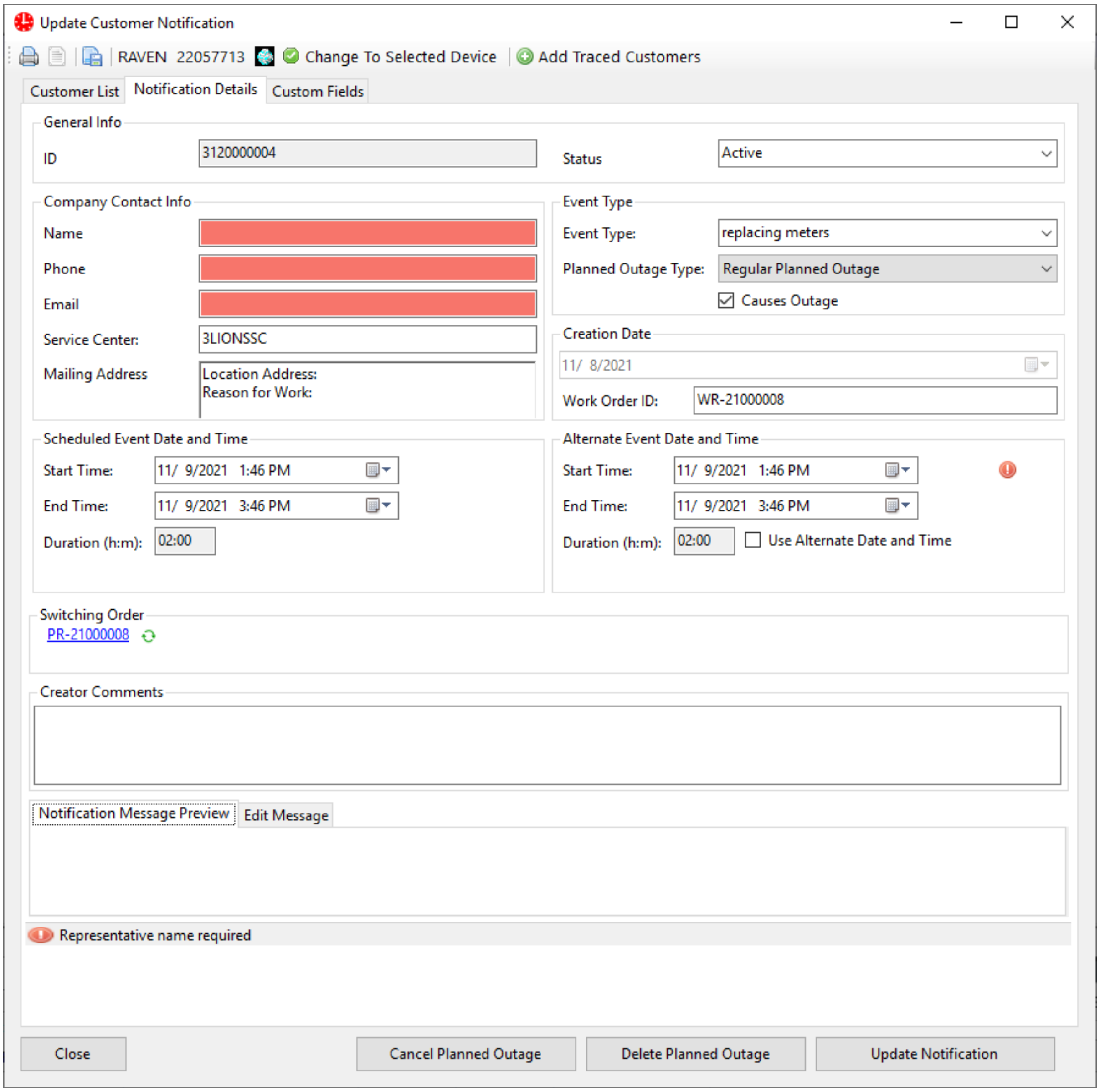
10. In the switching order form, click the Create Planned Outage button under the Switching Steps section. This button only appears if the Causes Outage option in the Work Details section is selected. The validation also checks that the scheduled date, isolation device, and affected customers are set.



11. In the switching order form, click the Open Planned Outage button under the Switching Steps section.



The associated planned outage opens.



12. Edit the planned outage and send the planned outage notification, as appropriate.

> **Note**
>
> When you delete a planned request document that is associated with an active planned outage, the system prompts you whether also you want to cancel the planned outage.

3.4. Saving a Switching Order

To save a switching order:

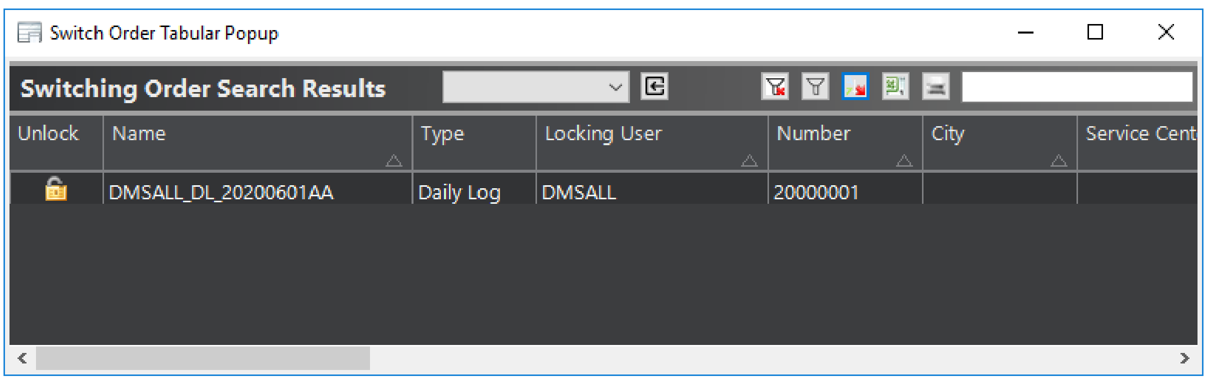
1. In the Name field, enter a unique name of the switching order.

   > **Note**
   >
   > If the Switching Order Name field is blank, the switch order is saved under the automatically generated switch order number shown in the Number field of the Switch Order Properties panel.

2. Click Save on the Switching Order Form toolbar.

   The Name field dims and becomes read-only. The switch order is now locked for parallel editing and can be simultaneously opened by other users in read-only mode.

   > **Note**
   >
   > A switching order is locked until closed by the operator. In case of emergency (for example, crash of the client) the switching order is not properly closed and remains locked. To unlock it, enter NETVIEW://SO.RT/POPUP/TABULAR/SWITCHORDERUNLOCK and click Unlock for the required switching order.



3.5. Renaming a Switch Order

Once a switching order is saved, editing the name field is disabled.

To rename the switching order:

1. Click Rename on the Switching Order Form toolbar.

   The Rename dialog box appears.

2. Enter a new unique name into the New Name field and click OK.

   If the new name is not unique and already exists in the system, the Failed Renaming Switch

Order message appears. To fix the problem, rename the switching order accordingly.

If the new name is unique, the Name field is updated with the new switching order name.

3.6. Retrieving a Switch Order

To retrieve a saved switching order:

1. Click Open File on the Switching Order Form toolbar.

   The Switching Order Search pop-up opens.

2. Enter the search options to filter switch orders and click Find.
3. From the Switching Order Search Results grid, select the desired switching order file.
4. Click Open to edit the unlocked switch orders or Read to view the read-only switch orders.

For more information about the Switching Order Search pop-up, refer to Switching Order Search.

3.7. Deleting a Switch Order

To delete a switching order:

1. Open the switching order that needs to be deleted.
2. Click Delete on the Switching Order Form toolbar.

   The Delete dialog box appears.

3. To delete the switching order, click Yes.

The switching order is now permanently deleted from the library.

3.8. Copying a Switch Order

To copy a switching order:

1. Open the switching order that needs to be copied.
2. Click Create a Copy of Switching Order on the Switching Order Form toolbar.

   The Create Switching Order Copy dialog box appears.

3. Enter a name for the new switching order and click Yes.

The switching order is copied and opened in a new tab.

3.9. Managing Corrupted Switch Orders

Once the system identifies a corrupted switching order or a daily log, it moves the document to the %IDMS_ROOT%\dynamics\SwitchOrder\GarbageDocs folder.

After the document is moved, you can create a new switching order with a similar name.

The corrupted documents are identified and moved to the GarbageDocs folder:

- When starting the Switching Order server
- Every SearchIndexUpdateInterval, which is set in the Switching Order Server configuration file (by default, 300 seconds).
- When attempting to open the document via the Switching Order Search display or from the Options page.
- When loading the Switching Order form and attempting to open a corrupted daily log.

> **Note**
>
> The GarbageDocs folder is only created when the system identifies a corrupted document.

3.10. Creating and Using Templates

A "template" is a type of switching order. Templates can be used to create multiple, identical switching orders.

Templates cannot be executed and are always in the Created state.

A template can be retrieved and edited as required. Steps from a template can be copied and paste to and from another switching order.

3.10.1. Creating a New Template

To create a new template:

1. Click New Template on the Switching Order Form toolbar or click the New Template shortcut on the Options page. A new blank template form opens.
2. Add switching steps to the template as described in Adding Steps to a Switching Order and Copying Steps into the Switching Order.
3. In the Name field, enter a unique name for the template and click Save. After clicking Save, the name field is no longer editable. You can change the name by clicking the Rename button.

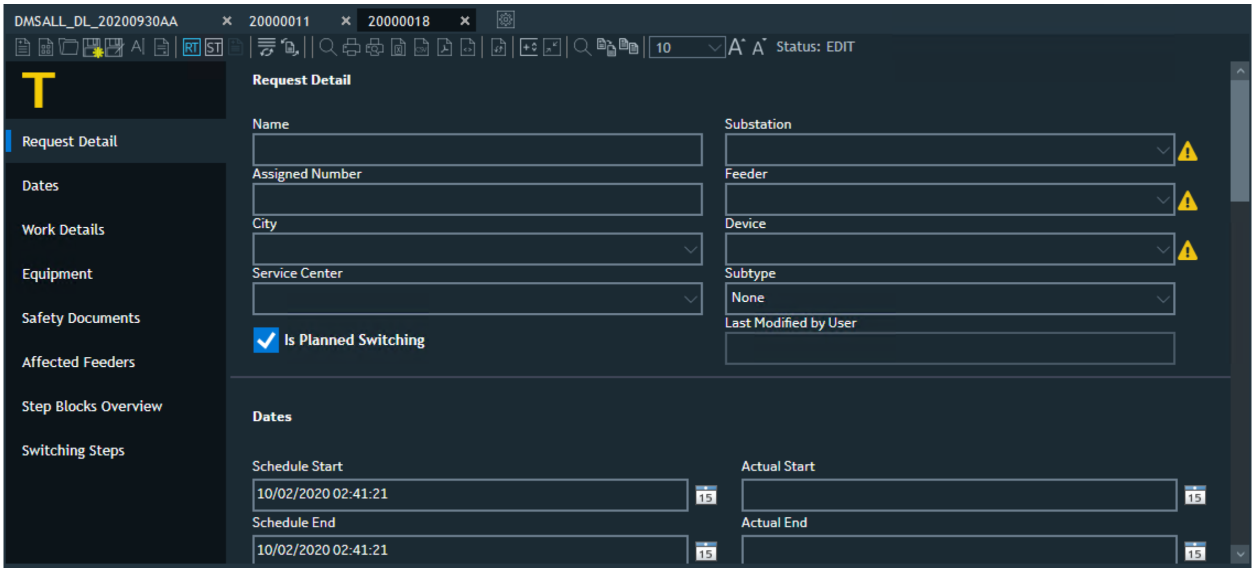
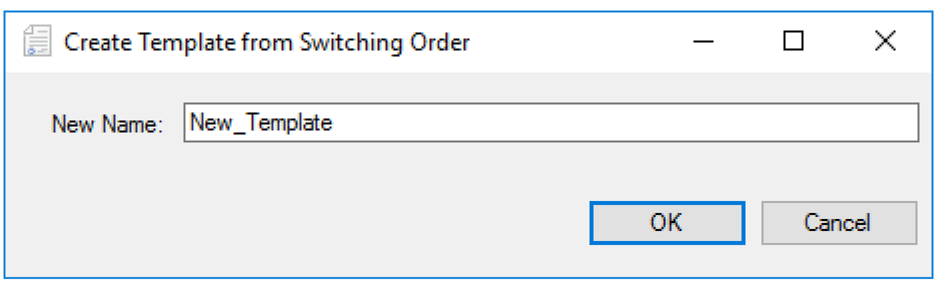


Figure 38. Creating a Template

3.10.2. Creating a Template from a Switching Order

To create a new template from a switching order:

1. Click Create Template From Switch Order on the Switching Order Form toolbar. The Create Template from Switching Order pop-up appears.
2. Enter the template name and click OK.



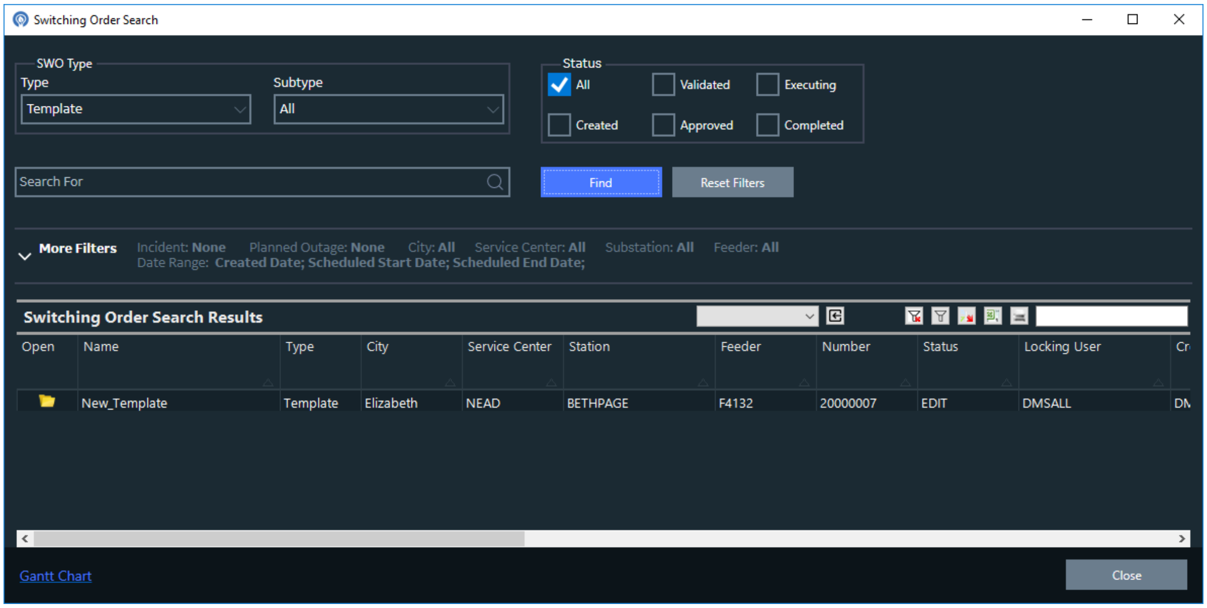
3. A template containing steps from the switching order opens.

3.10.3. Retrieving a Template

To retrieve and use a template:

1. To open the Switching Order Search pop-up, click Open File on the Switching Order Form toolbar.
2. From the Type drop-down list, select the Template option and set other filter options as needed to filter templates.
3. Click Find.

The search results appear in the Switching Order Search Results grid.

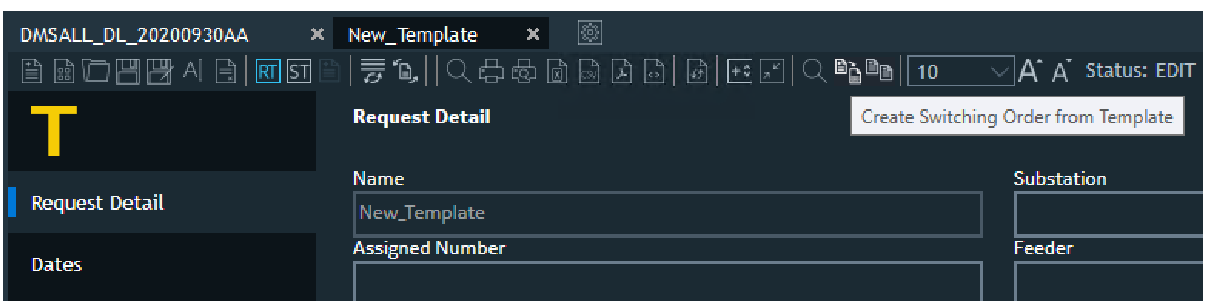


4. From the Switching Order Search Results grid, select the template to open.
5. Click Open. The template is now open.

3.10.4. Creating a Switching Order from a Template

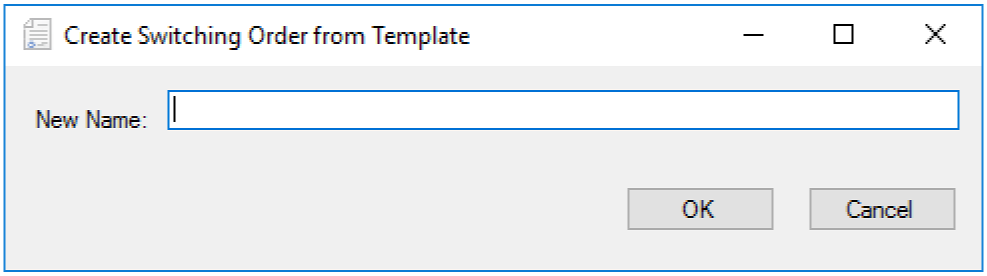
To create a switching order from a template:

1. Open a template.
2. Click the Create Switching Order from Template button on the template toolbar.



The Create Switching Order from Template window appears.

3. Enter a unique name for the new switching order in the text box and click OK.



The new switching order is created.

4. Enter other information into the corresponding fields.

   The switching order is in edit mode and can be changed as required before being approved and executed.

5. Using the Copy and Paste buttons on the Switching Order Form toolbar, copy steps to the new switching order as necessary.
6. Click Save.

   The new switching order is saved. It can now be retrieved, edited, and executed as described in Executing and Completing a Switching Order.

3.11. Exporting a Switching Order

You can export a switching order document into a file in the Excel, CSV, HTML, or PDF format. During the export, an XML file with all switching order data is created and then transformed into the required format using the XSL template file that is defined in the SwitchOrderClientConfig.xml configuration file.

> **Note**
>
> The output file from Excel is in XML format and has an Excel-specific schema, which means it does not use the non-ASCII `.xls` or `.xlsx` formats that are proprietary to Excel. Therefore, the file may not be opened by double-clicking it, and you will need to open it from within Excel.

To export a switching order document:

1. Open a switching order document.
2. Click the appropriate export button on the Switching Order toolbar (for example, click Export Switching Order to PDF).

   Switching Order files are created in the output folder defined in the `SwitchOrderClientConfig.xml` configuration file (by default, %IDMS_ROOT%).

1 of 1

SWITCHING ORDER

DOCUMENT VERSION: 8
DATE: Oct 02, 2020
SUBSTATION: RAVEN
NAME: 20000006
RELATED INCIDENT: 2760000010
PLANNED OUTAGE: RAVEN 6-7512-4121-0-4

TIME ISSUED	TIME COMPLETE	ISSUED TO	ISSUED BY	CUSTOM VALUES	STEP	SWITCHING PREPARED BY: DMSALL APPROVED BY: DMSALL APPROVED BY: DMSALL	SAFETY INSTRUCTIONS
10/2/2020 2:56 AM	10/2/2020 2:56 AM	John Smith	DMSALL	DNOMX: 933352.625000 SITE: 7663 West St RAVEN	1	Disconnect Transformer	
10/2/2020 2:56 AM		John Smith	DMSALL	DNOMX: 933352.625000 SITE: 7663 West St RAVEN	2	Connect Transformer	
		John Smith		DNOMX: 933905.000000 SITE: 6-7512-5418-0-0	3	OPEN 10908-1	
		John Smith		DNOMX: 933905.000000 SITE: 6-7512-5418-0-0	4	CLOSE 10908-1	

ISOLATING DEVICES		
Device ID	Device Name	Station Name
17824867	6-7512-4121-0-4	RAVEN
17466121	10908-1	RAVEN

NO TEMPORARY GROUND DEVICES
NO SAFETY DOCUMENTS

Figure 39. Switching Order PDF Document

3.12. Customizing Switching Order

This section describes how to customize the appearance of the Switching Order displays.

3.12.1. Adjusting the Font Size

You can adjust the font size on the Switching Order sections and the Switching Steps grid using the following controls:

- **Change the Font Size:** Opens a drop-down list to select the font size.
- **Increase the Font Size:** Increases the font size.
- **Decrease the Font Size:** Decreases the font size.

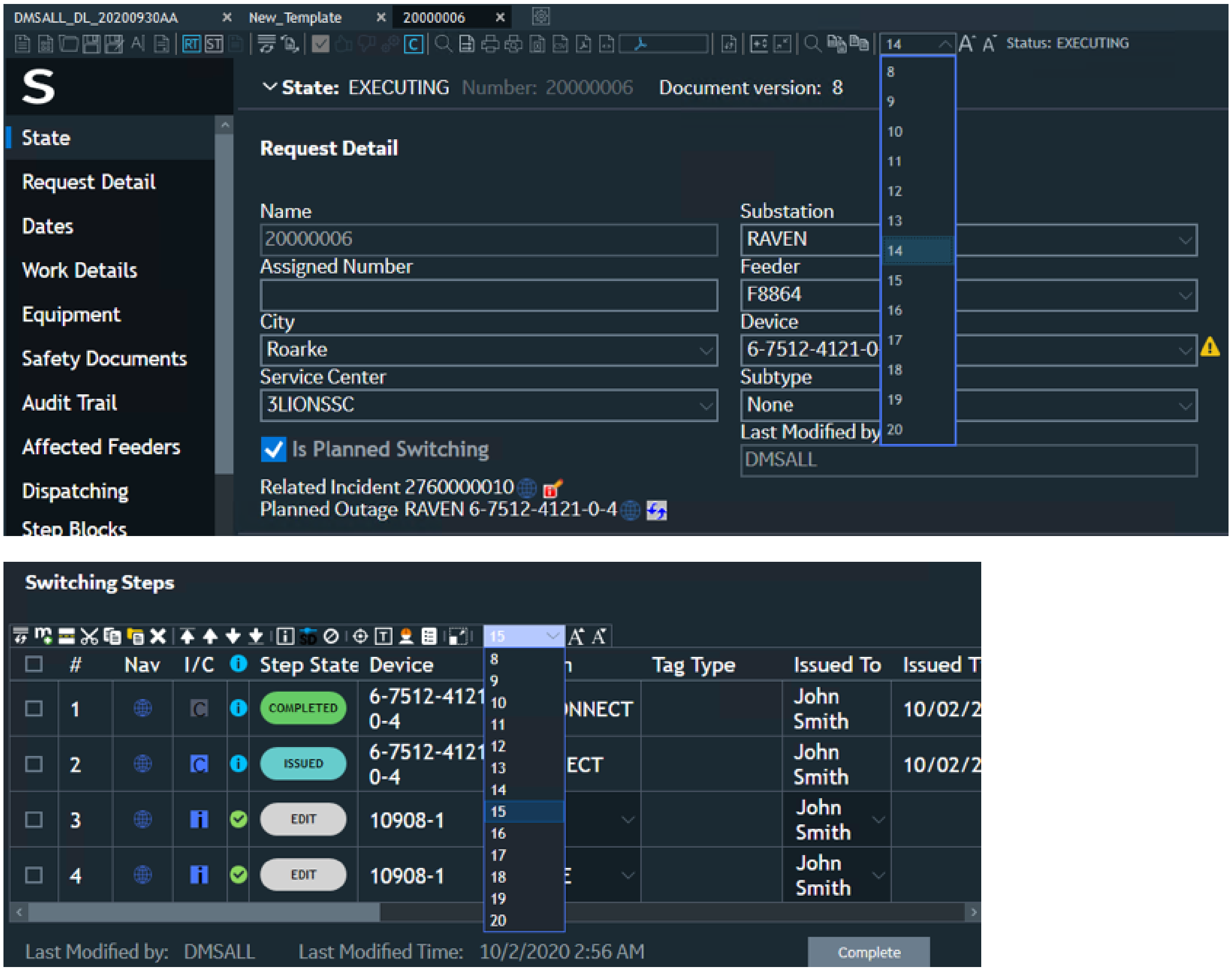


Figure 40. Adjusting the Font Size

Font size user preferences are automatically stored in an XML file and are applied to all Switching Orders operated by the user. The XML files are stored in the `C:\Users\<User>\AppData\Roaming\General Electric Technology GmbH\Advanced Presentation Framework\<Version>` directory with the `RecentList_SwitchOrder.RecentFontSize.<User_Name>.xml` file name.

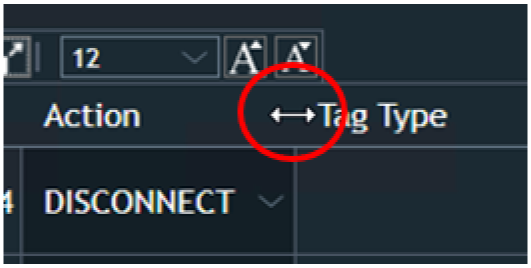
3.12.2. Configuring the Column Width and Order

You can configure column width and column order in the Switching Steps grid.

To set column width:

1. Hover the mouse cursor over the column header border.

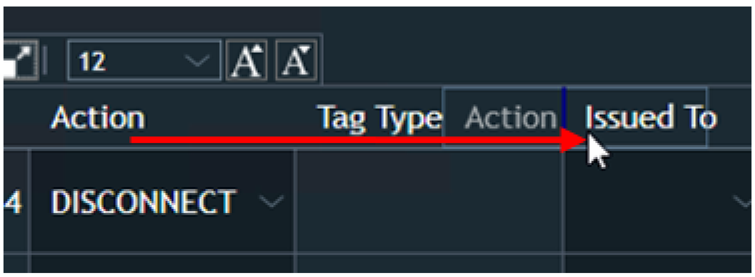
   A double-sided arrow appears.



2. Drag the arrow to adjust the column width.

To configure column order:

- Drag the column header to a different position and release the mouse button.



The column order user preferences are automatically stored in an XML file and are applied to all Switching Orders operated by the user. The XML files are stored in the C:\Users\<User>\AppData\Roaming\General Electric Technology GmbH\Advanced Presentation Framework\<Version> directory with the SwitchingOrder_RecentColumnOrder.<User_Name> file name.

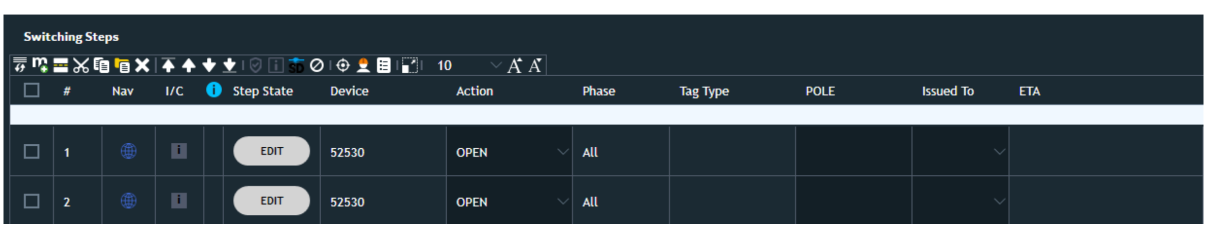
3.12.3. Add and Label a Step Separator

Separators help organize blocks of steps within a Switching Order, making the order easier to read. A separator is represented by a blank line with a header text and can be inserted at any point in the Switching Order.

To insert a separator:

- Select the step immediately above the desired position and click the Separator button on the Switching Steps toolbar.

  A thick line appears above the selected switching step, as shown below.

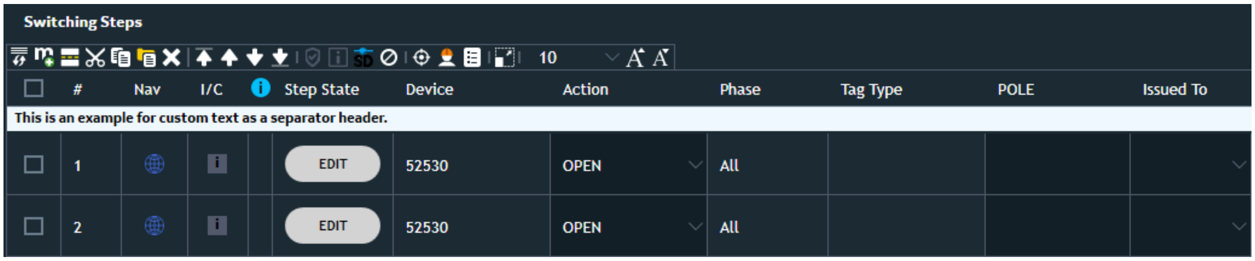


3.12.3.1. Labeling a Separator

Two options are available for labeling a separator. These options are configured in the *Switch Order Client Configuration* > *Switch Steps Tab* > *Switching Step Separator*.

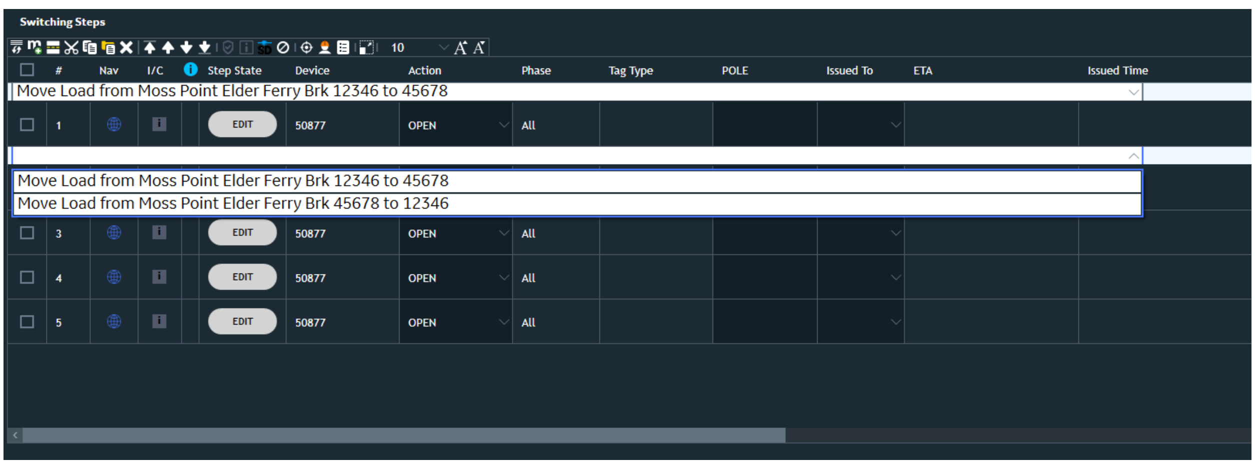
Option 1: Custom Text

- Check the **Use Custom Text for Switching Step Separator** option.
- When checked, the separator can be clicked to enter custom text.

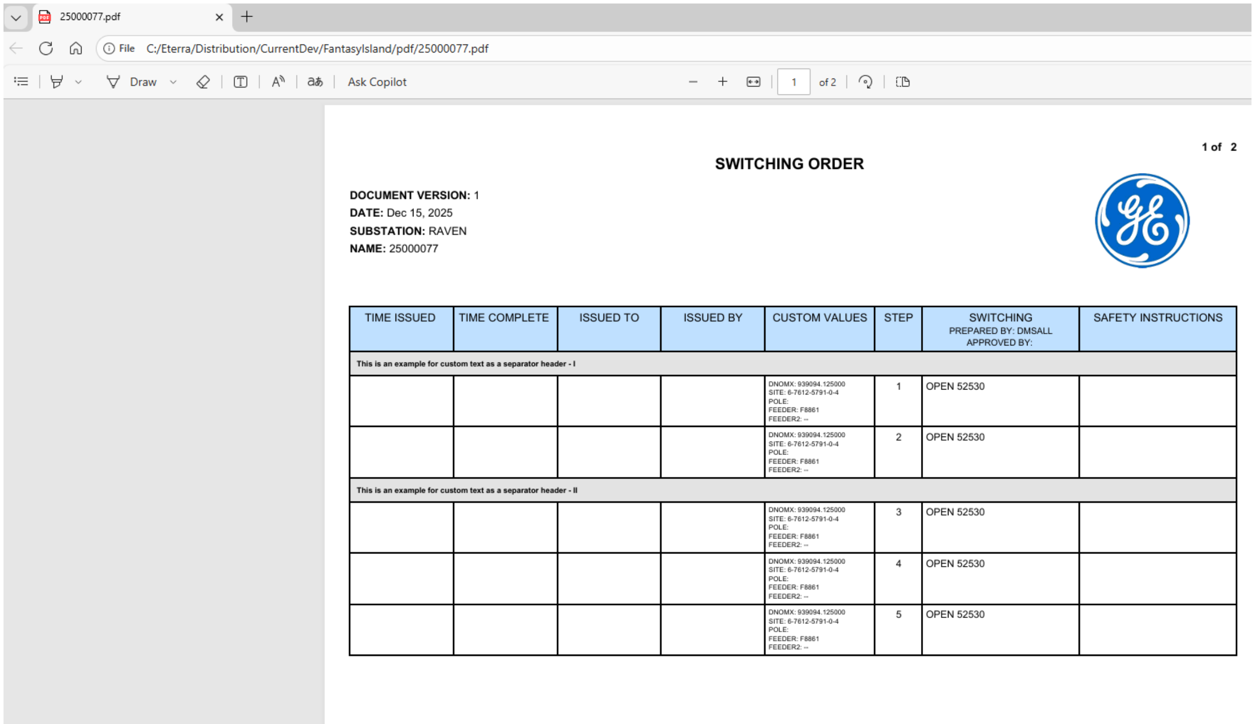


Option 2: Predefined Separator Pairs

- Uncheck the **Use Custom Text for Switching Step Separator** option.
- When unchecked, predefined separator pairs must be selected.
- Each pair includes **Separator Text** and **Header Text**, which are displayed in a dropdown list when adding a separator.



> **Note**
>
> - Separators move with their associated step and can be deleted using the **Remove Separator** button on the Switching Steps toolbar.
> - Regardless of configuration, the **Header Text** displayed in the separator is included in all exported formats, such as **XLS, CSV, PDF, HTML, Print Preview**.



1 of 2

SWITCHING ORDER

DOCUMENT VERSION: 1
DATE: Dec 15, 2025
SUBSTATION: RAVEN
NAME: 25000077

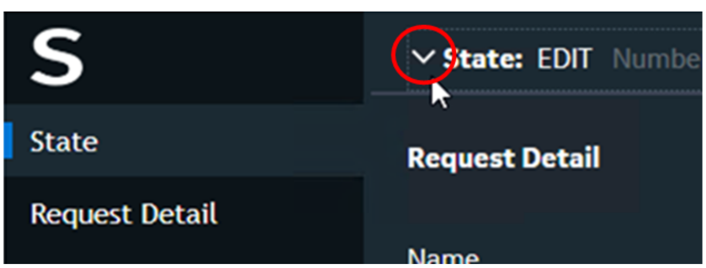
TIME ISSUED	TIME COMPLETE	ISSUED TO	ISSUED BY	CUSTOM VALUES	STEP	SWITCHING PREPARED BY: DMSALL APPROVED BY:	SAFETY INSTRUCTIONS
This is an example for custom text as a separator header - I							
				DNOMX: 939094.125000 SITE: 6-7612-5791-0-4 POLE: FEEDER: F8861 FEEDER2: --	1	OPEN 52530	
				DNOMX: 939094.125000 SITE: 6-7612-5791-0-4 POLE: FEEDER: F8861 FEEDER2: --	2	OPEN 52530	
This is an example for custom text as a separator header - II							
				DNOMX: 939094.125000 SITE: 6-7612-5791-0-4 POLE: FEEDER: F8861 FEEDER2: --	3	OPEN 52530	
				DNOMX: 939094.125000 SITE: 6-7612-5791-0-4 POLE: FEEDER: F8861 FEEDER2: --	4	OPEN 52530	
				DNOMX: 939094.125000 SITE: 6-7612-5791-0-4 POLE: FEEDER: F8861 FEEDER2: --	5	OPEN 52530	

3.12.4. Expanding and Collapsing Sections

You can collapse and expand the Switching Order State and Switching Steps sections at run time.

To expand the Switching Order State section, do one of the following:

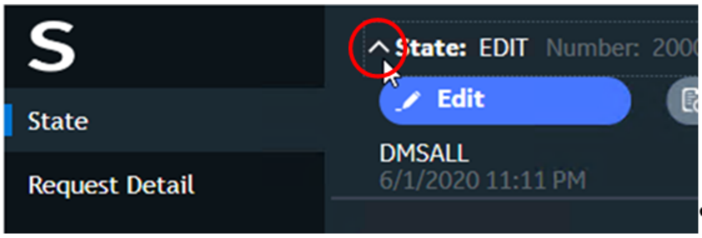
- Click the Expand button near the section header.



- Click the Expand Header Information button on the Switching Order toolbar.

To collapse a Switching Order section:

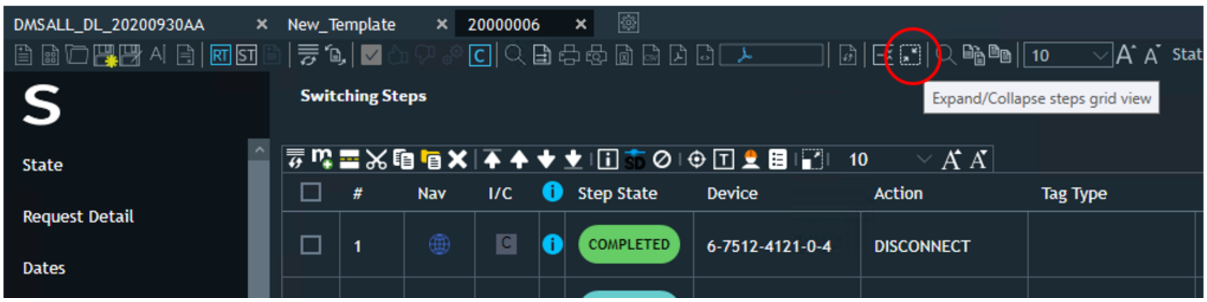
- Click the Collapse button near the section header.



- Click the Collapse Header Information button on the Switching Order toolbar.

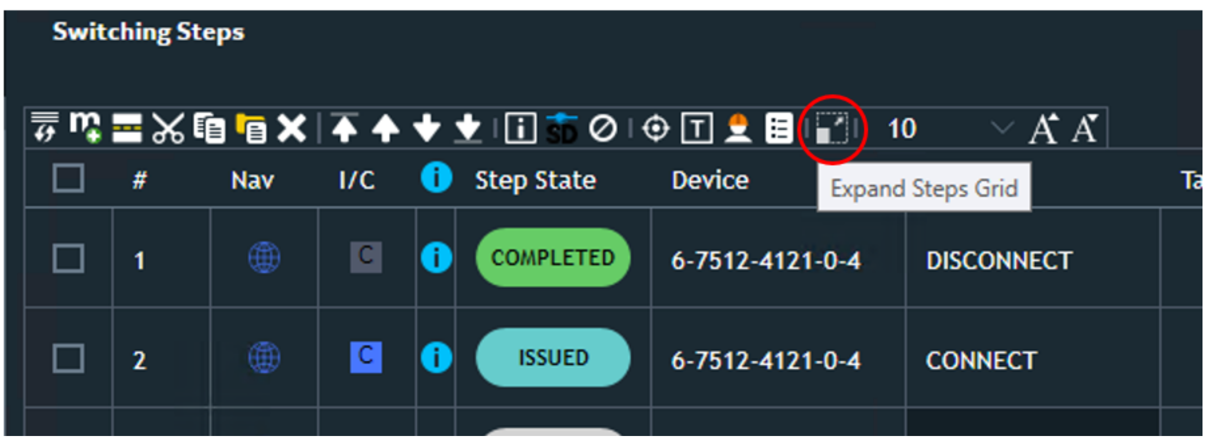
To expand/collapse the Switching Steps section:

- Click the Expand/Collapse Steps Grid View button on the Switching Order toolbar.



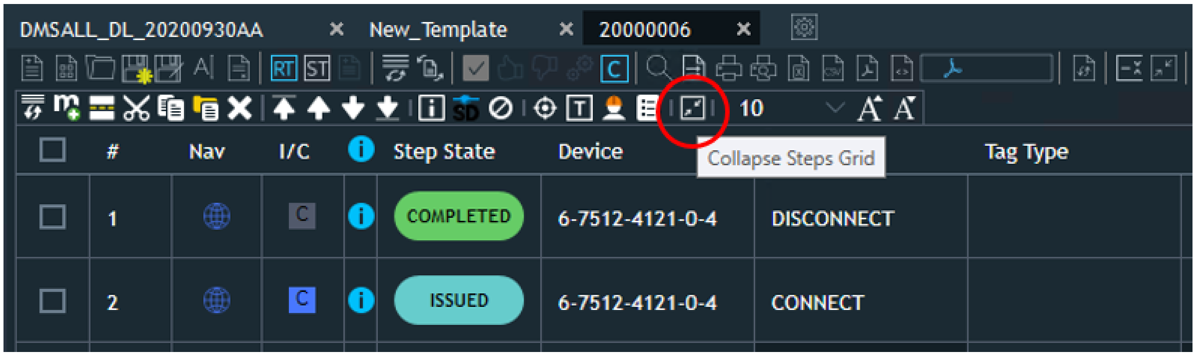
To hide Switching Order details and maximize the Switching Steps grid:

- Click the Expand Steps Grid button near the section header.



To show Switching Order details and minimize the Switching Steps grid:

- Click the Collapse Steps Grid button near the section header.


DMSALL_DL_20200930AA
New_Template
20000006
10
Collapse Steps Grid
#
Nav
I/C
Step State
Device
Tag Type
1
COMPLETED
6-7512-4121-0-4
DISCONNECT
2
ISSUED
6-7512-4121-0-4
CONNECT

4. Manually Creating a Switching Order

Switching orders can be created in real-time mode or in study mode. However, it is more usual to start in study mode, so that the effects of the planned switching can be properly examined.

4.1. Initializing Study Mode

To run study mode on an ADMS client:

1. Start the study server from the Windows Start> Programs> Eterra> Distribution> Client menu.
2. Open a viewport in real-time mode with the required substation.
3. Click Study Case Setup on the network view common toolbar or the Switching Operations form toolbar. The Study Case Setup pop-up appears (see the below image).
4. Initialize the study environment to the required station either with the real-time state or the normal state by clicking the corresponding button.

   The station model files are copied into the study case folders. The study environment is now initialized.

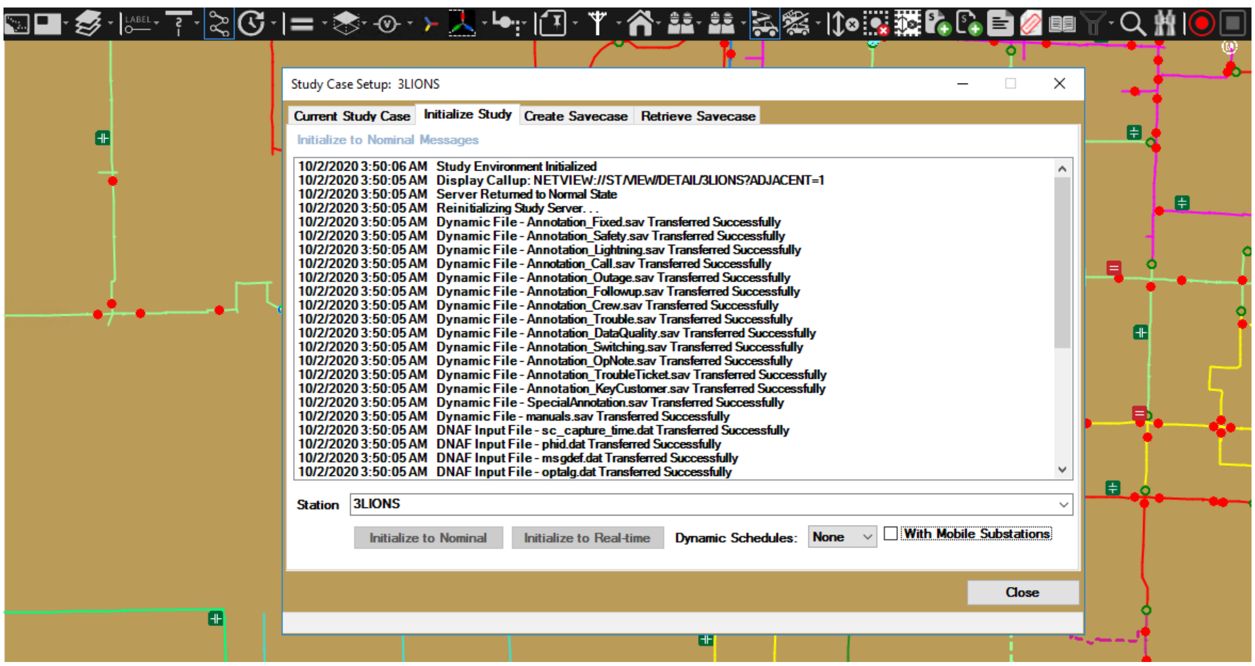


Figure 41. Initializing the Study Environment

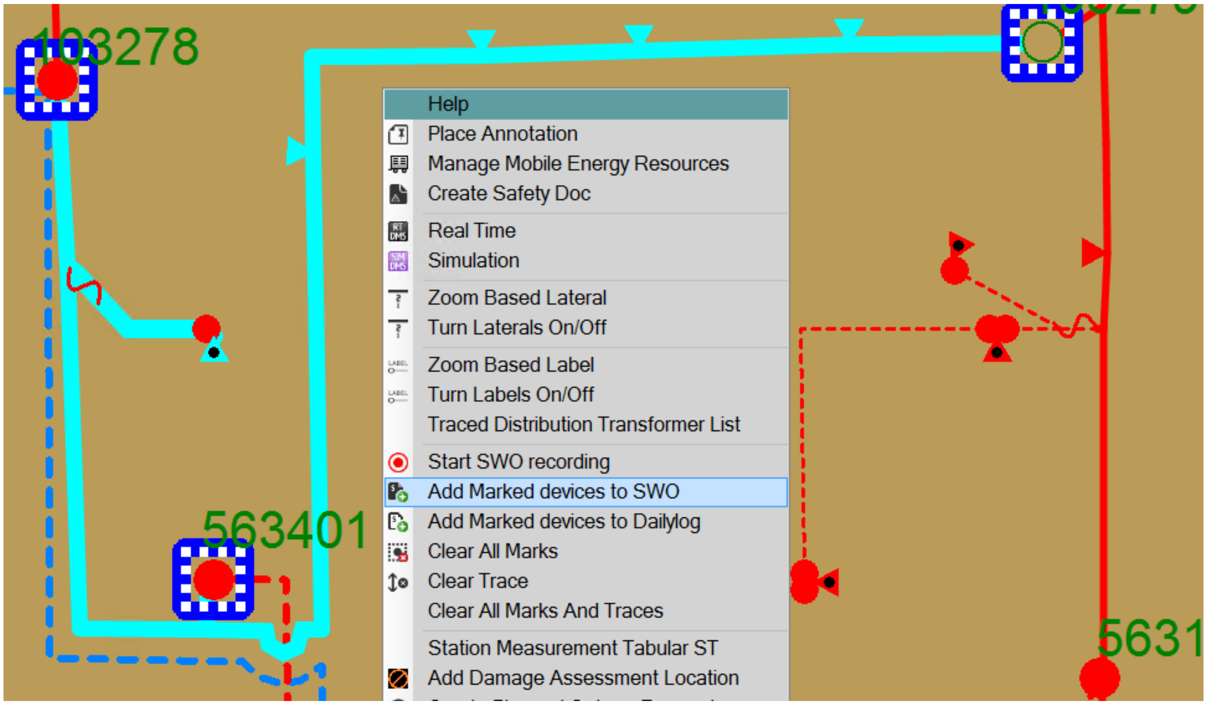
4.2. Adding Steps to a Switching Order

To add a switching step to a switching order:

1. On the network view, select a component and click the Add to Switching Order button or the

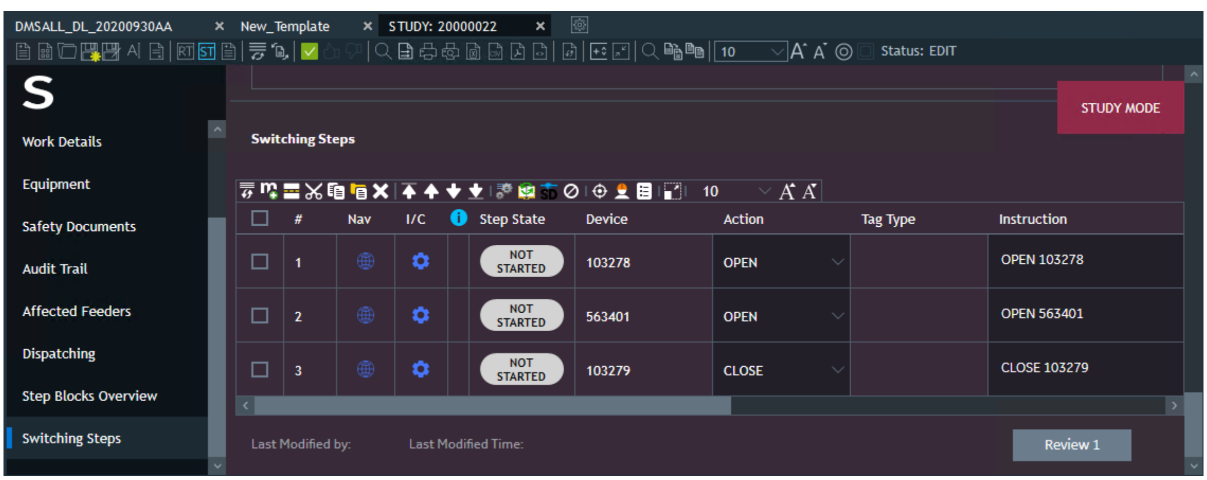
Daily Log button on the device toolbar (or right-click menu).

To add multiple components, mark the components on the network view, and then select Add Marked Devices to SWO from the network view context menu.



This opens a new switching order with the selected component as the first step, or adds the step to the previously open document.

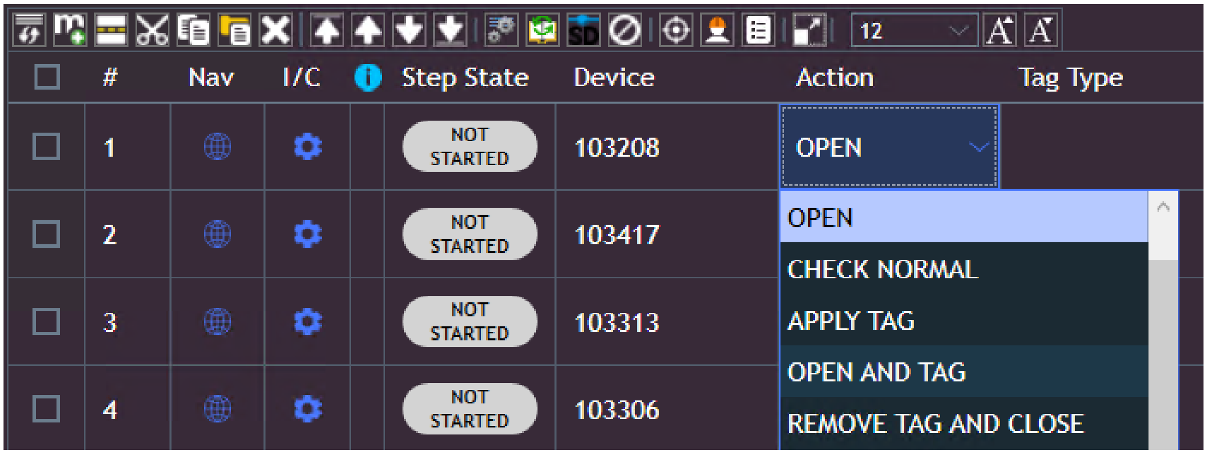
Switching steps for SCADA devices can be added from both the network view and the device's SCADA display.



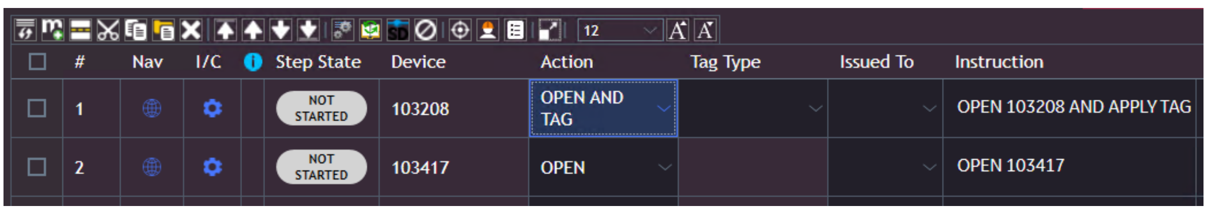
2. Select the action for the steps.

Tip

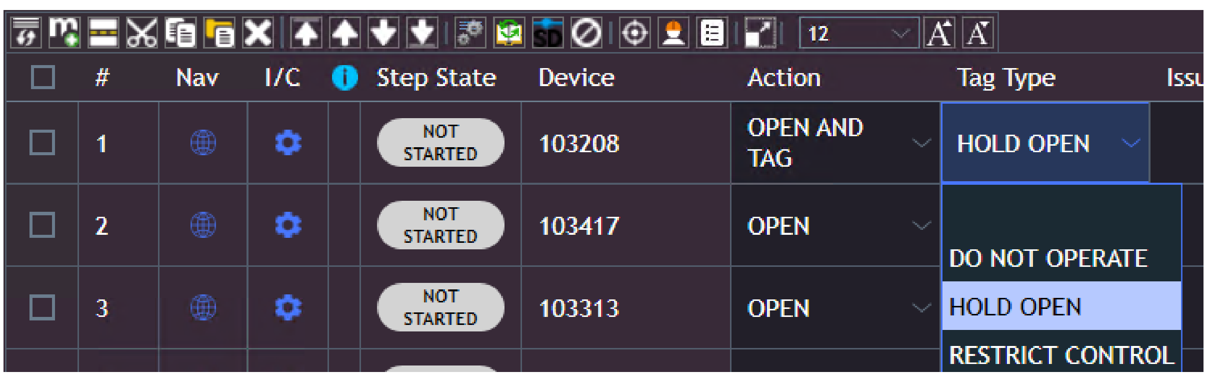
You can toggle the switch on the study mode network diagram to verify the desired switching action.



The Instruction field is populated with the corresponding action text for the selected device.



3. Add the tag type to be applied, if required.



Note

The step cannot be completed until the proper tag type is applied on the network view.

4. Add a comment to the step, if required.
5. Select the next switch on the network view and click the Add to Switching Order button to add the next step in the switching order.
6. Select the action and add a tag type as required.
7. Modify the instruction as needed.

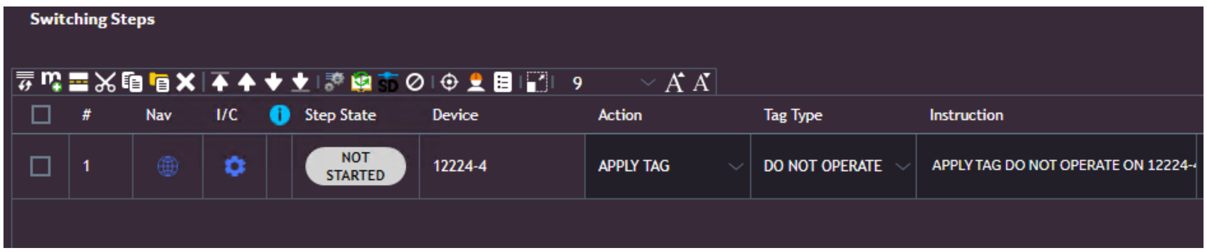
8. Add a comment to the step, if required.
9. Continue until all the required switching steps are created.

4.2.1. Adding Tagging Steps

"Tagging steps" are used to apply tags to or remove tags from switching devices. Tagging steps can also include actual switching operations such as Check Open and Tag.

To create a tagging step, use the same method as creating a switching step:

1. Select the switch and click the Switching Order button on the toolbar.
2. Select the tagging operation in the Action field.
3. Select the actual tag type in the Tag Type field.



> **Tip**
>
> If a switching step is created and a following tagging step for the same switch is required, it is more efficient to copy the step as described in Copying Steps into the Switching Order.

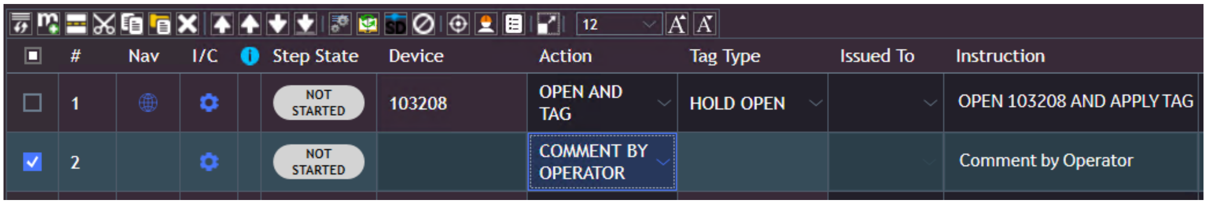
4.2.2. Adding Manual or Comment Steps

In addition to changing the order of steps that directly operate or tag switches, there is often a need to include other types of instructions within the switching order, such as the instructions to place safety barriers around the work site. This is done by using manual steps, also called "comment steps".

Comment steps are text instructions that can be used for any operation that does not directly reference a switching device.

To insert a comment step:

1. Select the step immediately above the desired position and click the Add Manual Step button  on the Switching Steps toolbar.



2. Enter the instruction text, and then add further information in the Comments field as required.

   The Action field is normally left blank.

> **Note**
>
> Issued switching steps for non-existent devices behave as manual steps. They can be completed without receiving step feedback from the system. For example, the operator added a temporary device (i.e., cut) to the switching order. If the cut is removed, the step behaves as a manual step.

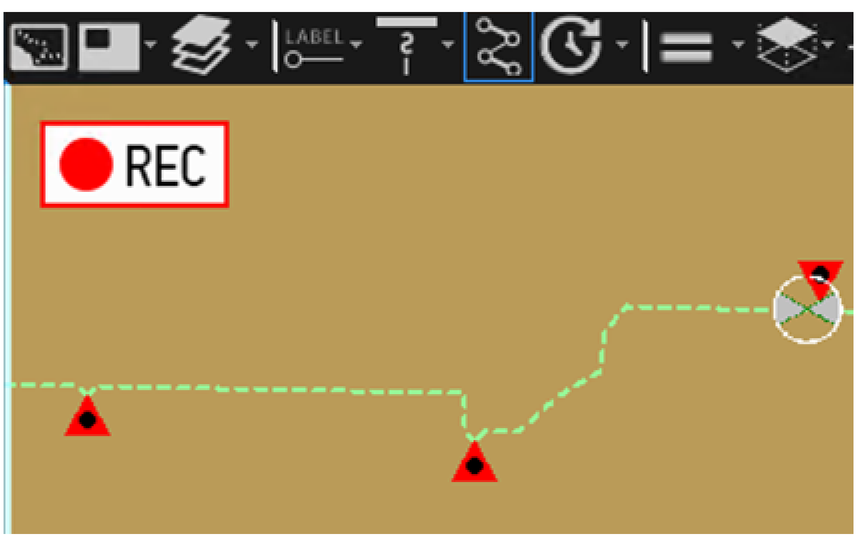
4.2.3. Recording Operator’s Actions as Switching Steps

In study mode, you can record device operations as switching steps. The recording is supported for all displays, such as geographic and schematic network views, station internals, cabinet internals, and URD loop schematic views.

To record an operator’s actions as switching steps:

1. Enable the Switching Order recording mode by clicking the Start Recording button on the switching order toolbar, on the common toolbar, or in the network view context menu.

   When the Switching Order recording mode is enabled, a recording mode heads-up display appears in the upper-left corner of the display. The Switching Order display appears.



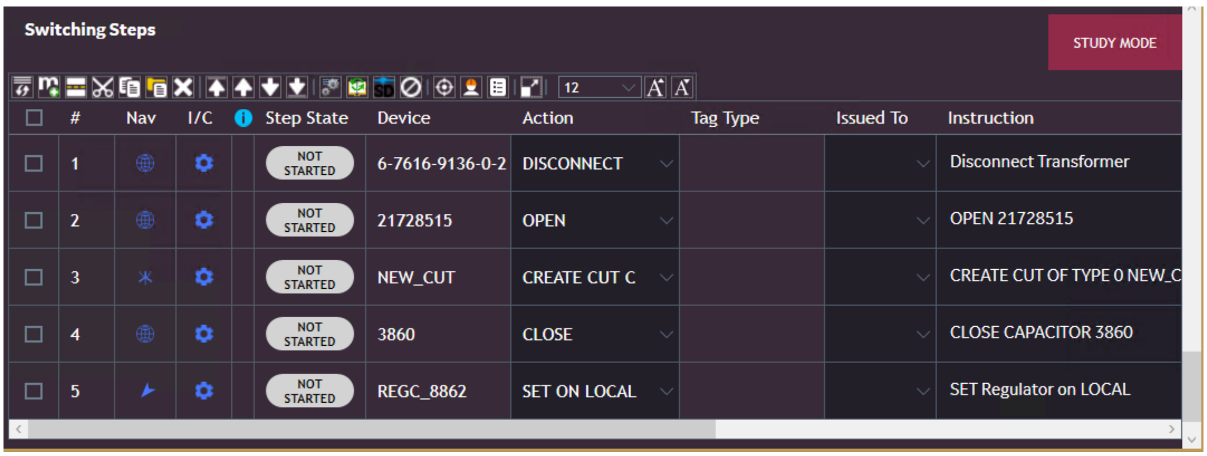
2. Perform the required device operations, such as opening or closing a switch,

connecting/disconnecting a distribution transformer, blocking/unblocking a regulator, placing/removing a cut, or placing/removing a mobile substation.

> **Note**
>
> Recording tagging operations is not supported.

> **Note**
>
> When moving a mobile substation, steps for removing and adding the substation are created. A connected substation cannot be modified.

A switching step is added each time you perform a device operation.



> **Note**
>
> When recording a non-configured device operation (for example, changing the capacitor status to operable/inoperable), the step is recorded with the default device action and an associated comment is added.

3. Stop recording by clicking the Stop Recording button on the switching order toolbar, on the common toolbar, or in the network view context menu.

   The Switching Order recording mode is disabled, and the Switching Order display appears showing the recorded steps.

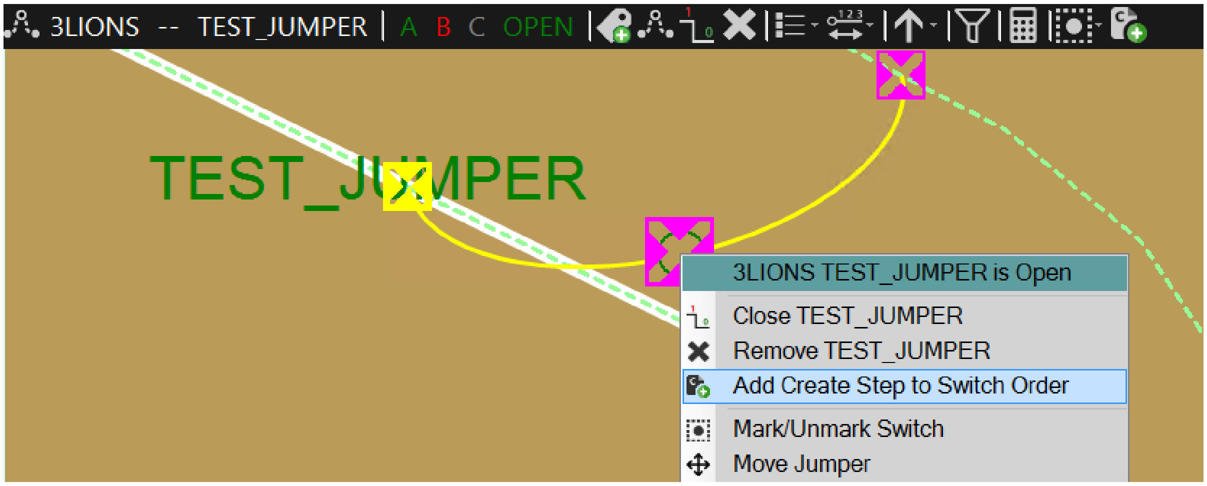
4.2.4. Adding Create and Remove Steps for Temporary Devices

You can add steps to create and remove temporary network modifications (cuts, temporary grounds, and jumpers).

To add "create" and "remove" steps for a temporary device:

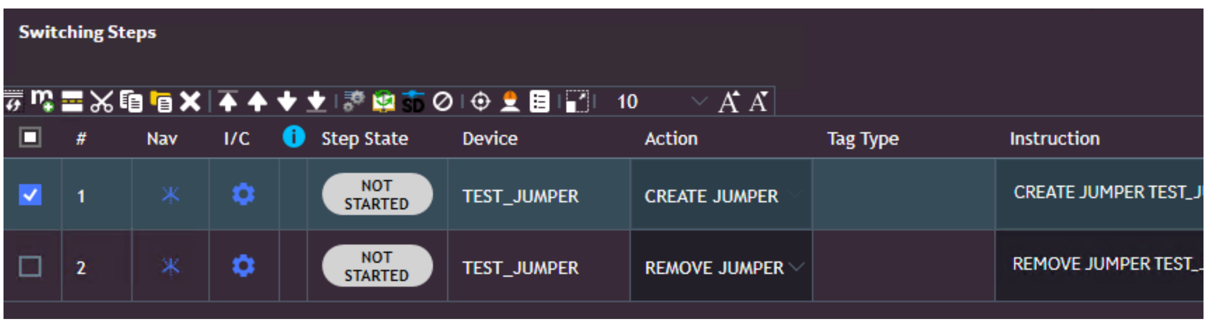
1. On the network view, place a temporary device.

2. Click the Add the Create Step to Switching Order button on the device toolbar (or right-click menu).



This opens a new switching order with the "create temporary device" step as the first step, or adds the step to the previously open document.

3. To add a "remove" step, select the "create" step and click the Add Return To Normal Steps button  on the Switching Steps toolbar.



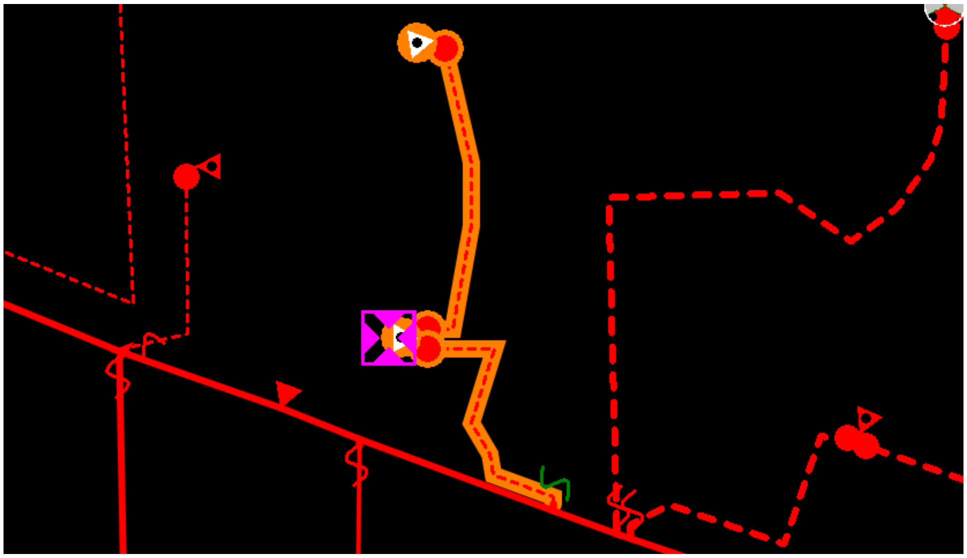
> **Note**
>
> When implementing a "create" step for a temporary device, the device must already exist on the network view or the option to auto-implement non-SCADA steps in real-time mode must be enabled in the Switching Order Client Configuration Editor. Otherwise, the Switching Order application will not be able to match the internally specified device location against the manually added device location.

4.2.5. Adding Commission and Decommission Steps for Devices

You can add steps to commission or decommission individual devices or all devices that are associated with a specific commissioning project ID and stage.

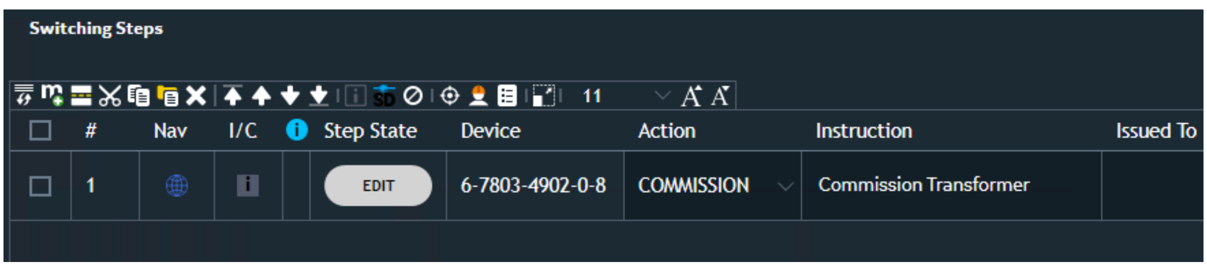
To add Commission and Decommission steps for a device:

1. On the network view, select a device under commissioning.

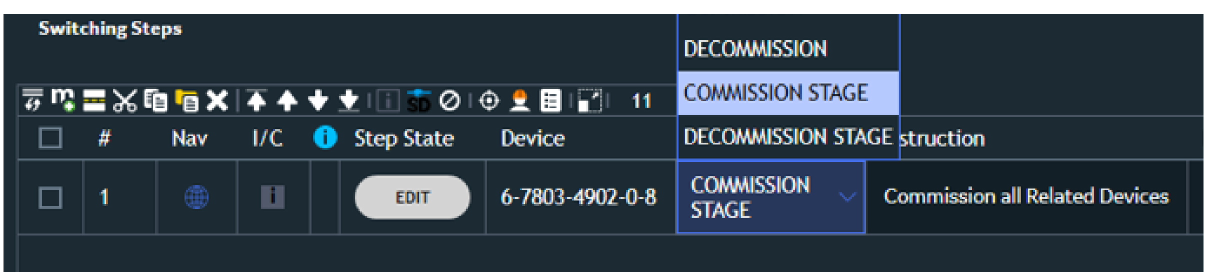


2. Click the Add to Switching Order button on the device toolbar (or right-click menu).

   This opens a new or existing switching order and adds a commissioning step for the device (the "Commission" action is selected for devices with "future" phases; the "Decommission" action is selected for devices with "future remove" phases).



3. To add a step to commission or decommission all devices that are associated with a specific commissioning project ID and stage, select the Commission Stage or Decommission Stage action for a device.



Note

To complete commissioning steps, you need to change the device commissioning state from the Equipment Commissioning/Decommissioning display. Commissioning and decommissioning steps are considered completed when there are no Future and Future Remove phases on a device.

Note

Commissioning and decommissioning steps do not support return to normal and clearance steps.

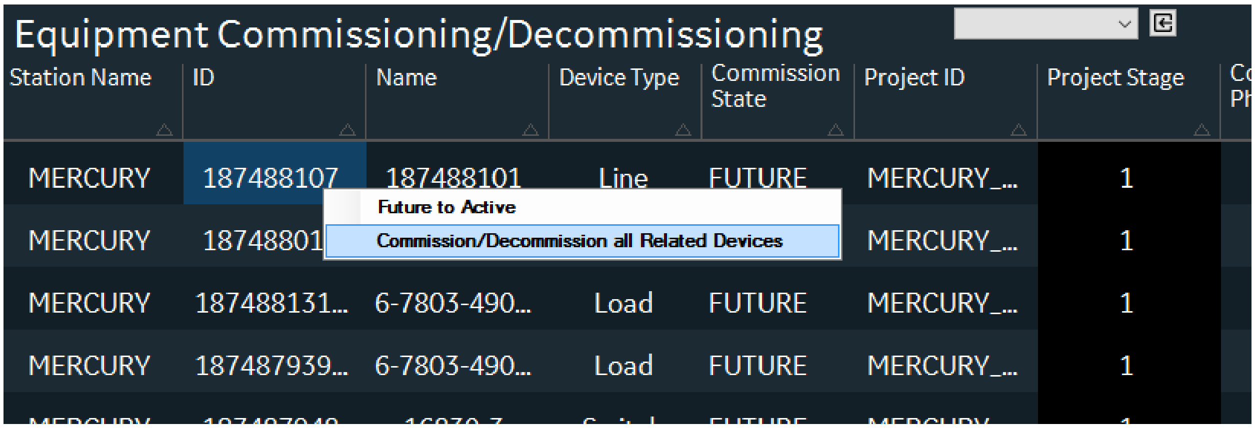


Figure 42. Updating Device Commissioning State

4.3. Adding Comments to Steps

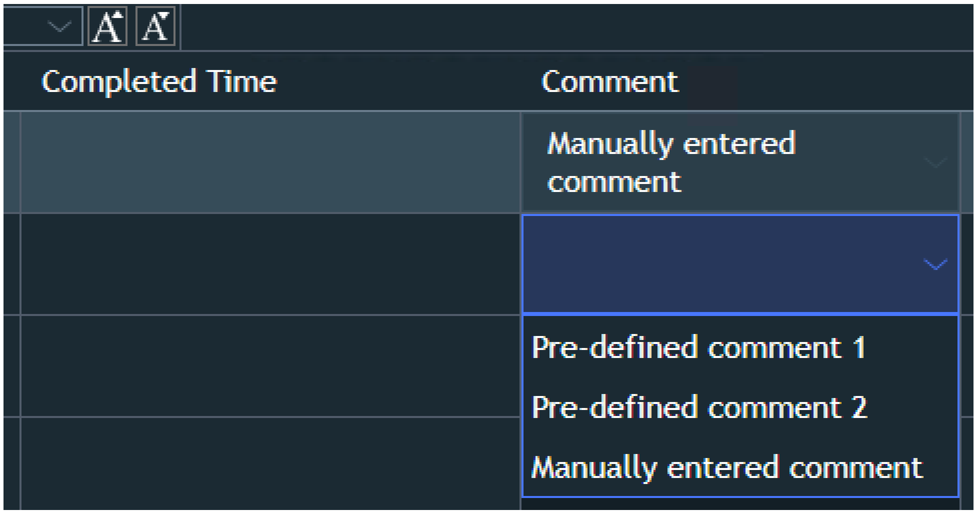
To add a comment, do one of the following:

- Manually enter comment text in the Comments field.

  Once entered, the comment is added to the Comments drop-down list, and can be selected for the current switching order.

- Select a comment from the drop-down list.

  The Comments drop-down list contains previously entered comments and comments that were pre-defined using the Switching Order Client Configuration Editor.



4.4. Copying, Moving, and Deleting Steps

Once a few switching steps have been created, it is common to decide that a new step must be inserted into the switching order, that the order of the steps must be changed, or that a step must be taken out in order to achieve the required switching sequence.

4.4.1. Copying Steps into the Switching Order

Use the Copy and Paste buttons on the Switching Steps toolbar to copy steps within the currently open switching order or between the Daily Log and Switching Order tabs.

To copy steps into the switching order:

1. Open the desired switching order.
2. Select the steps you need to copy.
3. To copy the selected steps, click the Copy button on the Switching Steps toolbar.

   Alternatively, to cut the selected steps, click the Cut button on the Switching Steps toolbar.
4. Open the switching order to paste the steps.
5. To paste the selected steps into the currently open switching order, click the Paste button on the Switching Steps toolbar.

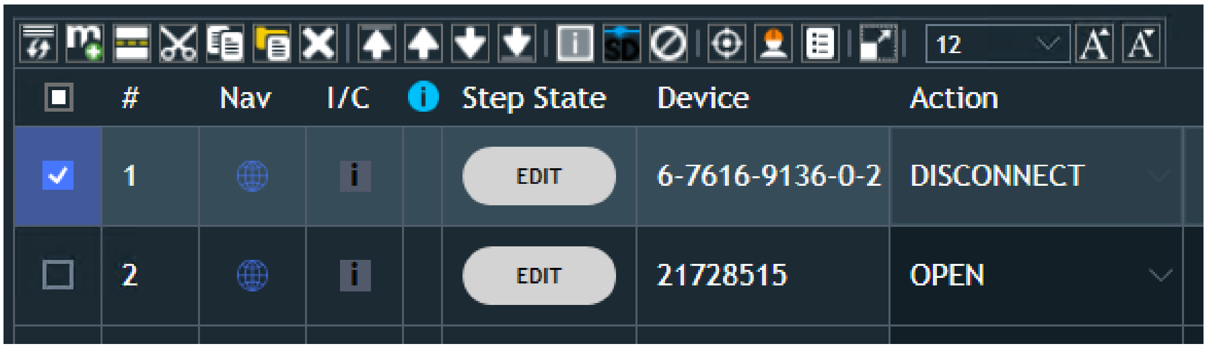
   The steps are pasted immediately after the last selected step in the list.
6. To rearrange the steps, use the Move buttons on the Switching Steps toolbar as described in Moving Steps within the Switching Order.

4.4.2. Moving Steps within the Switching Order

To move steps within the switching order:

1. Select the step to be moved.

   The selected step is highlighted.



2. On the Switching Steps toolbar, click the Move Up button or the Move Down button to move the step up or down one position.

   Or

3. To move steps directly to the top or bottom of the switching order, click the Move To Top or Move Current Step To Bottom buttons.

4.4.3. Deleting Steps from the Switching Order

To delete a step from the switching order, select the step and click the Delete button on the toolbar.

4.4.4. Manipulating Steps Using the Right-Click Menu

In addition to the toolbar options, the context menu options can also be used to manipulate steps.

To access the context menu (see below image), right-click on a step.

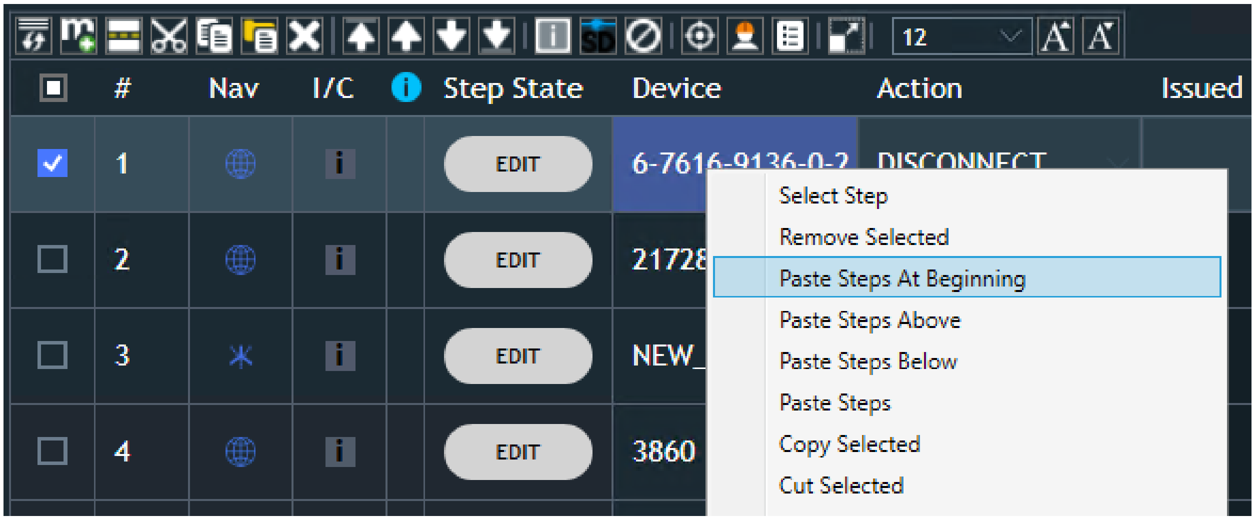


Figure 43. Manipulating Steps with the Right-Click Menu

For more information about these right-click menu options, see Switching Steps Context Menu.

5. Automatically Generating a Switching Order

In addition to the manual creation of switching orders, it is possible to automatically generate switching orders as the output of switching reconfiguration applications of the Distribution Network Analysis Function (DNAF) or from the Outage Management System (OMS).

5.1. Generating a Switching Order from DNAF

5.1.1. Automated Restoration Plans

In the case of an unplanned network outage, the Fault Isolation and Service Restoration (FISR) function can be invoked. For more information about the operation of the FISR function, refer to the *Distribution Network Analysis Functions (DNAF) User Guide*.

The output of the FISR function is one or more switching plans (see the below image). The switching plans consist of basic switching steps only; they do not include tagging or other types of steps.

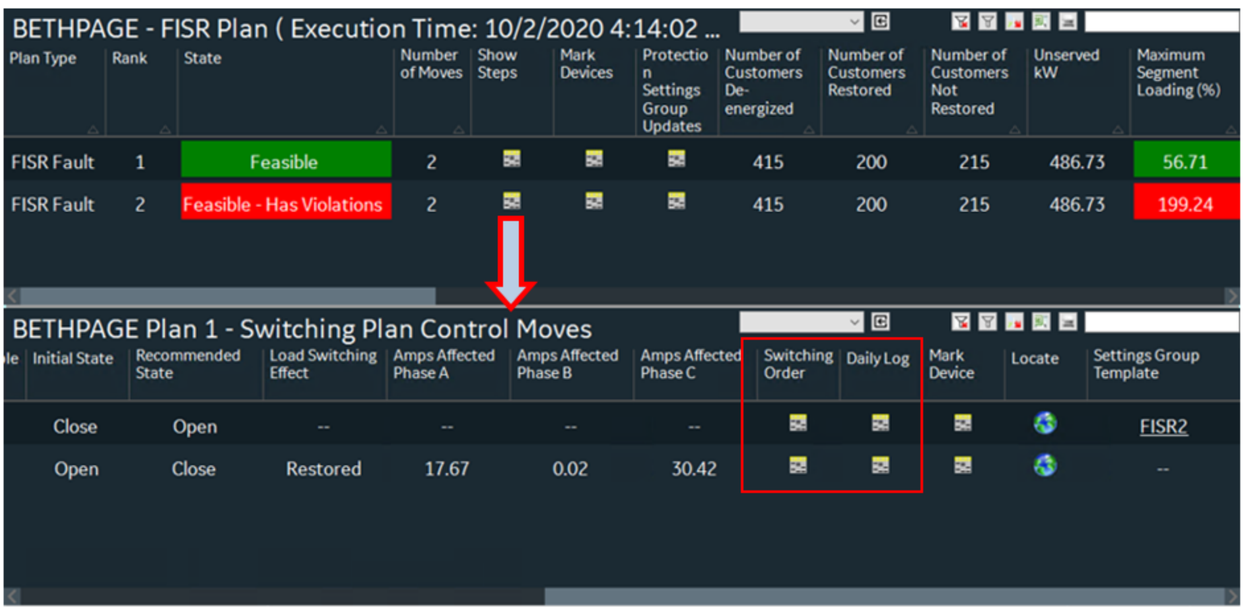


Figure 44. FISR Output Switching Plan

When the desired switching plan has been selected, click the Switch Order or Daily Log button for any step of the selected plan to automatically generate a formal switching order or add the switching steps to a daily log. The switching order form and the associated network display are opened. The steps from the switching plan are automatically transferred to the switching order form.

The steps can now be edited or modified as required, to create a complete switching order ready for approval and execution.

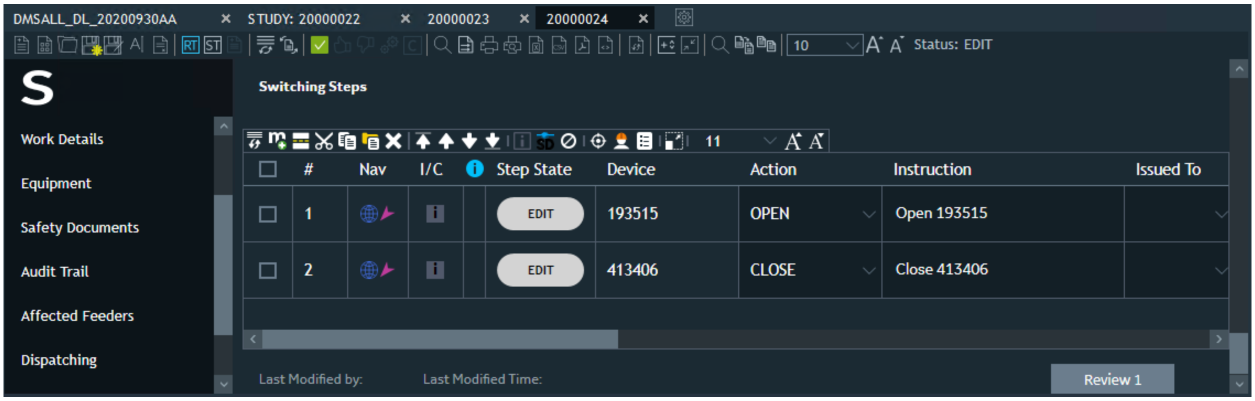


Figure 45. Switch Order Generated from FISR

5.1.2. Planned Outage Study

The Planned Outage Study (PLOS) function is used to develop switching plans for the de-energization of a specified network component while maintaining supply to the maximum number of customers. This work is normally done well in advance of the time when the network switching is required.

Figure 46. Planned Outage Switching Plan

The PLOS function can only be run in study mode. For more information about the operation of the PLOS function, refer to the *Distribution Network Analysis Functions (DNAF) User Guide*.

The output of the PLOS function is a tabular display of suggested switching plans, as shown in the above image. The switching plans consist of basic switching steps only; they do not include tagging or other types of steps.

To add the PLOS plan to the switching order form:

1. On the PLOS Plan display, select the desired switching plan and then click the Show Steps button.

   The Switching Plan Control Moves display appears.

2. On the Switching Plan Control Moves display, click the Switch Order or Daily Log icon for any step of the selected plan to automatically generate a formal switching order or add the switching steps to a daily log.

   The steps from the switching plan are automatically transferred to the switching order form (see the below image).

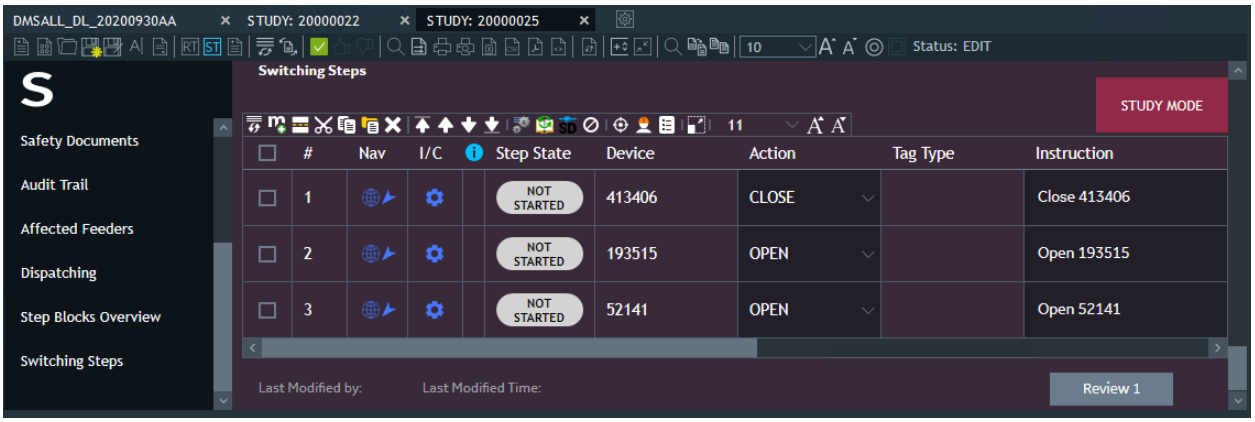


Figure 47. Planned Outage Switching Steps

The steps can now be edited or modified as required to create a complete switching order. The operation of the switching order can be verified by executing it in study mode and then saving it for transfer to real-time operation.

5.1.3. Return to Normal

In many cases, after an unplanned outage the customers may be restored but there can still be a number of switches across multiple feeders that are left in abnormal states. Therefore, a procedure is required to safely return the switching to its normal state while avoiding any further de-energization of customers.

The Automatic Feeder Reconfiguration option's Return to Normal function is used to develop this switching procedure. For more information about the operation of the Return to Normal function, refer to the *Distribution Network Analysis Functions (DNAF) User Guide*.

The Return to Normal function can be run from the real-time mode or from the study mode initialized to real time. The output of the Return to Normal function is one or more switching plans. The switching plans consist of basic switching steps only; they do not include tagging or other types of steps.

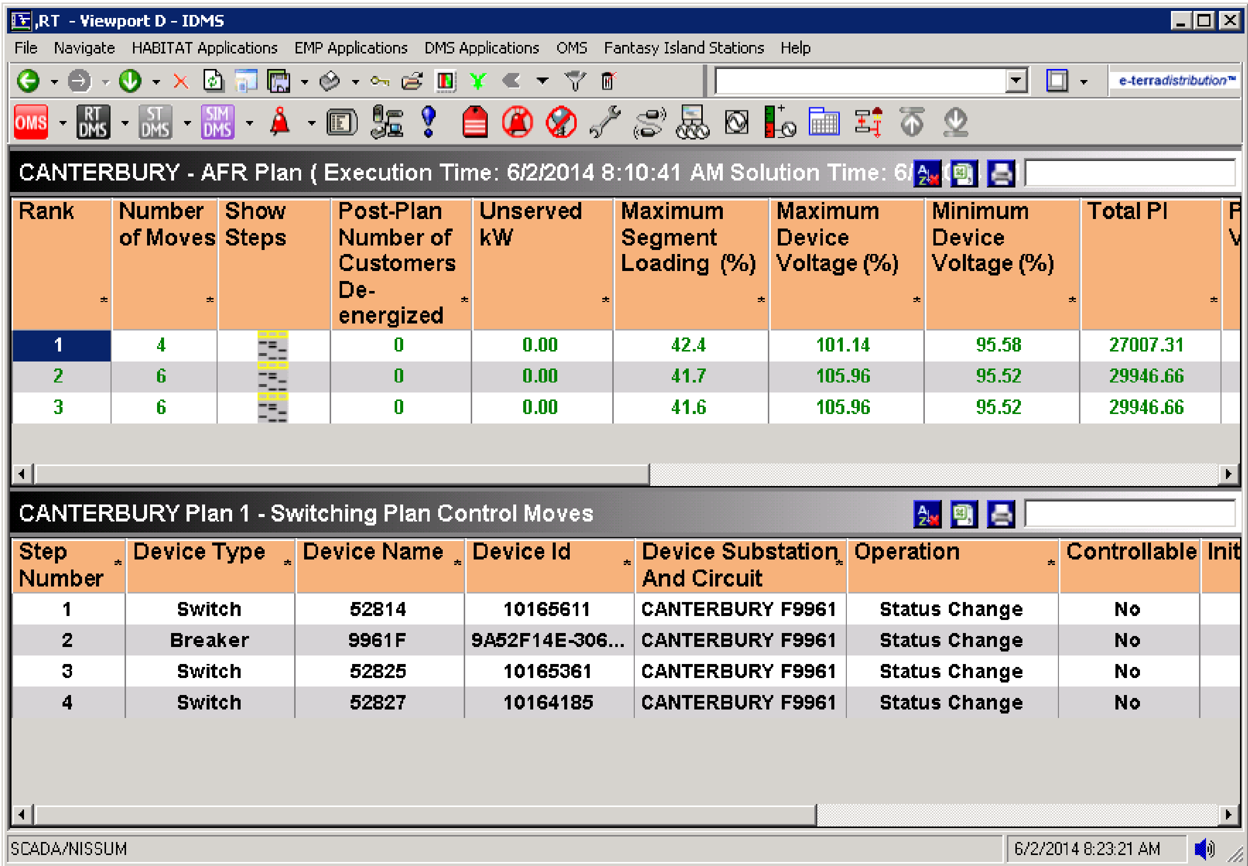


Figure 48. Return to Normal Output Tabular

When the desired switching plan has been selected, click the relevant Switching Order or Daily Log button on the Return to Normal output display for any step of the selected plan to automatically generate a formal switching order or add the switching steps to a daily log.

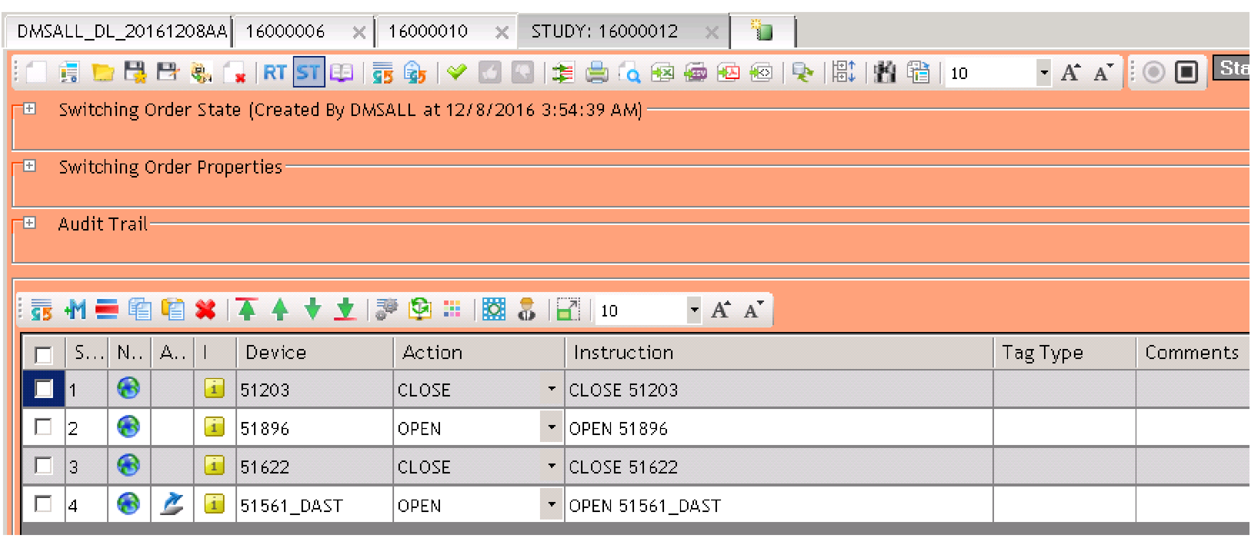


Figure 49. Return to Normal Switching Steps

The steps can now be edited or modified as required to create a complete switching order ready for approval and execution.

If the Return to Normal function has been run in study mode, the switching order can be fully evaluated prior to being transferred to real-time mode.

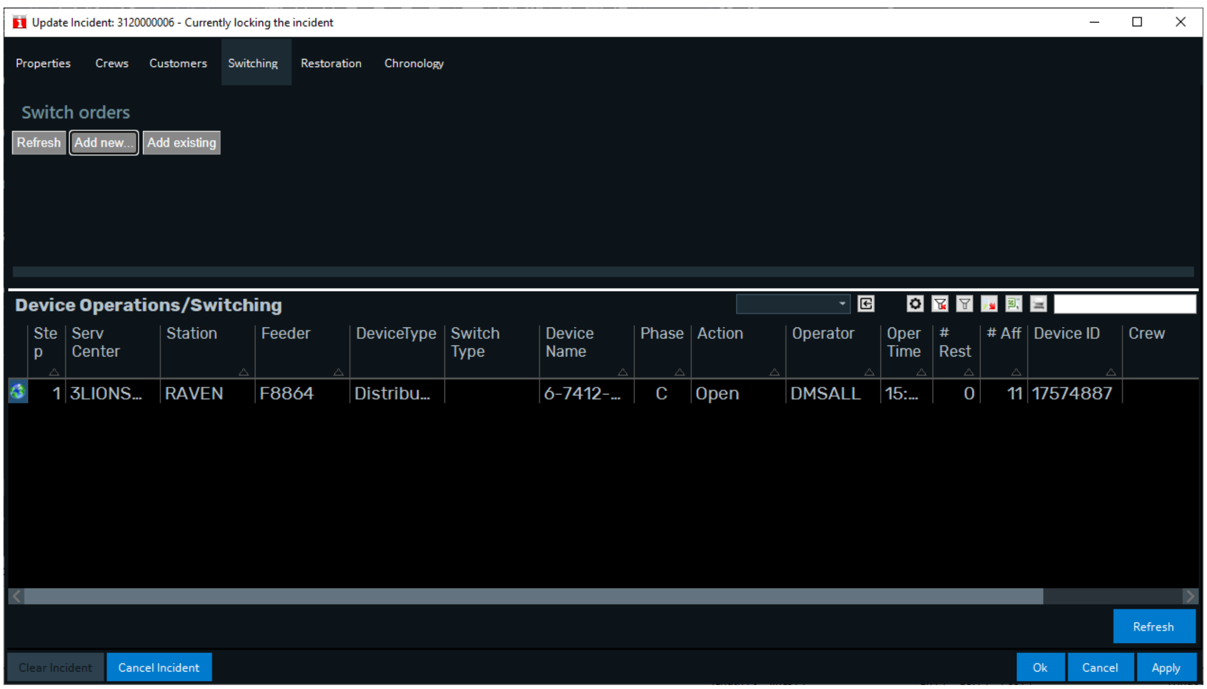
The Return to Normal function can also be started on the Switching Order Form using the Open New Return to Normal Switching Order button on the Switching Order toolbar. For more information, see Creating a Go Back Order.

5.2. Generating a Switching Order from OMS

5.2.1. Generating a Switching Order for an Incident

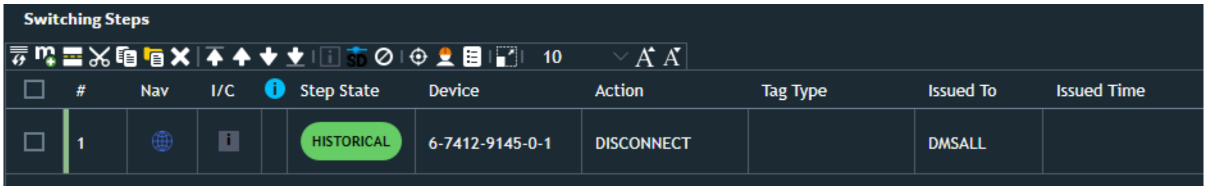
To create a switching order for an incident:

1. Open the Update Incident display for an incident.
2. Open the Switching tab.



3. In the Switching Orders section, click the Add New button.

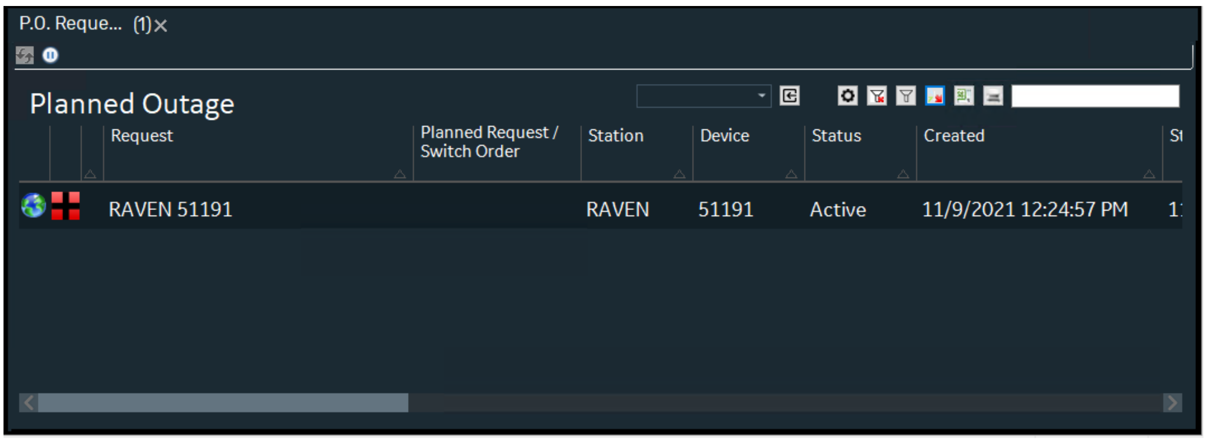
   The Switching Order window opens. The switching order is populated with the incident device details. The Switching Steps section includes switching steps for already completed switching actions, and these steps have the Historical (read-only) state.



5.2.2. Creating a Switching Order for a Planned Outage

To create a switching order for a planned outage:

1. Open the OMS Display plugin.
2. Open the P.O. Requests tab.

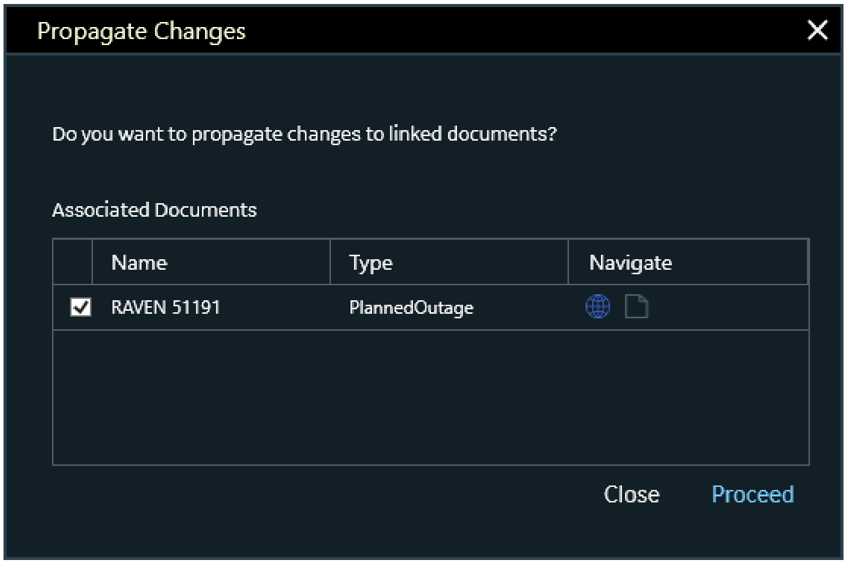


3. Click the Create Switching Order bitmap to the left of the Request field for the required planned outage.

   The Switching Order window opens. The switching order is populated with the planned outage device details. The Switching Steps section includes steps to open and close the device of record.

4. Edit the switching order as needed.
5. Click Save.

   The confirmation dialog appears, where you can navigate to the planned outage on the network view, open the planned outage, or confirm creating a switching order for the planned outage.



The switching order is created and associated with the planned outage.

6. Validating and Approving a Switching Order

Once a switching order is fully edited and it is ready to proceed toward execution either immediately or sometime in the future, it must be validated and approved.

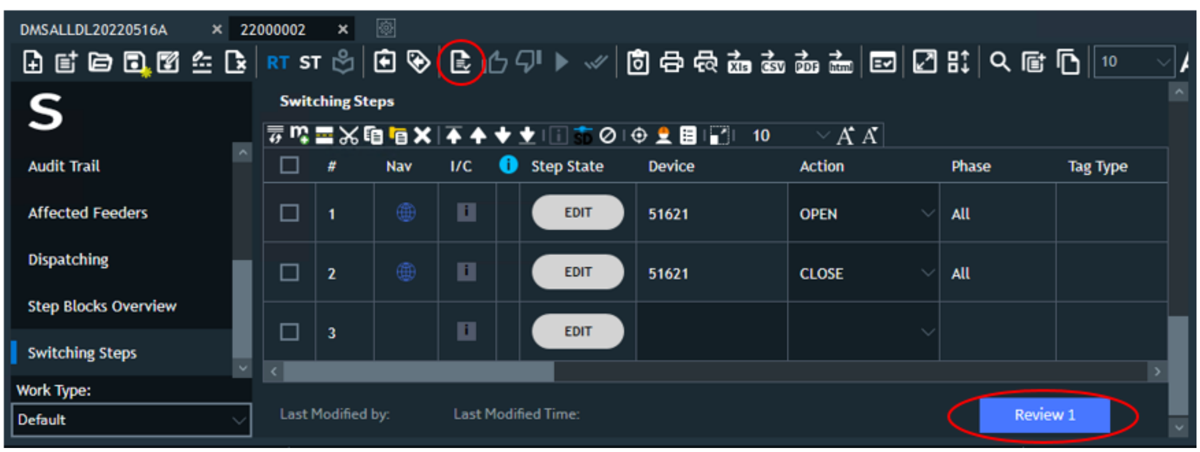
> **Note**
>
> The switching order workflow can be configured to exclude the review and approval stages and hide the related statuses and controls in the Switching Order form. For more details, refer to the *Switching Order Server Configuration Editor User Guide*.

6.1. Validating a Switching Order

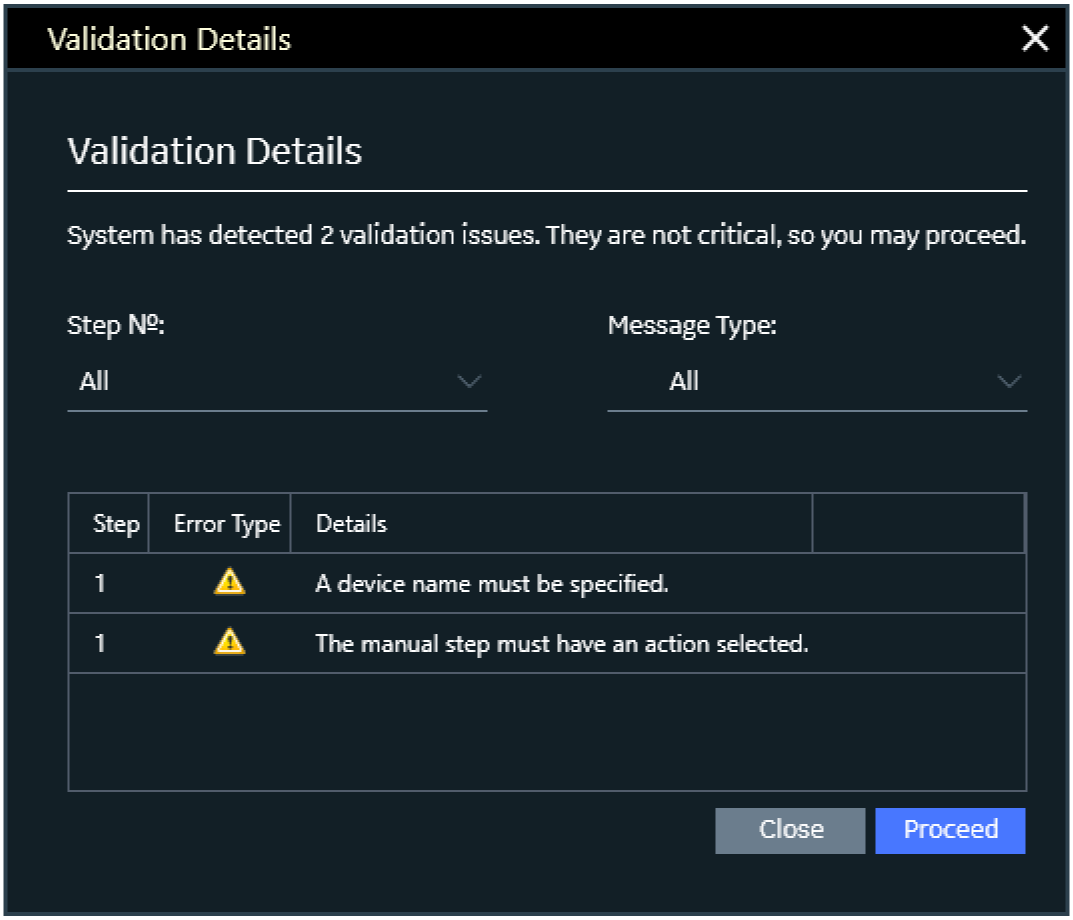
If the switching order has been prepared in the study mode, it must be saved and loaded into the real-time mode for the validation and approval process. If the switching order has been prepared in the real-time mode, it can proceed directly to validation.

To validate the switching order:

1. Create a new switching order or open an existing switching order in the Edit state.
2. Click the Validation ✅ button on the Switching Order toolbar, or click the Review 1 button at the bottom of the switching order form.



The validation process checks that the switching order and switching step properties are correctly populated. If any error or warning is detected, the Validation Details display appears.



The Validation Details display shows the validation summary and validation details. You can filter the validation details by step number and error type.

3. To proceed with the validation, correct the validation errors, if any. The validation warnings can be corrected or skipped as needed.

> **Note**
>
> The Online Validation button on the Switching Order toolbar allows you to validate the switching steps for errors at any time until the switching order status is Completed.

When the Switching Order Status field shows VALIDATED, the switching order is ready for approval and the Approve button on the Switching Order toolbar is enabled.

6.2. Approving a Switching Order

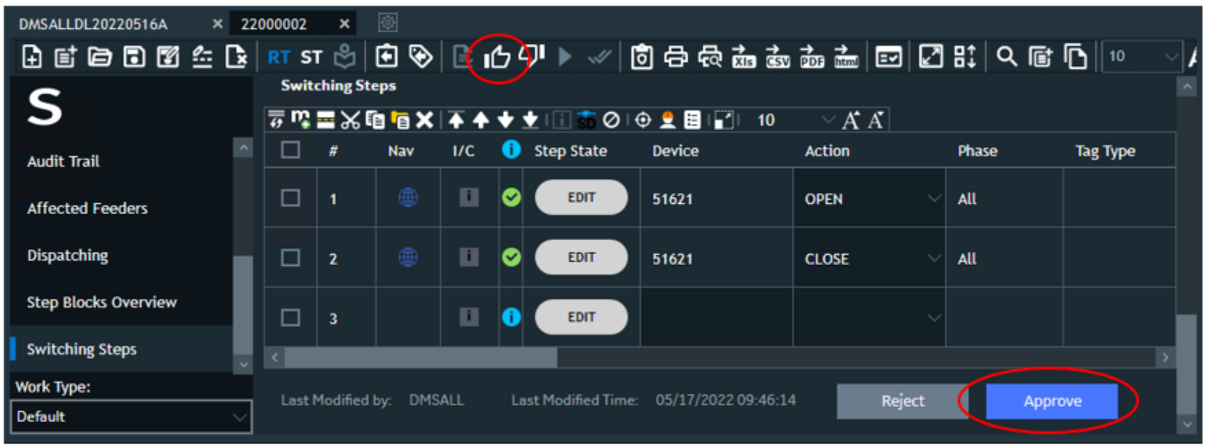
A good practice is to have another operator (other than the operator who prepared the order) open the switching order from the library, check that all the steps are correct for the required operation, and then approve it following the steps below.

A switching order can be configured to require more than one approval by the same or different users. In this case, the Approve button on the toolbar remains enabled until the switching order is approved by the required number of users. For more information about configuring the number of approvers, refer to the *Switching Order Server Configuration Editor User Guide*.

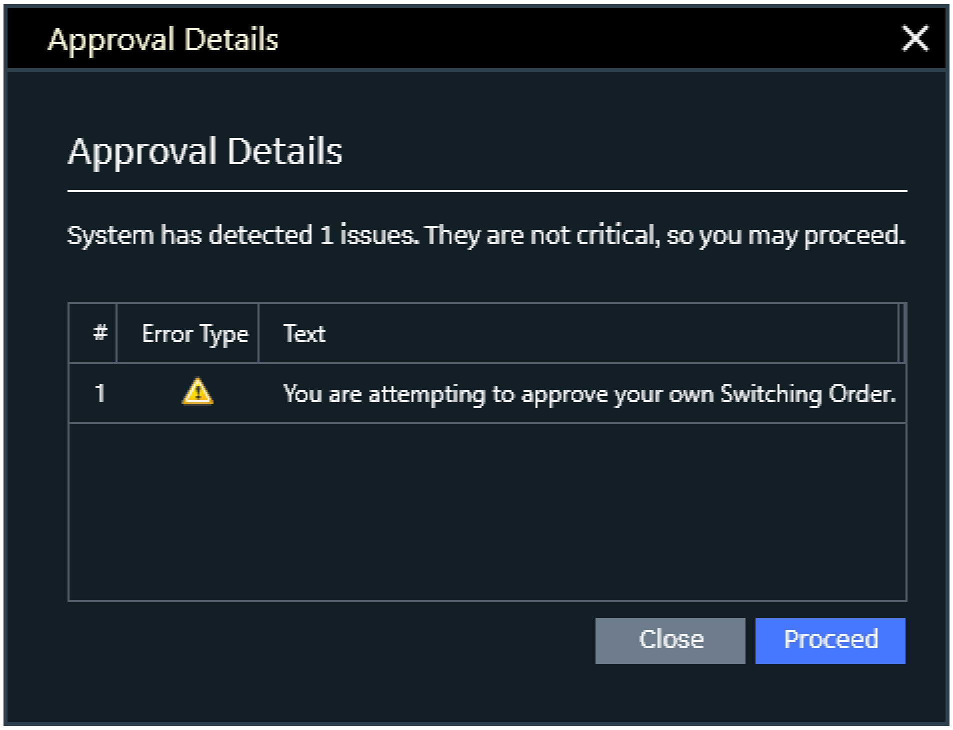
The Reject button is enabled on the switching order toolbar and the Reject button appears at the bottom of the Switching Order form once the switching order is validated. When the Reject button is clicked, all provided approvals are removed and the relevant record is added on the Audit Trail pane. The switching order can be rejected any number of times until the switching order status is updated to Executing.

To approve the switching order:

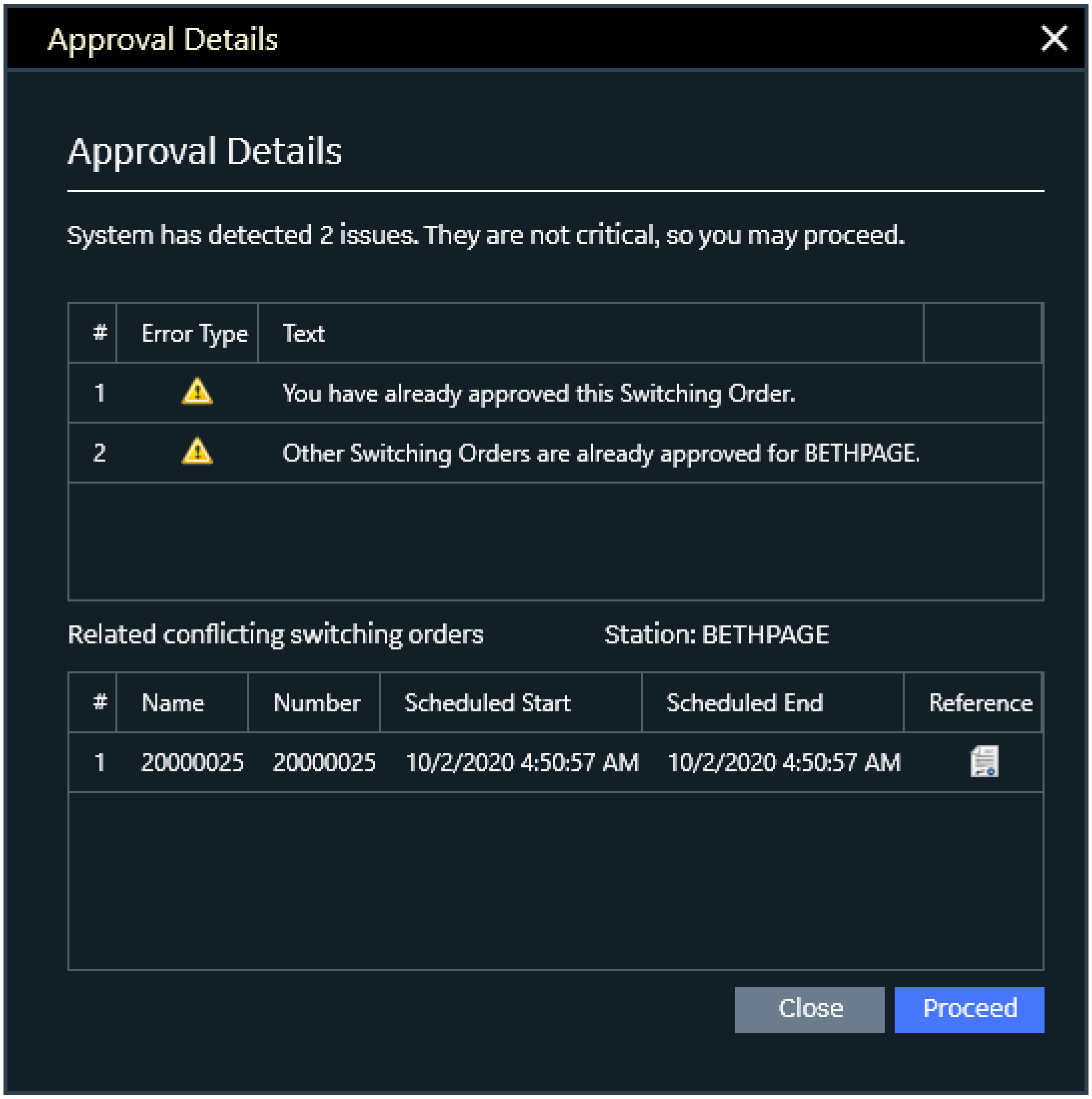
1. For a validated switching order, click the Approve button on the Switching Order toolbar, or click the Approve button at the bottom of the switching order form.



If the operator approving the switching order is the same person who prepared the switching order, a warning message is displayed.



In addition, when approving a switching order, the application checks if there are approved or executing switching orders associated with the same substation and feeder within the scheduled start and end dates (as specified in the Switching Order Properties section). If such switching orders already exist, the summary information is shown for the related conflicting switching orders.



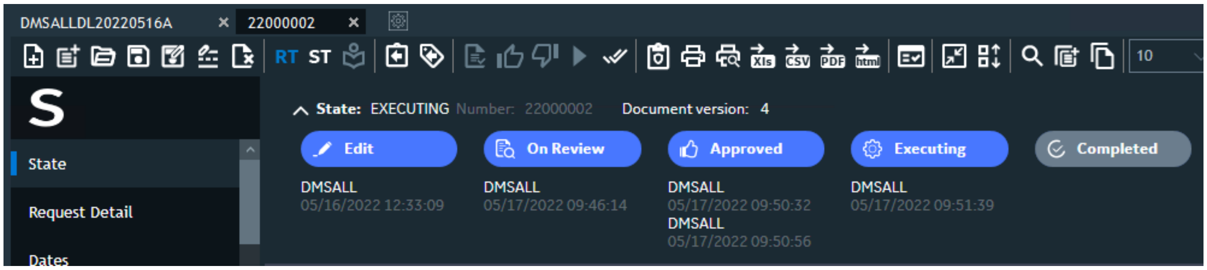
This warning ensures that the operator is aware of a possible conflict. The operator may then create a study simulation that includes execution of the conflicting switching orders.

Clicking the Reference button opens a conflicting switching order so you can view its details.

2. Click Close to cancel approving the switching order, or click Proceed to acknowledge the warning message and proceed with the approval.

   Now the next approver needs to reject or approve the switching order. Until all the approvals are made, the Execute Switching Order button is disabled.

3. After all approvals are made, click Execute to set the status to Executing, or click Reject to turn the document back to the first approver. After a document is rejected, users in both the corporate zone and the secured zone can modify the document. (If rejected, enter a reason for the rejection in the Reject Switch Order pop-up.)



All the actions and comments are displayed in the Audit Trail section of the Switching Order form.

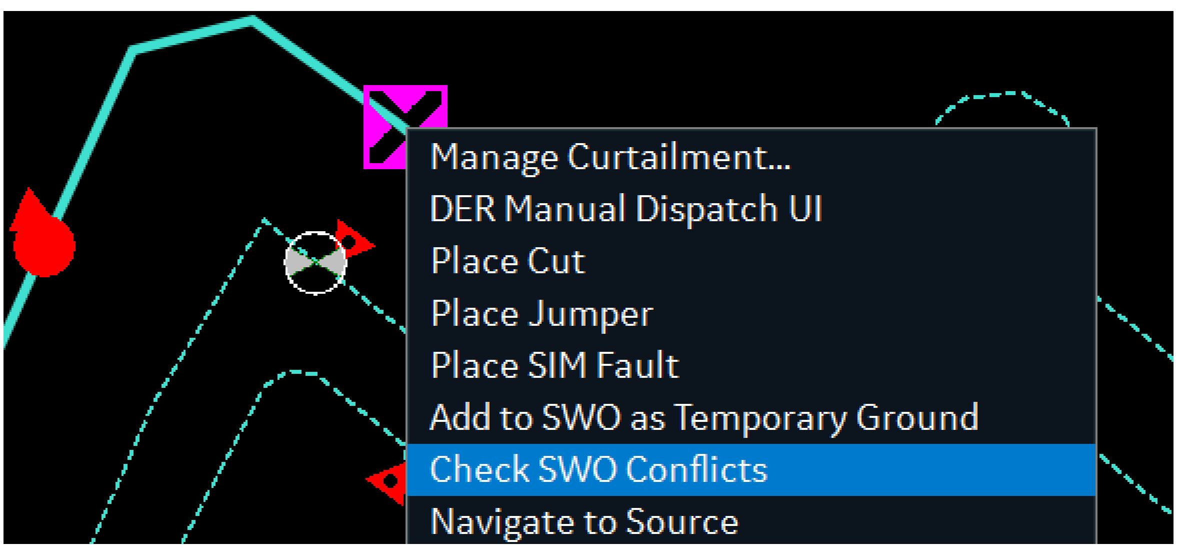
As part of the checking process, it is possible to open the switching order in study mode so that the steps can be individually verified. For more information about using the study mode, see Study Mode.

6.3. Checking for Conflicting Switching Orders

You can check whether switching orders associated with the same substation and feeder are scheduled for similar time periods. The check is performed based on the substations and feeders that are listed in the Affected Feeders section.

To check for conflicting switching orders:

1. Navigate to the line that is associated with the required substation and feeder on the network view.
2. Select the Check SWO Conflicts option from the line context menu.

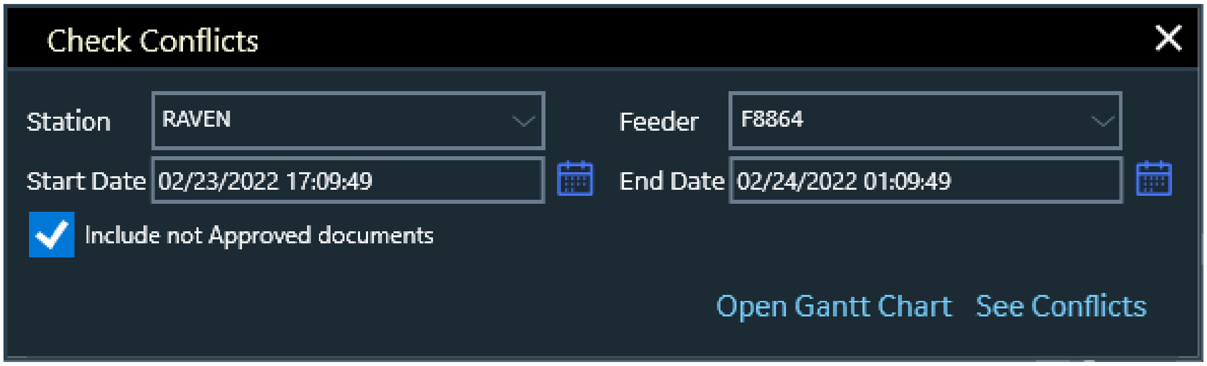


Optionally, execute the following Netview command:

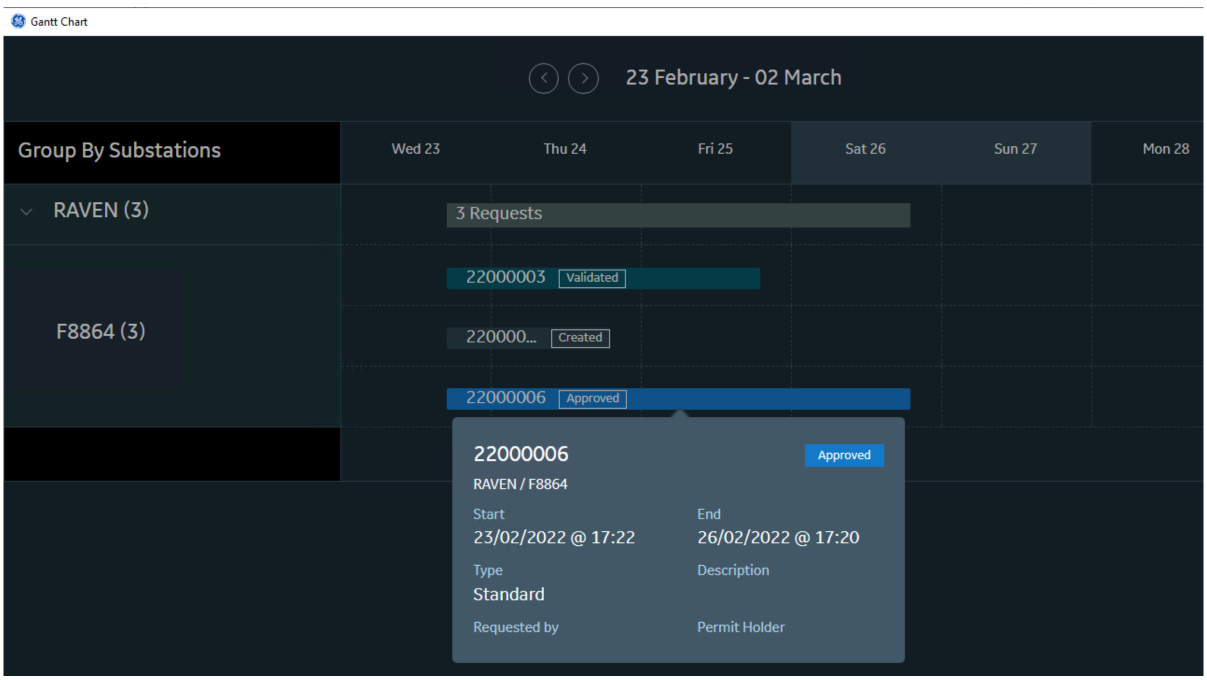
```
NETVIEW://SO.RT/POPUP/CHECKCONFLICTS?STARTDATE= &ENDDATE= &STATION= &FEEDER=
&INCLUDEALL=TRUE&EXCLUDEFROMRESULTS=
```

The Check Conflicts popup appears.

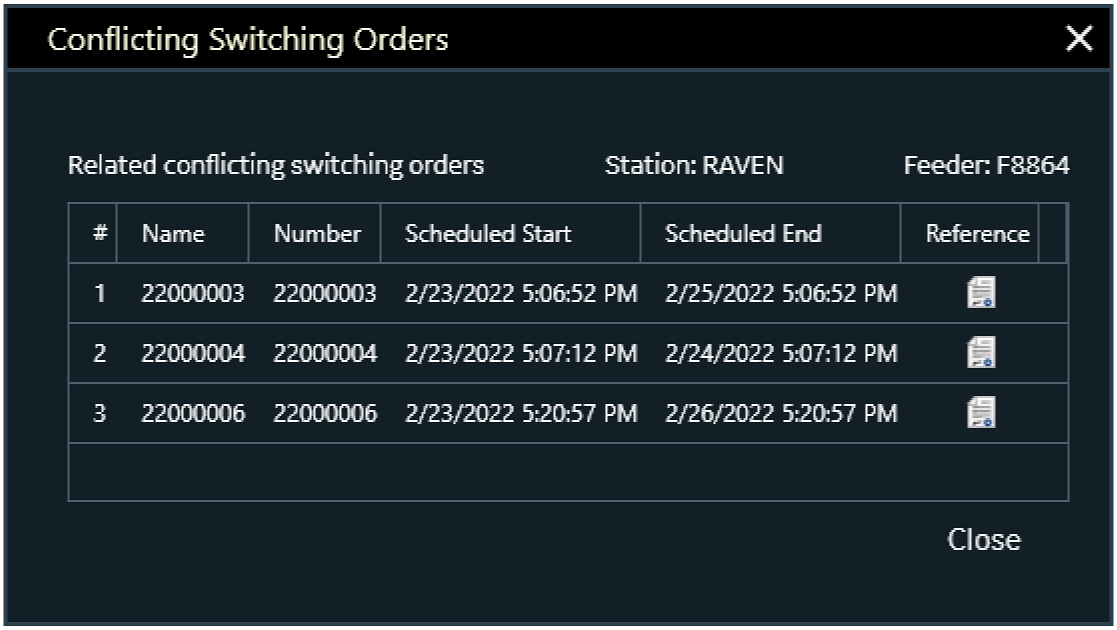
3. Select the required station, feeder, and start and end dates.



4. Specify whether to check not approved documents (in the Edit or Validated state).
5. Optionally, click Open Gantt Chart to show the Gantt chart with switching order timeline for the specified substation and feeder.



6. On the Check Conflicts popup, click See Conflicts.



The Conflicting Switching Orders popup shows switching orders with the same substation and feeder and with the start or end date within the selected time period. Clicking the Reference button opens the conflicting switching order.

7. Creating a Planned Outage with Detailed Interruptions from a Switching Order

The steps defined for a switching order can be used to record detailed de-energization and energization information. This can then be used to generate detailed information for a planned outage that records the different groups of customers that are expected to be interrupted, when they'll be affected and for how long.

To do this, the following steps must be carried out for a switching order:

1. Record the customers affected by each switching step in study mode; and
2. Create a planned outage from the switching order.

7.1. Create a Switching Order

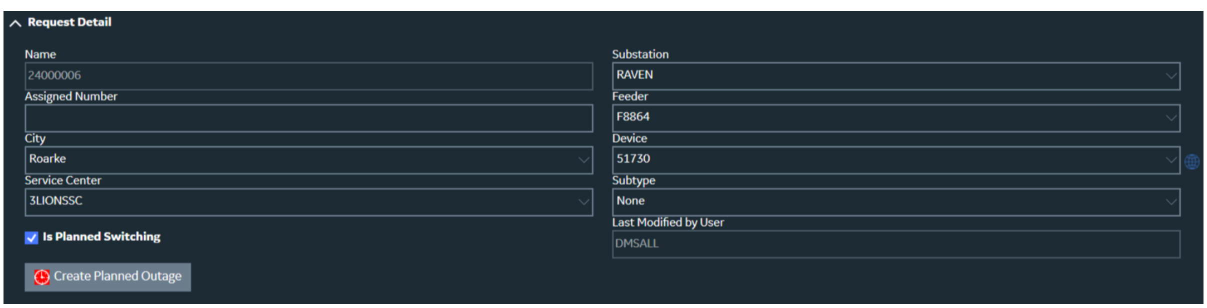
If there's no switching order, create one by following these steps:

1. Click the New Switch Order button on the common toolbar or on the Switching Order Options page.

   A new, empty switching order document appears.

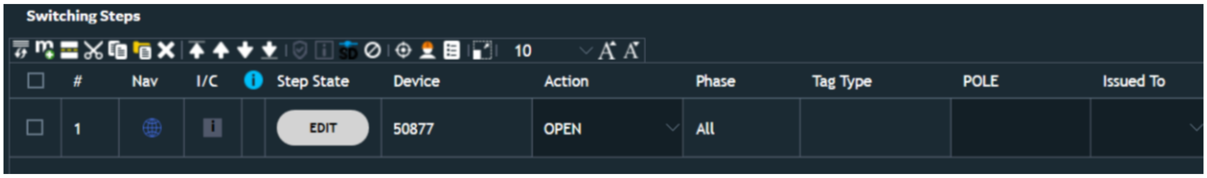
2. On the network view, navigate to the device of record (device to be associated with the switching order document).
3. Select the Add to Switching Order option from the device toolbar or right-click menu.

   The Request Details section is populated with device details.



4. In the switch order form, set the planned outage start and end dates in the Priority/Date section.

5. Add the steps for changing the state of the device of record to the Switching Steps section.



6. Add more steps to create an order that deenergizes customers and then reenergizes them.

> **Note**
>
> Changes to switching steps are propagated to linked documents.

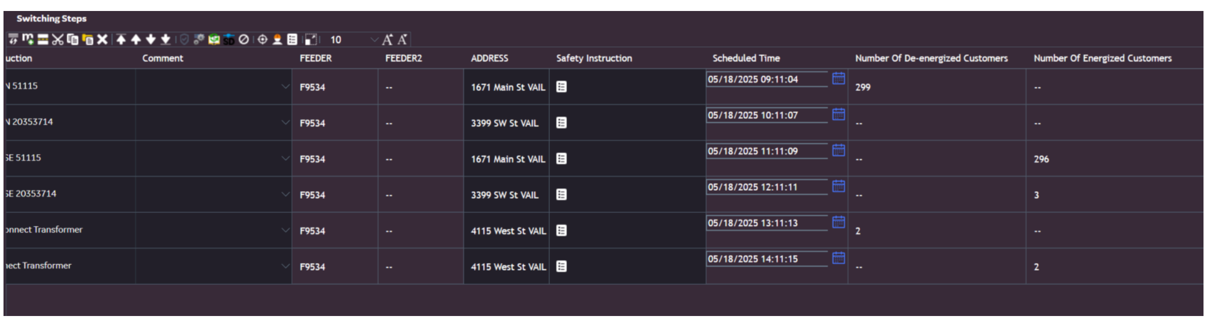
7.2. Recording Customers Affected by Steps in a Switching Order

The customers affected can be recorded for each switching step by following the below steps:

1. From the geographic view, enter study mode and initialize the station of the device of record to real-time.

   Then, in the switching order form, switch to ST mode.

2. Enter the scheduled time for the switching steps that de-energize or energize customers.
3. Arrange the switching steps according to their scheduled times.
4. Select all the steps, and click 'Implement Selected Steps' on the toolbar.
5. The system will record the customers affected by each switching action, and a count of affected customers will be shown for every step that energizes or deenergizes customers.

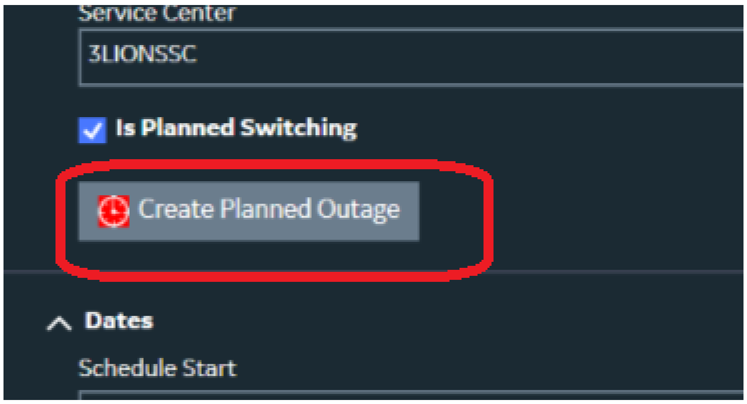


> **Note**
>
> If the switching steps are modified, the study should be reset and the steps reimplemented.

7.3. Creating a Planned Outage from a Switching Order

Once the customers affected are recorded for each switching step, a planned outage can be created that includes detailed records of the interruptions as a result of the switching steps.

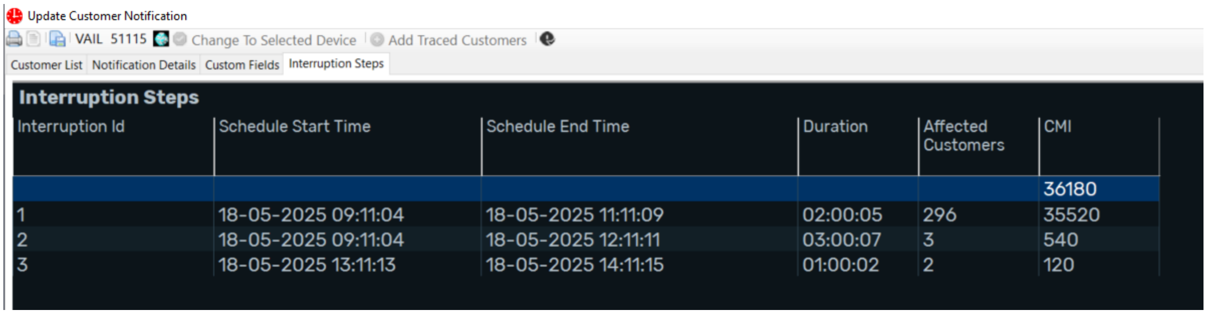
1. In the switching order form, click the Create Planned Outage button in the Request details section. This button appears only if the `Allow Planned Outage from SWO` is enabled in the switching order client configuration. Validation also ensures that the scheduled start date is in the future.

   > **Note**
   >
   > A planned outage will not be created for a switching order if there are customers left de-energized after the switching steps have been implemented.

   > **Note**
   >
   > All steps in the switching order must be implemented to ensure that all affected customers are accurately recorded for the order.

2. The planned outage will include detailed entries that capture the expected interruptions resulting from executing the switching steps at their scheduled times.
3. Each interruption has a start and an end time, a list of the customers who will be interrupted for that period of time, a duration and a corresponding CMI value.
4. In the switching order form, click the Update Planned Outage button in the Request details section. The associated planned outage opens.



> **Note**
>
> Details on how interruption records are displayed for planned outages can be found in the *Interruptions tab* section of *OMS User Guide*.

5. Edit the planned outage and send the planned outage notification, as appropriate.
6. If any changes are made to the switching order steps and saved, the corresponding details of the planned outage are automatically updated.

> **Note**
>
> When you delete a planned request document that is associated with an active planned outage, the system prompts you whether also you want to cancel the planned outage.

8. Executing and Completing a Switching Order

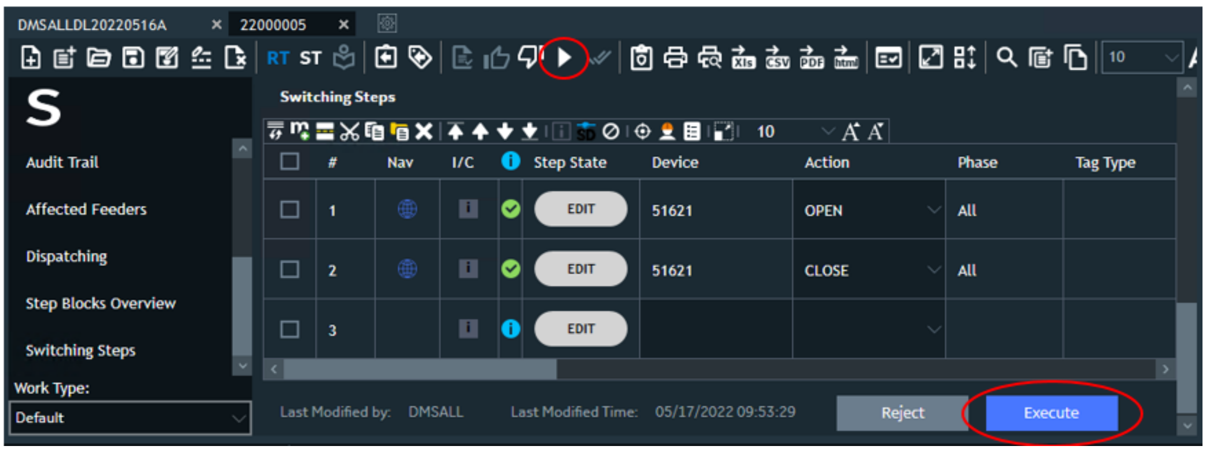
Once a switching order has been validated and approved, it can be executed on the real-time network.

To execute a switching order:

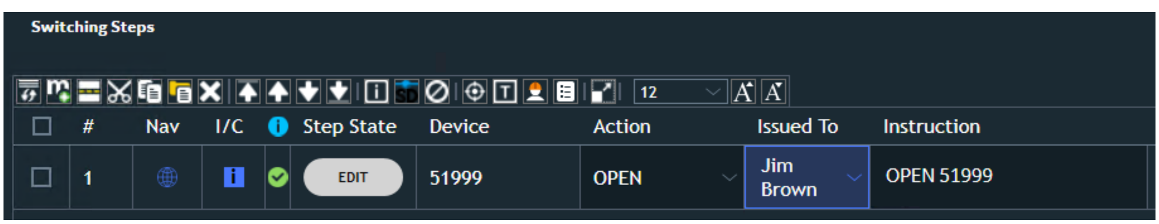
1. Open a switching order in the Approved state.

   The Execute button on the toolbar is enabled and the Execute button appears at the bottom of the switching order form, which indicates that the switching order is approved.

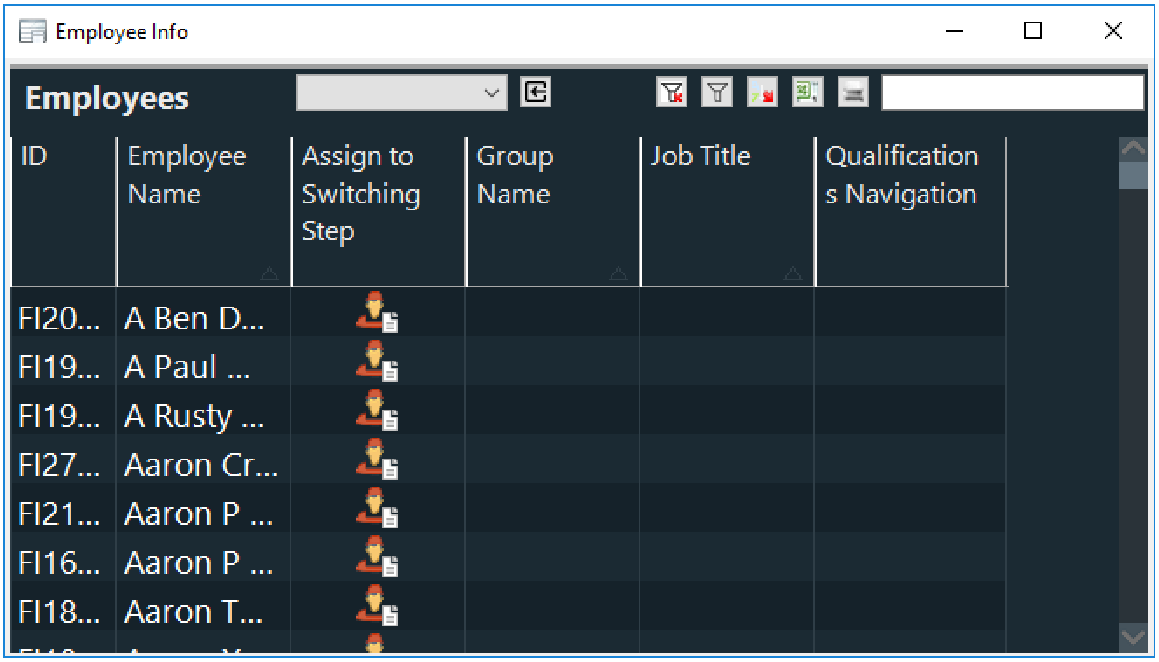
2. Click the Execute button to put the switching order into execute state.



3. To select an employee to perform the step's task, do one of the following:
   - Select an employee from the Issue To drop-down list.
   - Manually enter an employee name in the Issue To field.



   - Click the Employee Info Pop-Up button, and then select an employee from the Employee Info display.



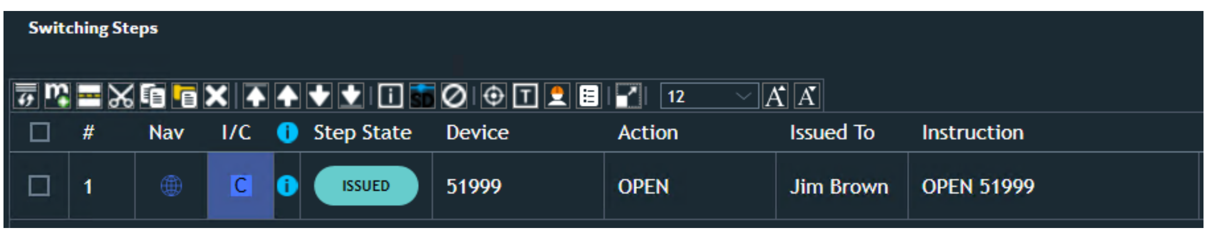
For more information about the Employee Info display, refer to Employee Info Display.

4. Issue a switching step by clicking the "i" button on the Switching Steps toolbar.

   If you issue a step before selecting an employee crew, a dialog box appears with a message to select the IssueTo option first. The issued step is highlighted with blue.

   The Issued Time field is populated automatically with the date and time when the step was issued.

   If a switching step results in de-energizing customers, a warning message appears with the number of customers.



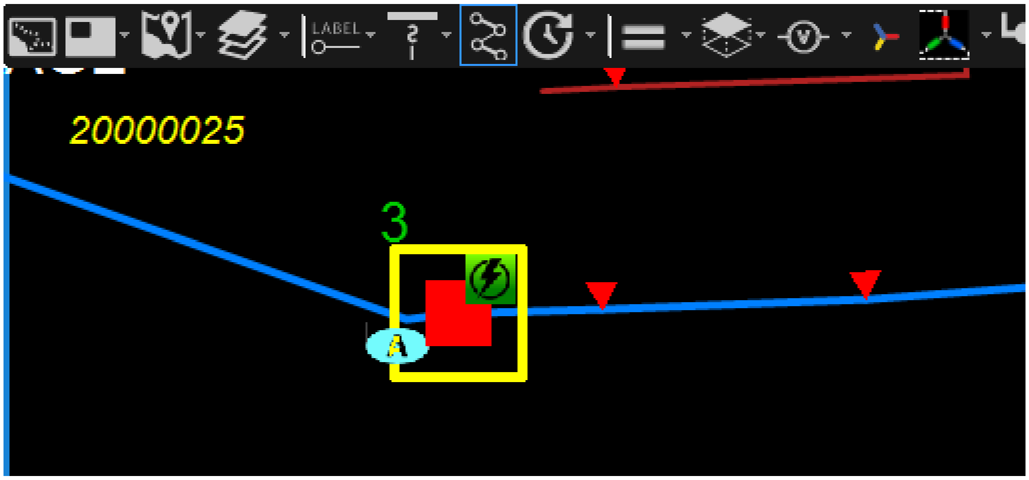
   "I" button for the switching order step changes to the "C" button, which indicates that the step is issued. The issued date and time are automatically entered. The Complete button for this step is highlighted.

   > **Note**
   >
   > The color coding of the Step State field background and text for different step states is configurable in the Switching Order Client Configuration Editor.

5. Make sure that the step action is completed in the field. If necessary, communicate with the

field crew that the switching device has been correctly identified and that the action to be performed is clearly understood.

Click the Track Devices button to show the devices on the network view. A new viewport is opened with the switching order number shown in the upper left corner. The devices are marked with yellow squares. Step numbers are green for devices where the switching action is Open and red for devices where the switching action is Close.



6. Implement the step action in the real-time network view.

   When the field crew reports that a step has been successfully completed, locate the device on the network view and update the device status accordingly.

   > **Tip**
   >
   > Steps for non-SCADA devices can be implemented directly from the Switching Order document in real-time mode. When the Complete button on the Switch Order form is clicked, the steps are automatically implemented and completed. (This is configurable.)

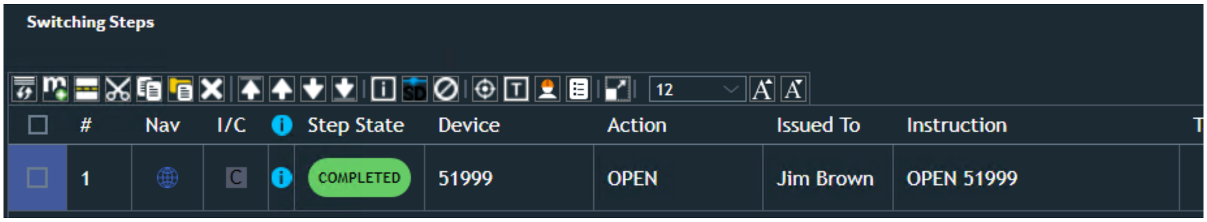
   To locate a non-SCADA device, click the Locate button .

   To locate a SCADA device, click the Locate button to open the network view, or click the Go to SCADA button to open the SCADA schematic view.

7. Mark the step as completed by clicking the "C" button C.

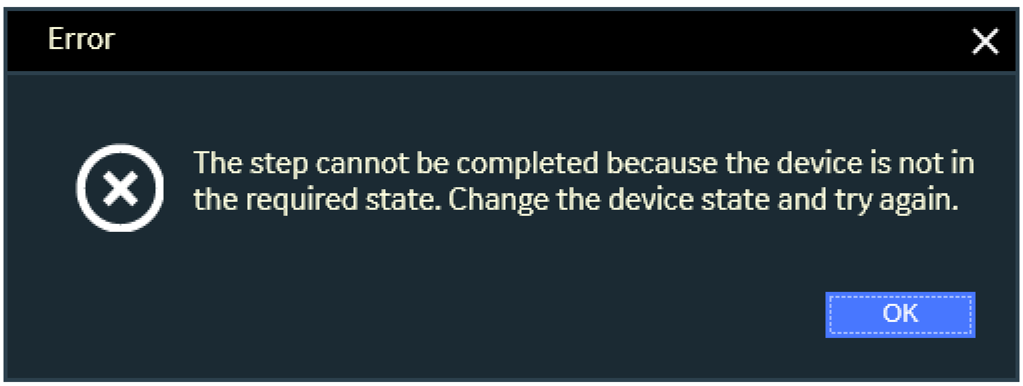
   For the completed step no further changes are possible.

   The completed time is filled in automatically with the date and time when the step was marked as completed.



To continue the process, select the next step in the switching order.

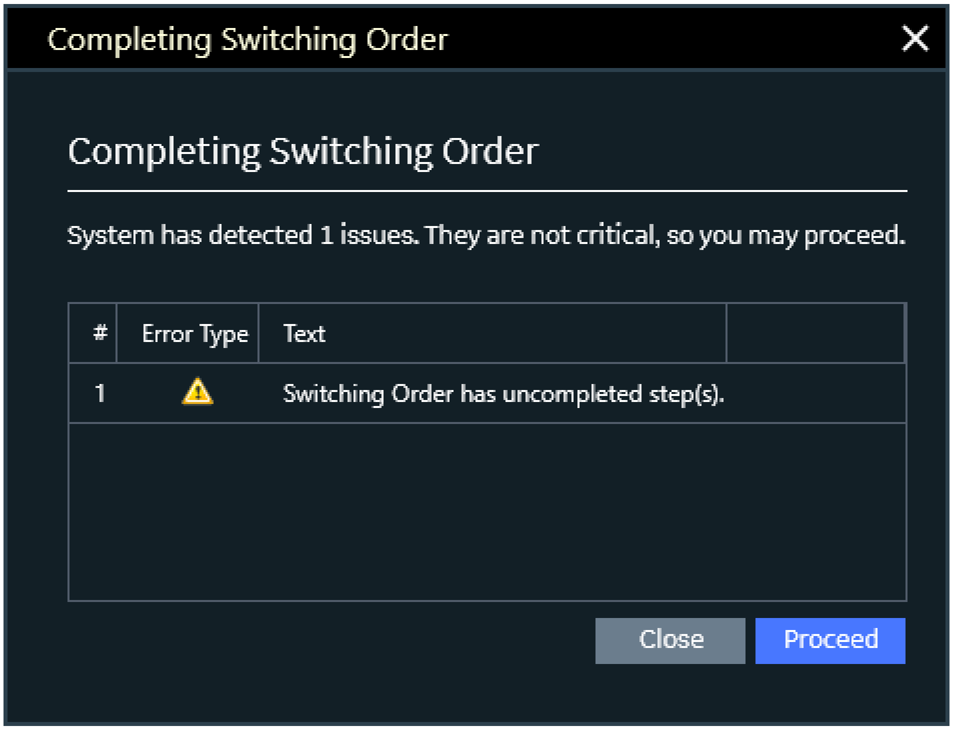
If there is an attempt to mark the step as completed without the switching being updated in the real-time network view, a warning message appears. (For non-SCADA devices, this depends on the option to automatically operate devices and complete steps by clicking the Complete button.)



For manually controlled SCADA devices, this behavior depends on the configuration setting that enables automatic operation and step completion. When enabled, clicking the C button updates the device status and completes the step automatically.

8. When all steps of the switching order are complete, update the switching order to Completed status by clicking the Switching Order Complete button on the Switching Order toolbar or the Complete button at the bottom of the Switching Order form.

   If there are steps not completed remaining, a warning message appears.



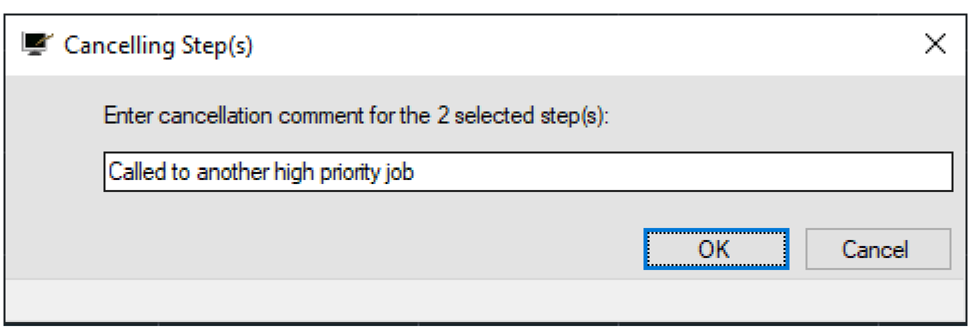
Once the switching order is updated to Completed status, no further changes are possible. The switching order becomes a read-only form. The Actual End field is populated with the date and time of completion.

8.1. Cancelling Steps

Steps in execution mode can be cancelled.

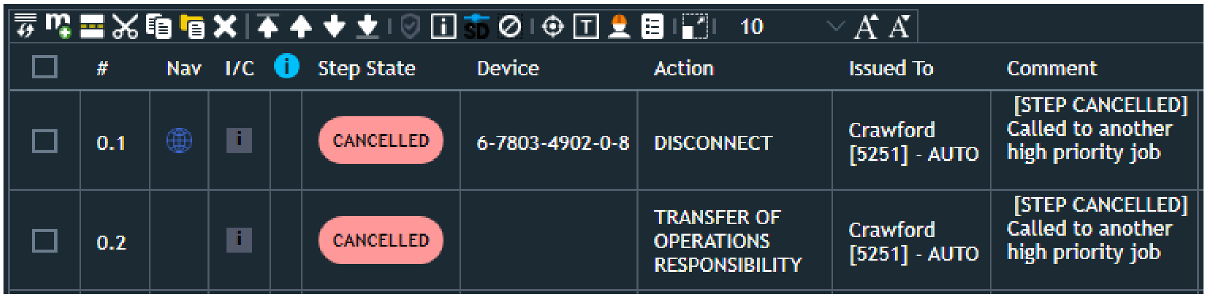
To cancel steps:

1. Select one or more uncompleted issued steps that needs to be cancelled.
2. Click Cancel on the Switching Steps toolbar or right-click the step and select Cancel Step from the menu.
3. Pop-up appears to enter a note describing the reason for the selected step's cancellation.



4. Click Ok.

The state of selected steps changes to Cancelled and no further changes are possible.



The note is updated in the comment field for all cancelled steps.

8.2. Tagging Steps

To complete a step that involves a tagging operation, manually place the tag in coordination with the physical placement of the tag by the field crew, and then mark the step as completed in the normal manner.

If the operator clicks the Complete button when the tagging action was not performed, the system issues the corresponding warning and doesn't allow the step to complete.

> **Note**
>
> When a tagging step is issued, there are options to automatically add or remove the specified tag type on the network view and bind a tag to a switching order or a responsible person by clicking the Complete button. For more information, refer to the *Switching Order Client Configuration Editor User Guide*.

8.3. Completing Comment or Manual Steps

To complete a comment or manual step, coordinate the actions with the field crew, update the real-time network view if necessary, and then mark the step as completed in the normal manner.

8.4. Resuming Execution of Saved Switching Orders

If there is a considerable time span between the completions of steps, it may be necessary to close the switching order form so that other operations can be performed.

To resume the execution of the switching order:

1. Click Switching Order Search on the Switch Order Form toolbar.
2. On the Switching Order Search form, set the Status filter to the Executing option.

   The switching orders appear on the Switching Order Search filtered by the Executing state.

3. Open the switching order that needs to be completed and proceed with the execution of the remainder of the order as usual.

8.5. Study Mode

Study mode allows network switching to be planned and evaluated in a separate environment that is independent of the real-time environment and is not visible by other users.

> **Note**
>
> To enable study mode functions, the study server may need to be manually started and initialized. For more information, see Initializing Study Mode.

Switching orders for planned operations are normally created in study mode. This allows the switching order to be created, tested, and modified as required, to verify that procedures are correct and that network limits are not exceeded. In addition, there is an option to record operator device operations as switching steps.

In study mode, the switching order can be executed directly. No approval step is required.

Study mode can be selected by clicking the Study Mode button ST on either the Switching Order or Daily Log toolbar. This connects the switching order form to the study mode server and opens a study view of the network (see the below image).

The switching order can be created and edited using the normal processes. However, execution of the steps can be performed directly by clicking the Implement button for a switching step or Implement Selected Steps button on the Switching Steps toolbar (these buttons appear in the study mode only). The switches in the study mode network view are automatically updated. Both SCADA and manual switches are operated.

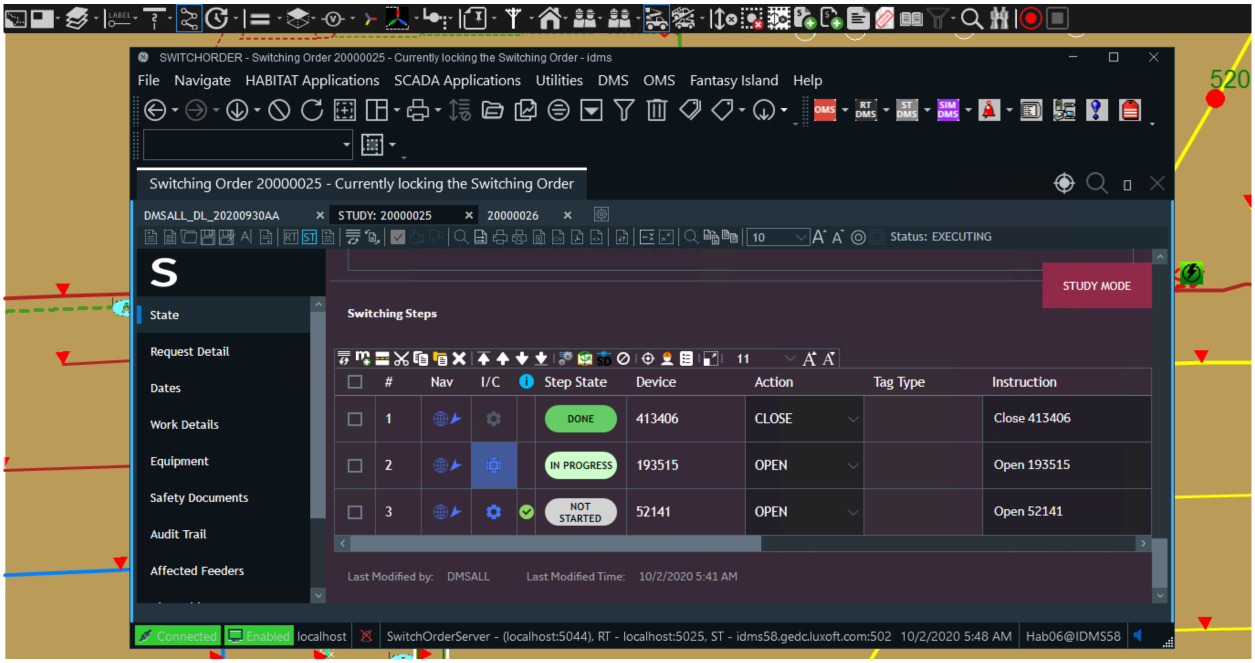


Figure 50. Study Mode Operation

The process executes one step in turn, to allow network analysis, such as running power flows, to be performed at any point in the procedure, to determine network loading and voltage levels.

The switching order effectively remains in edit mode, so that changes can be easily made and the steps can be re-run to check for problems. Once the switching order is complete, you should save it to the library so that it can be retrieved in the real time mode for the validation and approval process.

> **Note**
>
> In study mode, there is no difference between SCADA and manually operated switches. They all behave as manual switches.

Completed and saved switching orders can be retrieved and replayed in the study mode. You can reset all steps to their initial states using the Clear Implemented Steps in ST Mode button on the switching steps toolbar and then, if needed, repeat the steps.

> **Note**
>
> The steps are reset within the switching order, but not in the ST Mode of the network view. To reset the steps in the network view, click the Initialize Study Environment button on the Switching Order toolbar.

8.6. Dispatching a Switching Order

> **Note**

This feature requires integration with GE's Mobile Enterprise solution.

In the Switching Order Dispatching section, you can dispatch switching orders to crews within the Mobile Enterprise system.

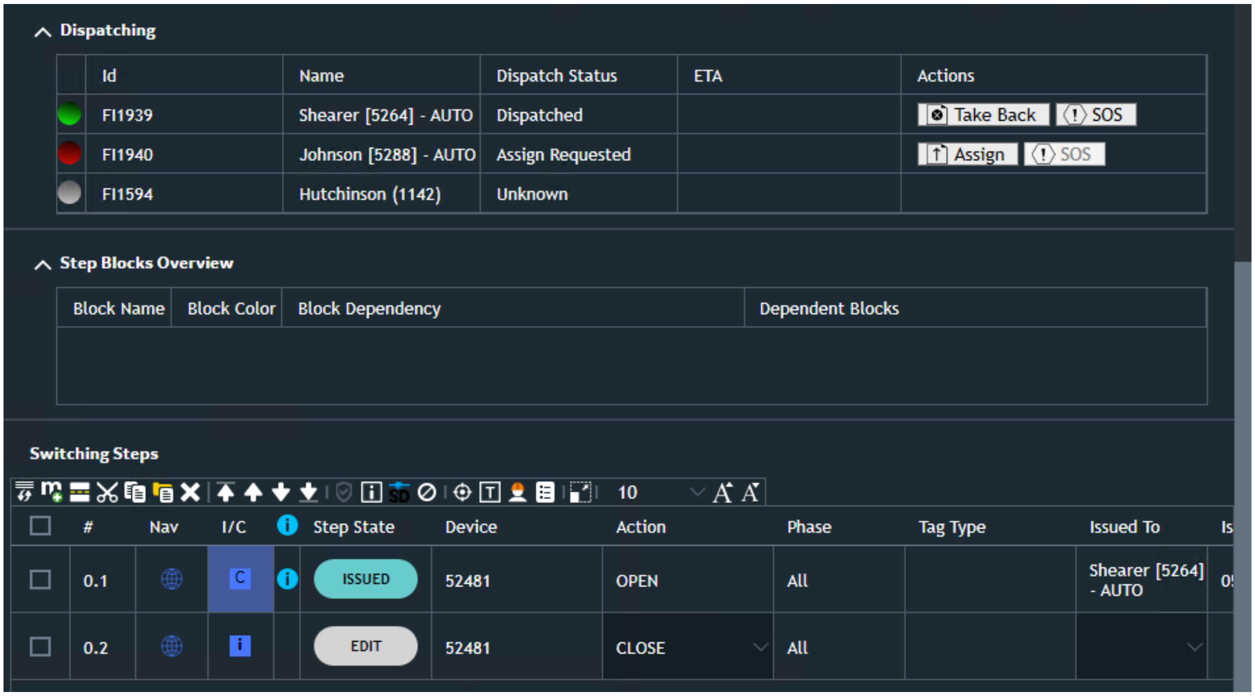


Figure 51. Switching Order Dispatching

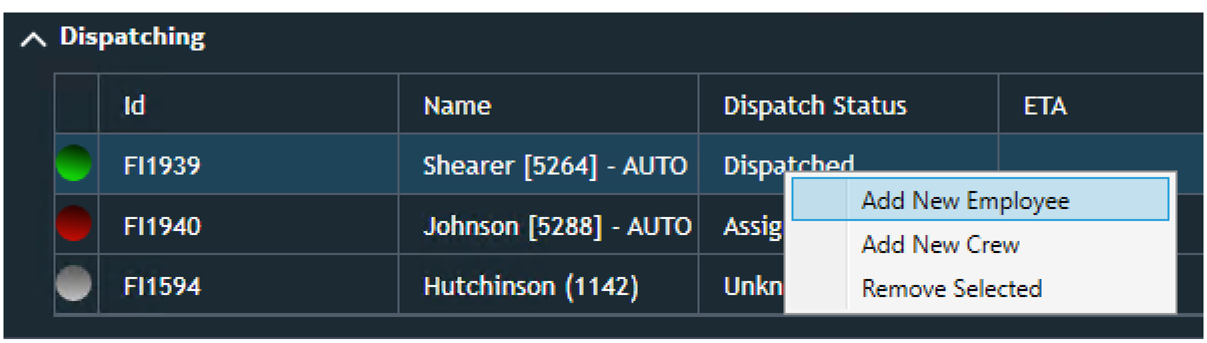
8.6.1. Adding and Removing Field Employees for Dispatching

To add an employee to the Dispatch Employee list from the Switching Steps section:

- Issue a step to an employee.

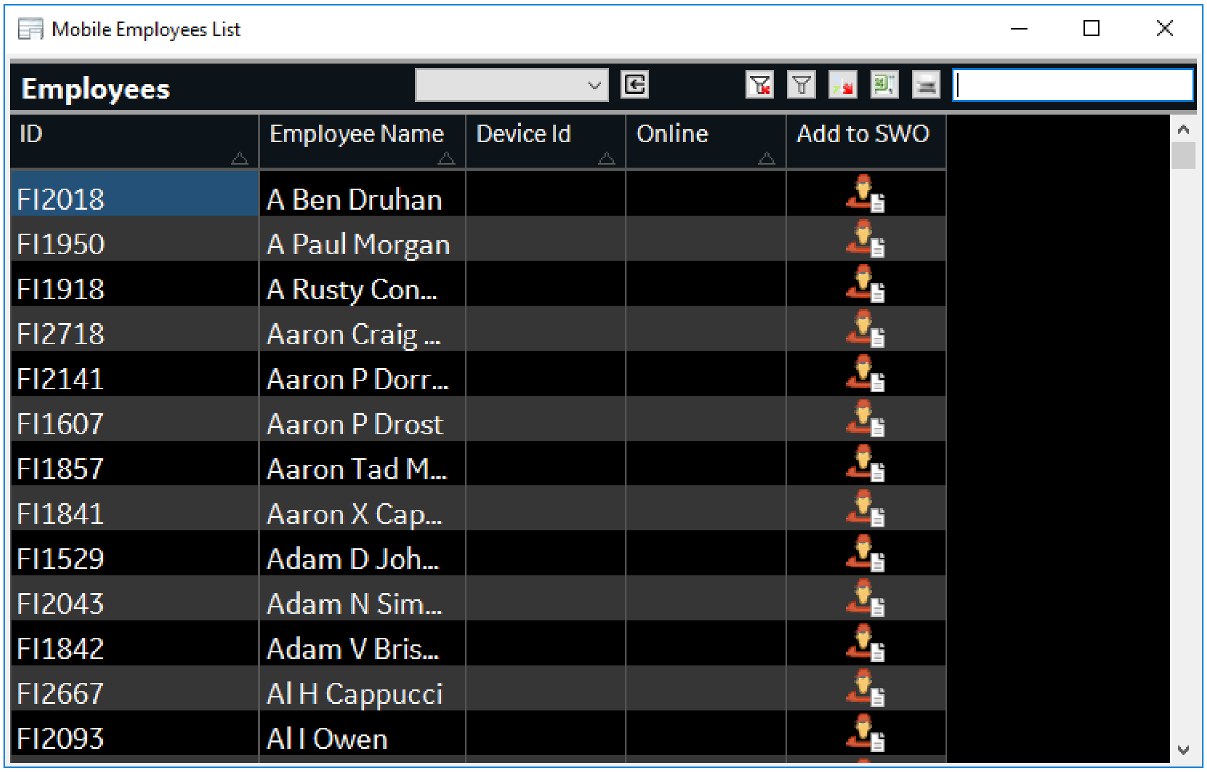
To add an employee to the Dispatch Employee list from the Switching Order Dispatching section:

1. Right-click the Dispatch Employee list.



2. From the context menu, select the Add New Employee option.

The Mobile Employees List display appears.



3. In the Mobile Employees List table, click the Add to SWO button for the required employee.

   The employee appears in the Dispatch Employee list in the Switching Order Dispatching section.

To add a crew to the Dispatch Employee list from the Switching Order Dispatching section:

1. Right-click the Dispatch Employee list.
2. From the context menu, select the Add New Crew option.

   The Mobile Crews List display appears.

3. In the Mobile Crews List table, click the Add to SWO button for the required crew.

   The crew appears in the Dispatch Employee list in the Switching Order Dispatching section.

To remove an employee or crew from the Dispatch Employee list:

1. Right-click an employee or crew in the Dispatch Employees list.
2. Select the Remove Selected option.

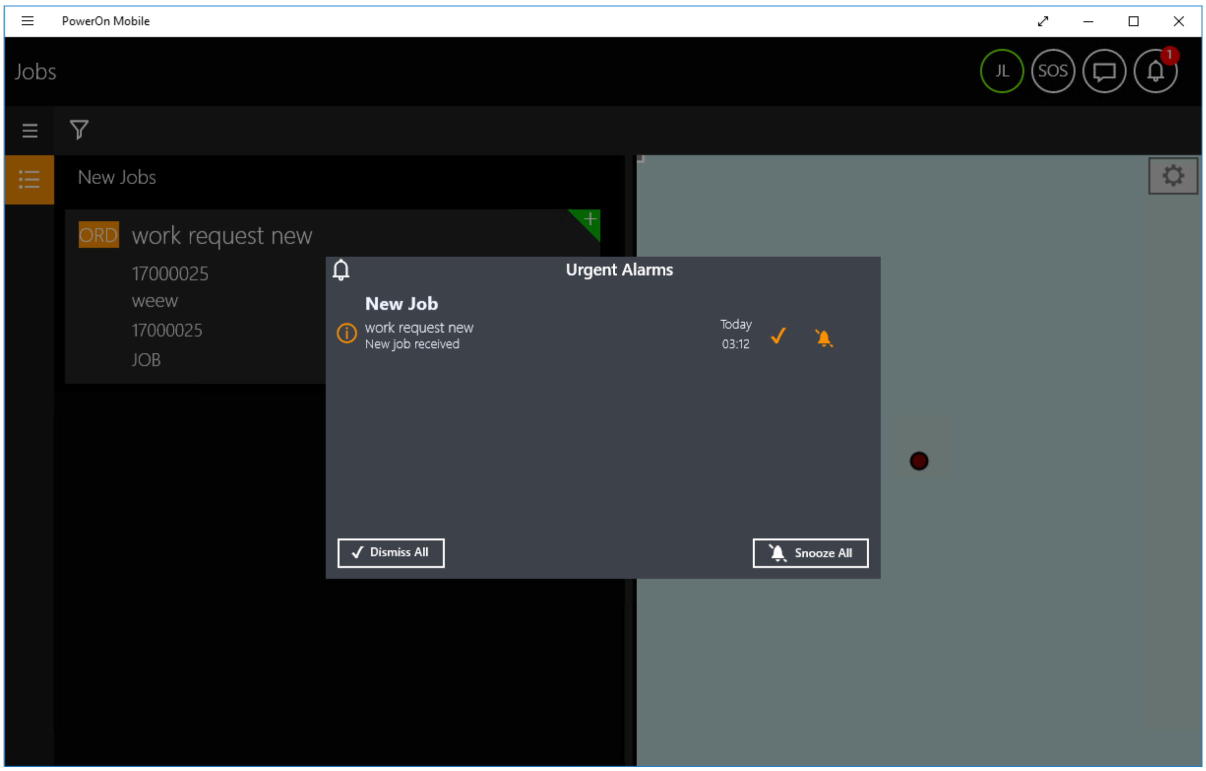
   The employee or crew is removed from the Dispatch Employee list in the Switching Order Dispatching section.

8.6.2. Dispatching and Taking Back a Switching Order

To dispatch a switching order to an employee:

1. In the Switching Order Dispatching section, add employees equipped with the Mobile application.
2. For the required employee, click the Dispatch button.
   - The Dispatch Status changes to Dispatched for logged on employees.
   - The Dispatch Status changes to Queued for Dispatch for logged off employees.

   When a switching order is dispatched, a notification appears in the employee's Mobile application.

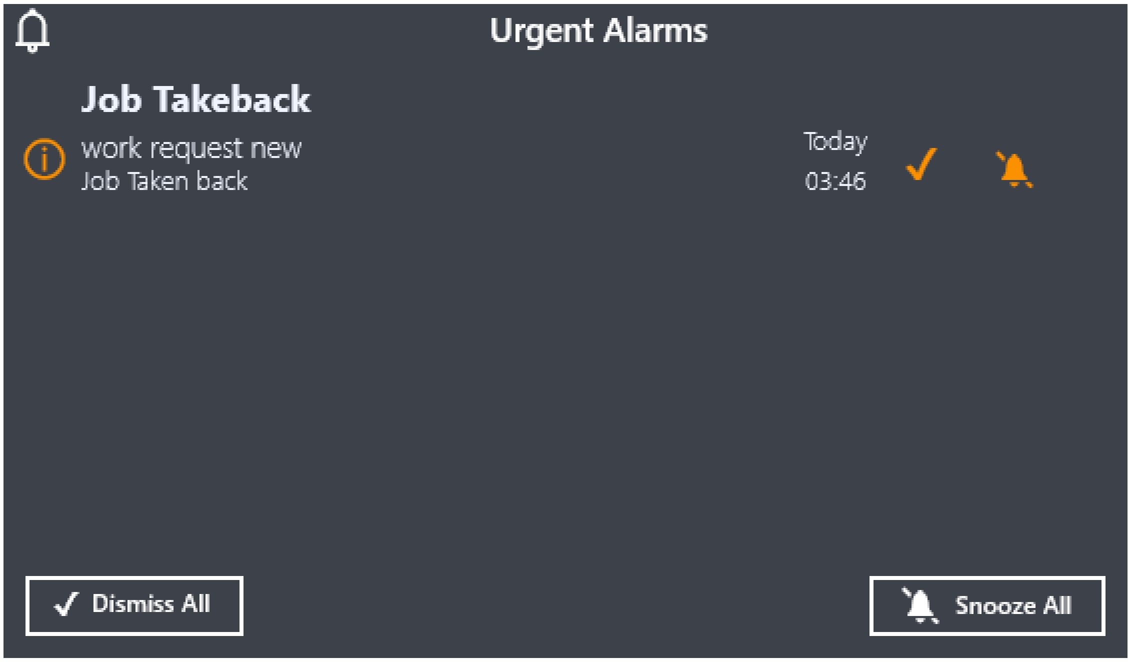


To take back a switching order from an employee:

- In the Switching Order Dispatching section, for the required employee, click the Dispatch button.

  The Dispatch Status changes to Taken Back.

  When a switching order is taken back, a notification appears in the employee's Mobile application.



8.6.3. Working with Dispatched Switching Orders

The Mobile application is designed to assist field crews in working with switching orders. For more information, refer to the Mobile Enterprise documentation.

8.6.4. Switching Order Dispatching Display

The Dispatcher display provides an overview of the current switching order dispatching state and allows the operator to dispatch and take back switching orders:

- The Switching Orders tab provides details for currently open switching orders
- The Employees tab provides details for employee dispatching

To access the display, click the Dispatcher Display field on the Switching Order Options tab. Alternatively, enter NETVIEW://DD.RT/ or NETVIEW://RT/DISPATCHER.

The filters on the Switching Orders and Employees tabs allow filtering the data by the specified criteria. You can also sort the data in the grid and change the column order as appropriate.

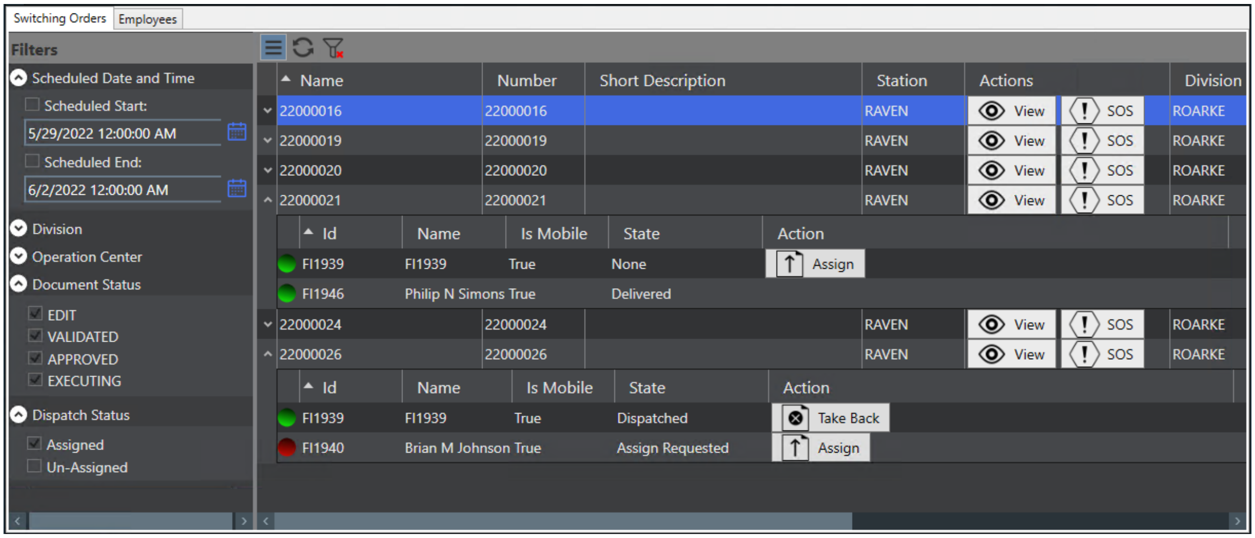


Figure 52. Dispatcher Display – Switching Orders Tab

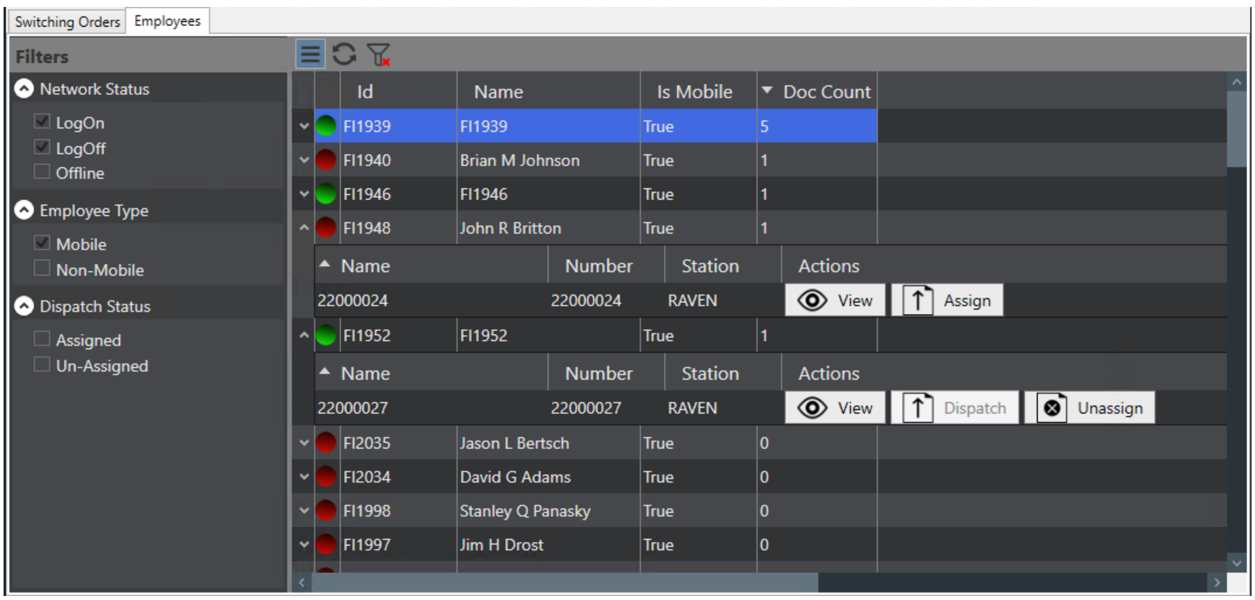


Figure 53. Dispatcher Display – Employees Tab

The Dispatcher display provides the following controls:

- **Dispatch:** Assigns a switching order to an employee.
- **Takeback:** De-assigns a switching order from an employee.
- **View:** Opens the Switching Order document for review.
- **SOS:** Issues the SOS signal for an employee to immediately stop working on the switching order.

8.6.5. Mobile Alarms

Mobile alarms appear as toast notifications in the lower-right corner of the ADMS client to inform the operator about switching order updates in the Mobile Enterprise system. For example, messages appear when a field crew is assigned to a switching order, completes a step, sends a message to the operator for feedback, or when a request failed to be delivered to the Mobile Enterprise system.

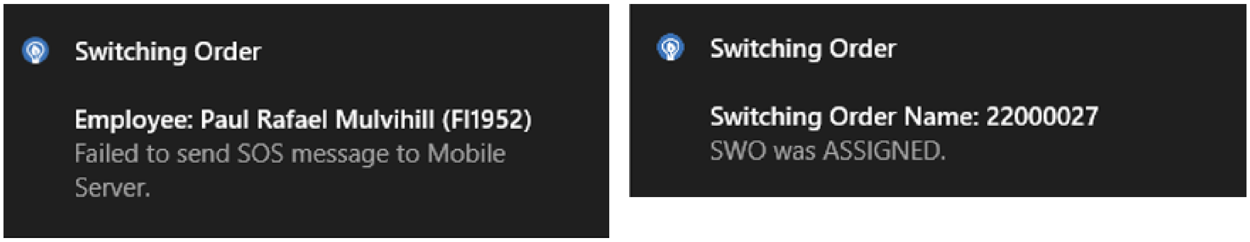


Figure 54. Mobile Alarms

9. Creating a Go Back Order

Normally, a switching order to restore a section of network that has been taken out of service as a planned outage is simply the reverse of the original switching order.

To create a go back switching order:

1. Ensure that the switching order to take the network components out of service is complete and loaded in the switching order form.
2. Select the steps to be included in the go back order.
3. Click the Open New Return to Normal Switching Order button on the Switching Order toolbar.

   A new switching order is created on a separate tab.
4. Review the switching order, and check that the action of each step has been correctly inverted and that the overall order of the steps has been reversed.
5. Add or modify steps as needed to complete the go back procedure.

To expedite the creation of a "go back" switching order, a second switching order can be created as a copy of the original, and then reversed with the step actions inverted. In most cases, only minor editing is required to complete the go back procedure.

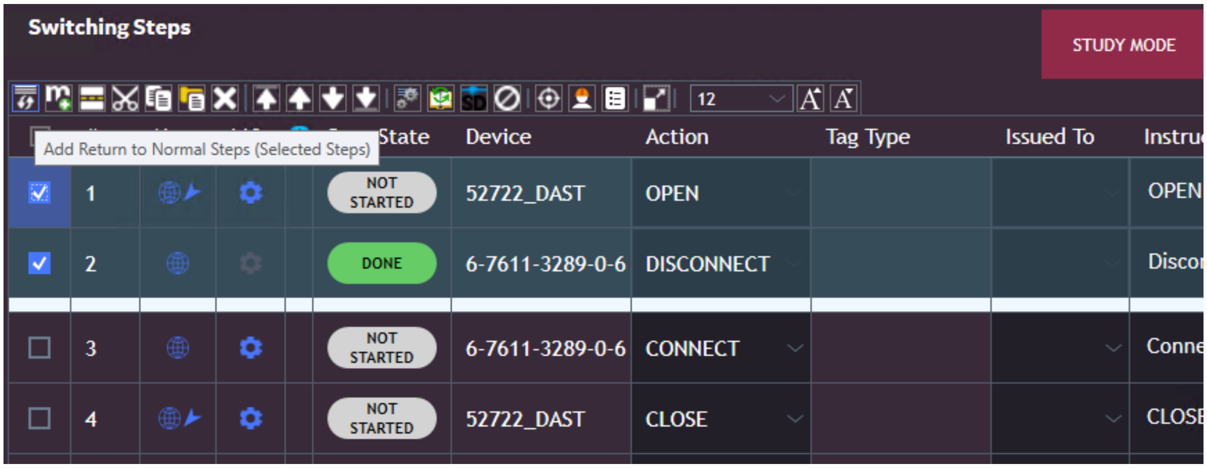
9.1. Adding the Return to Normal Steps

After completing a switching operation, you may need to add the return steps to restore the normal state of the network.

To create the Return to Normal steps and add them to the currently opened Switching Order:

1. Ensure that the switching order to take the network components out of service is complete and loaded in the switching order form.
2. Select the steps to be reversed and added to the current order.
3. Click the Add Return to Normal Steps button on the Switching Steps toolbar.

   The selected switching steps are copied and added with the reverse action to the same switching order.



4. Review the switching order, and check that the action of each step has been correctly inverted and that the overall order of the steps has been reversed.
5. Add or modify steps as needed to complete the Return to Normal procedure.

9.2. Clearance Steps for Actions with Tags

In some cases, you may need to only reverse actions that include tagging operations (i.e., close the switch and remove the applied tag, or restore the removed tag).

The Open New Switching Order with Clearance Steps option is different than the Open New Return to Normal Switching Order option, as the steps without tagging actions are ignored and not included in the new clearance switching order.

To create a clearance switching order:

1. Ensure that the switching order to take the network components out of service is complete and loaded in the switching order form.
2. Select the steps to be included in the new clearance order.
3. Click the Open New Switching Order with Clearance Steps button on the Switching Order toolbar.

   A new switching order is created on a separate tab. This switching order contains reverse steps only for steps that include tagging actions. Steps without tagging actions, even if selected, are not included in the clearance switching order.
4. Review the switching order and check that the action of each step has been correctly inverted and that the overall order of the steps has been reversed.
5. Add or modify steps as needed to complete the clearance procedure.

10. Daily Log

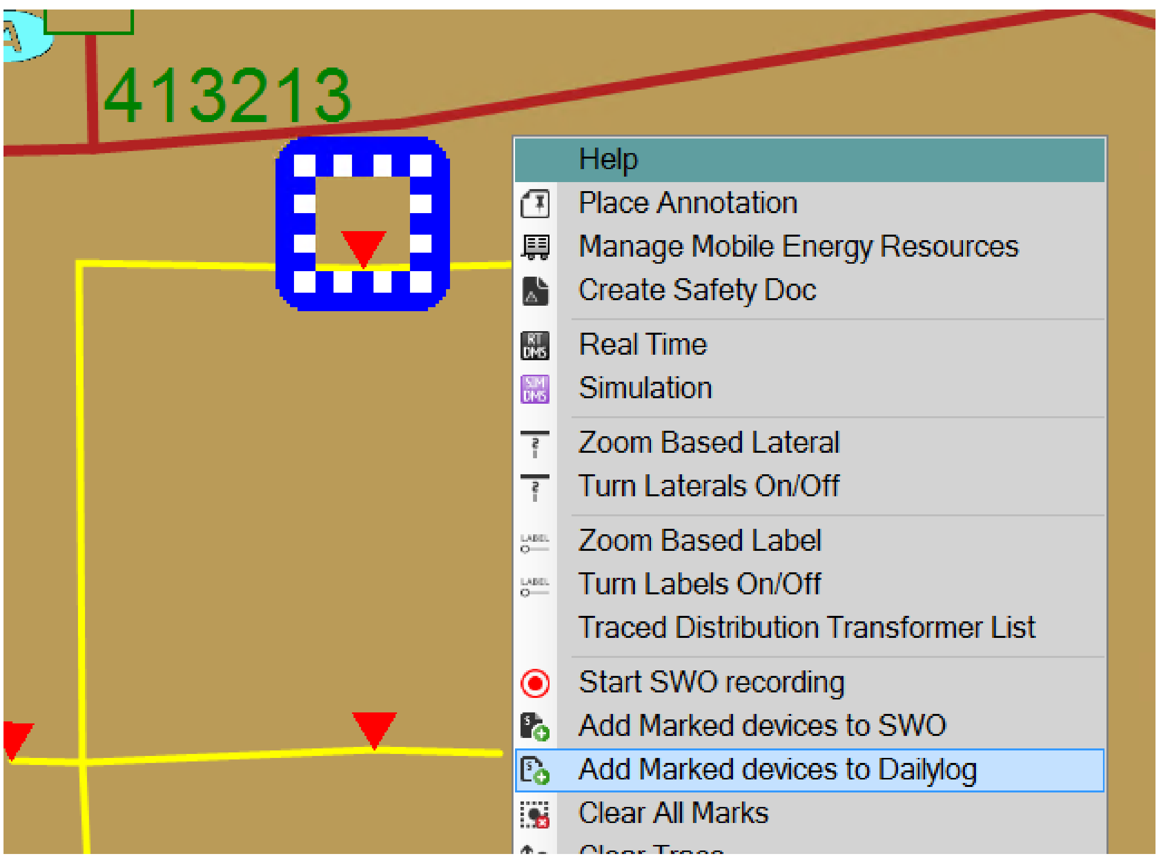
The daily log allows you to record and manage individual switching operations on an adhoc basis. These are switching operations that are not part of a formal switching order. The switching steps can be placed in the daily log one by one and executed immediately in any order. There is no validation or approval process.

The daily log appears on a separate tab on the Switching Order form. Therefore, it can be used concurrently with formal switching orders.

The daily log is opened automatically when the first switching step is selected to be added to the daily log. The log is maintained in the name of the operator and continues until that operator clicks the Complete button on the Daily Log toolbar. When the Complete button is pressed, the Daily Log becomes a read-only form. To add and manipulate more steps, a new Daily log must be started.

To execute the Daily Log operations:

1. Add switching steps to the daily log:
    a. Select a device on the network view and click Add to Daily Log on the toolbar or from the right-click menu.
    b. Mark devices on the network view using the Mark Switch or Mark All Potential Isolating Devices option. Right-click anywhere on the network view and select the Add Marked Devices to Daily Log option.



c. Manually enter steps into the daily log.

d. Copy and paste switching steps from a switching order into the daily log.

2. Before issuing the step, select the crew to perform the step's task from the IssueTo drop-down list.

3. To issue the required step, click the "I" button.

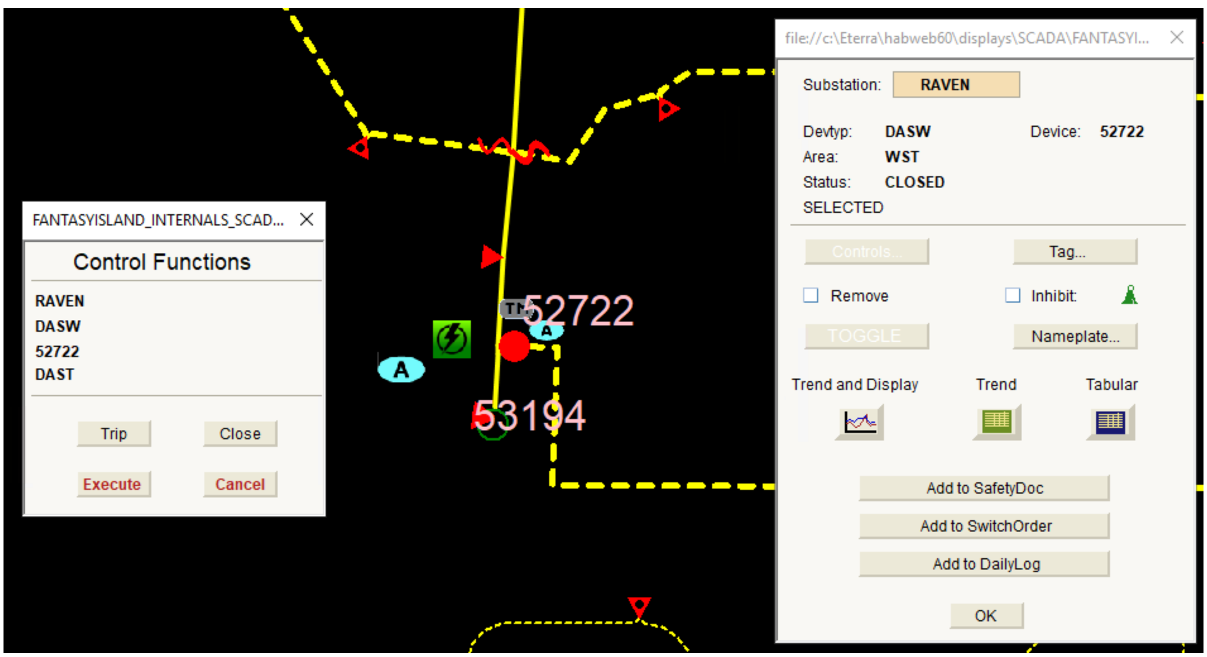
   If you click the "I" button before you select the crew, a dialog box appears with a message to select the IssueTo option first.

4. If necessary, communicate with the field crew that the switching device is correctly identified and that the action to be performed is clearly understood.

   The "I" button is shaded to indicate that it is issued. The issued date and time are automatically entered.

5. When the field crew reports that the step is successfully completed, update the real-time network view accordingly.

   In the case of a SCADA-controlled switch, the operator performs the operation directly in the real-time network view.



6. If the actual completion time needs to be corrected to allow for a delay in the field crew reporting the operation, after the step is issued, click the Actual End field to modify the date and time to correspond to the actual completion time.

7. Enter the new value and click OK.

8. Mark the step as complete by selecting the "C" box.

   No further changes to this step are possible.

If the operator attempts to mark the step as complete without the switching being updated in the real-time network view, a warning message appears.

9. At the end of a shift or when the Daily Log is overpopulated with completed steps, update the Daily Log to Completed status by clicking the Complete button on the Daily Log toolbar.

   Once the Daily Log is updated to Completed status, no further changes to this Daily Log are possible.

   To add and edit more switching steps, open a new Daily Log as described in step 1.

10.1. Study Mode

You can retrieve and replay completed and saved daily logs in study mode.

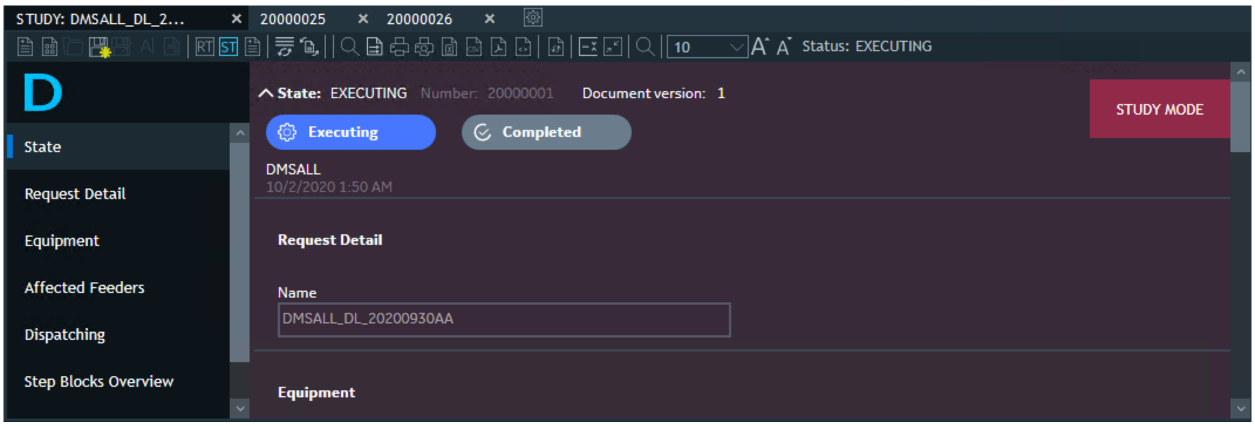


Figure 55. Daily Log in the Study Mode

You can reset all steps to their initial states by clicking the Reset Executed Steps button on the Switching Steps toolbar and then, if needed, re-executing the steps.

11. Creating a Safety Document

Safety documents are the formal means of passing jurisdiction for a defined section of the network from the operator to another group such as a maintenance crew.

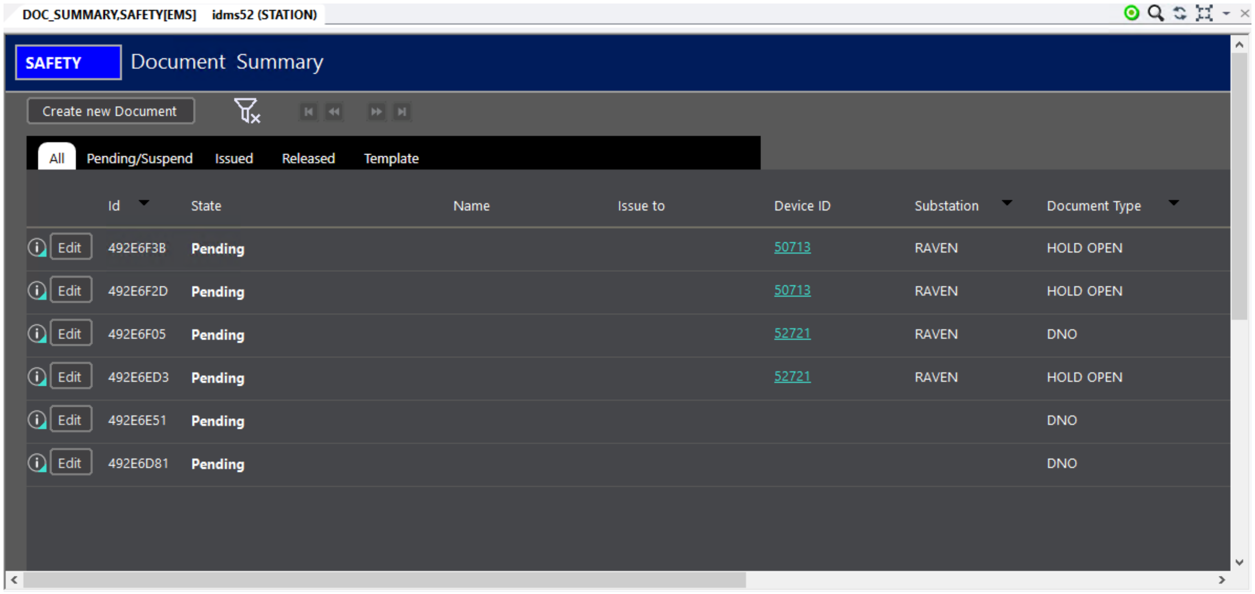


Figure 56. Safety Document Summary

When the switching is performed to isolate the required section of the network and make it safe for a planned maintenance activity, tags are applied to switches and other devices to prevent unauthorized re-energization. These tags are registered on the safety document to define the bounds of the section of network that is being passed to the field crew for the period of the outage to ensure that the tags cannot be removed until the safety document is released.

This function is only available after the system is configured to interact with the Safety Documents application.

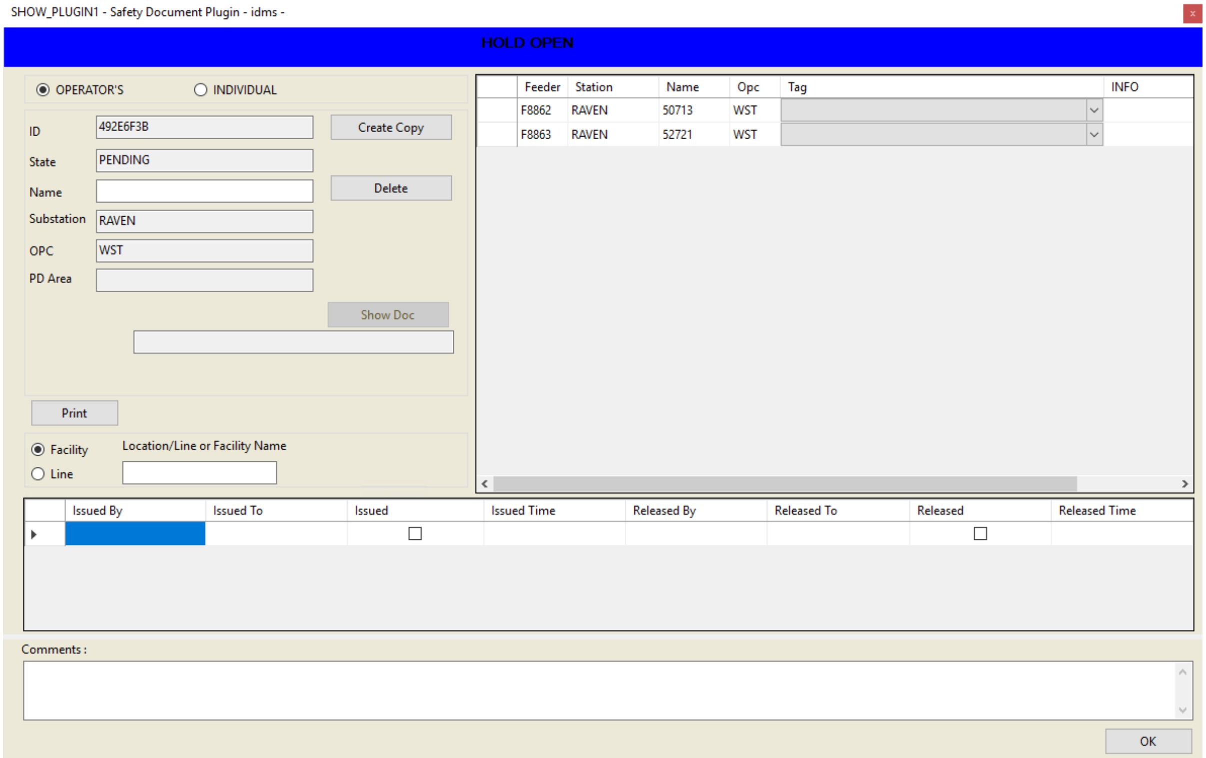
To support the requirements of various maintenance and testing tasks, a number of types of safety documents are defined.

To copy selected steps on the Switching Order Form to a Safety Document application:

1. Select the steps that require tagging and are to be copied to the Safety Document application.
2. Select the tagging type to be applied to each step.
3. Click the Create Safety Document from Switching Order button. If no tagging type is selected, a warning message appears.

   Multiple selected steps must have the same tag type.

   The selected steps are now copied to the safety documents.
4. The Safety Document form automatically opens and the required tag type is populated with the devices that were selected in the Switching Order form.


SHOW_PLUGIN1 - Safety Document Plugin - idms -
HOLD OPEN
OPERATOR'S
INDIVIDUAL
ID
492E6F3B
Create Copy
State
PENDING
Name
Delete
Substation
RAVEN
OPC
WST
PD Area
Show Doc
Print
Facility
Line
Location/Line or Facility Name
Feeder
Station
Name
Opc
Tag
INFO
F8862
RAVEN
50713
WST
F8863
RAVEN
52721
WST
Issued By
Issued To
Issued
Issued Time
Released By
Released To
Released
Released Time
Comments :
OK

12. Employee Info Display

When issuing a switching step, the Employee Info display allows an operator to choose from a much larger number of employees using the sorting and filtering capabilities of the grid tabular displays. Alternatively, depending on the configuration, a Crew List display is called, which lists crews from the OMS crew model and allows an operator to assign a crew for a switching step.

The Employee Info button visibility is configurable. For more information, refer to the *Switching Order Client Configuration Editor User Guide*.

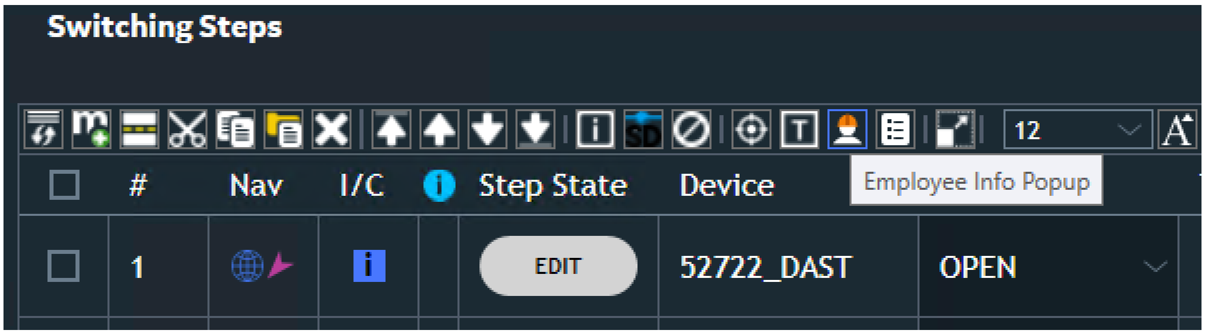


Figure 57. Employee Info Pop-Up Button

Click the Employee Info Popup button to view the Employee Info display or Crew List display (depending on the configuration).

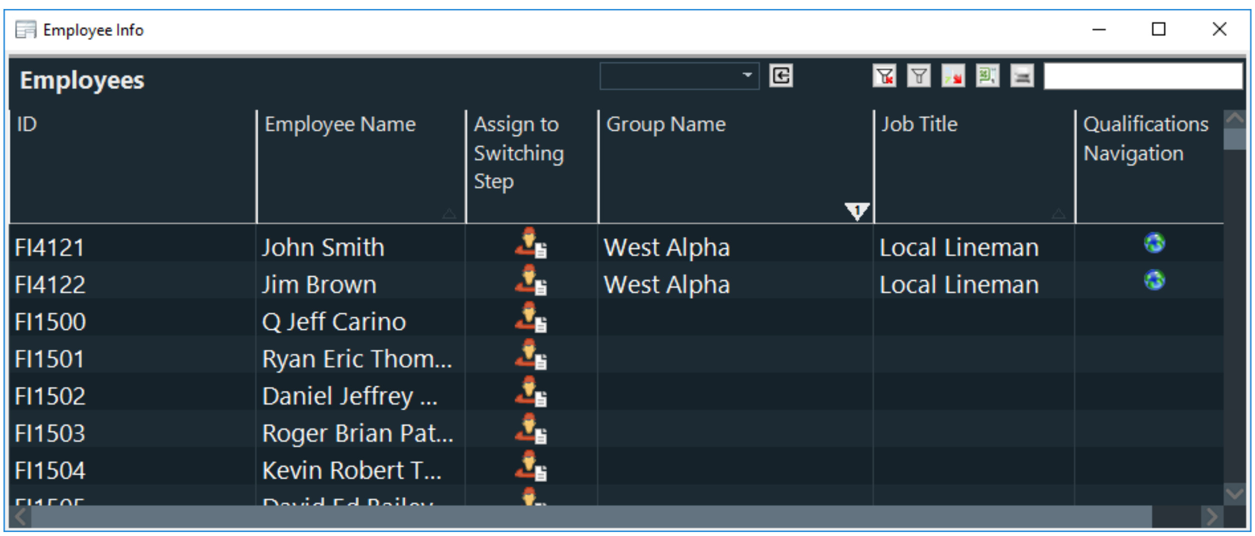


Figure 58. Employee Info Display

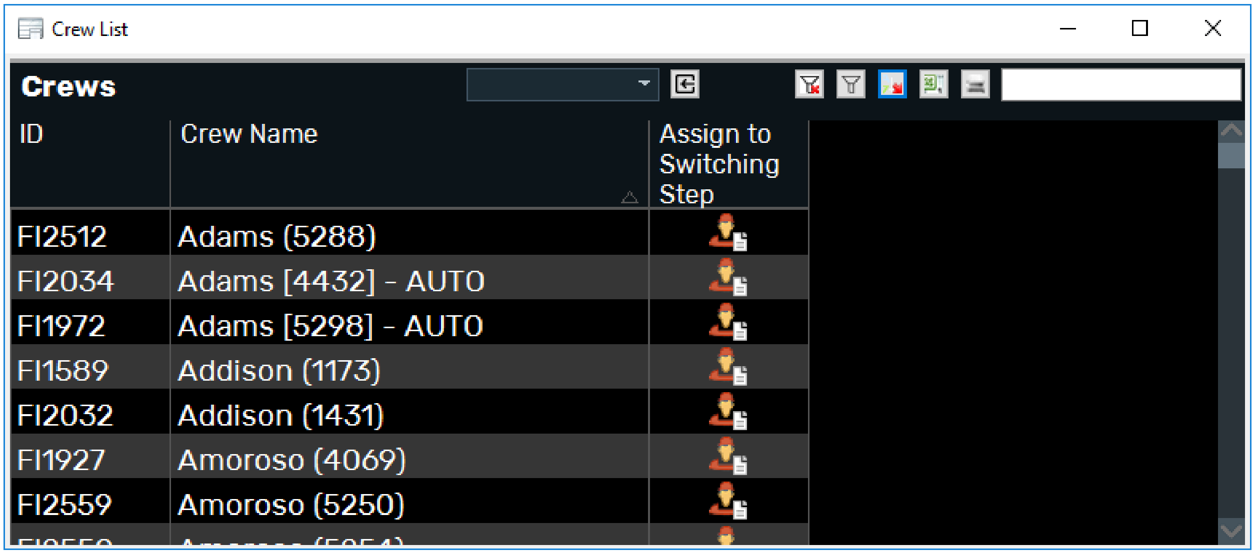


Figure 59. Crew List Display

The Employee Info display contains the following fields:

- **ID:** Employee ID.
- **Employee Name:** Full name of the employee.
- **Assign to Switching Step:** Click this button to assign the employee to the switching step.
- **Group Name:** Name of the working group the employee belongs to.
- **Job Title:** Employee's job title.
- **Qualifications Navigation:** Navigates to the Employee Qualifications display.

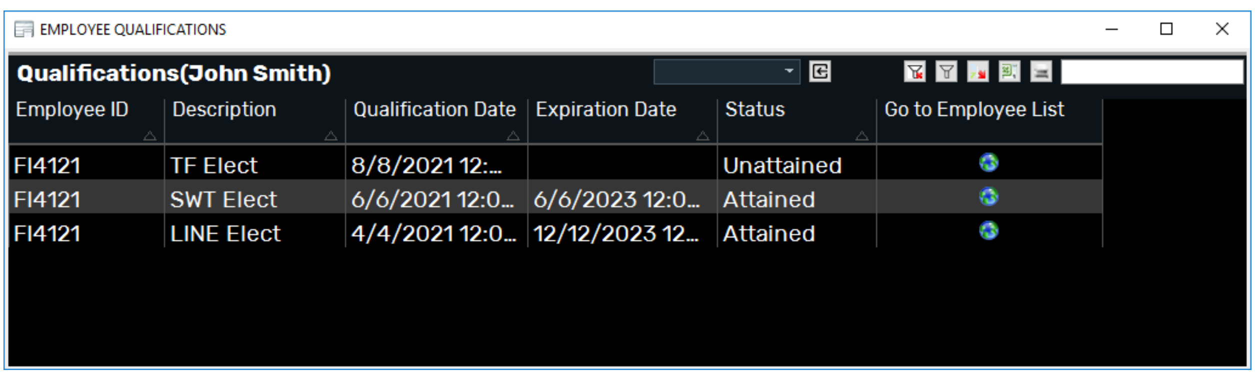


Figure 60. Employee Qualifications Display

The Employee Qualifications display contains the following fields:

- **Employee ID:** Employee ID.
- **Description:** Name of the qualification.
- **Requalification Date:** Date and time when the qualification was passed.
- **Expiration Date:** Date and time of the qualification expiration.

- **Status:** Status of the qualification (Attained/Unattained).
- **Go to Employee List:** Navigates back to the Employee Info display.

For more information about configuring and populating the Employee Info and Employee Qualifications displays, refer to the *Switching Order Server Configuration Editor User Guide*.